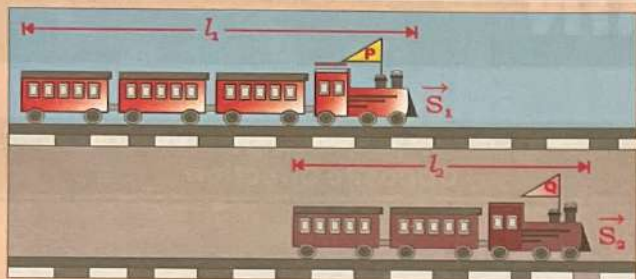


Train

Two trains crossing each other in same direction



Two trains, P and Q of length ' l_1 ' m and ' l_2 ' m respectively, are moving with respective speed of ' S_1 ' m/s and ' S_2 ' m/s (where, $S_1 > S_2$).

Time taken by faster train P to pass slower train Q when they are moving in same direction:

$$\text{Time} = \frac{l_1 + l_2}{S_1 - S_2}$$

Ex. Two trains of length 200 m and 250 m are running on parallel rail tracks at the speed of 68 km/h and 50 km/h, respectively. In how much time will the faster train cross the slower train if both trains are running in the same direction?

HINTS $l_1 = 200$ m, $l_2 = 250$ m

$$S_1 = 68 \text{ km/h}, S_2 = 50 \text{ km/h}$$

$$\begin{aligned} \therefore \text{Time} &= \frac{l_1 + l_2}{S_1 - S_2} \\ &= \frac{200 + 250}{(68 - 50) \times \frac{5}{18}} \\ &= \frac{450}{18 \times \frac{5}{18}} = 90 \text{ seconds} \end{aligned}$$

Ex. The lengths of two trains are 380 m and 220 m. The faster of these two trains takes 20 seconds to overtake the other, when travelling in same direction. The trains take 12 seconds to cross each other, when travelling in opposite directions. What is the speed (in km/h) of the faster train?

$$T_1 \xrightarrow{380 \text{ m}} x \text{ km/h} \rightarrow 20 \text{ sec}$$

$$T_2 \xrightarrow{220 \text{ m}} y \text{ km/h}$$

$$T_1 \xrightarrow{380 \text{ m}} x \text{ km/h}$$

$$y \text{ km/h} \xleftarrow{220 \text{ m}} T_2$$

$$x - y = \frac{600}{20} = 30 \text{ m/s} \dots (i)$$

$$x + y = \frac{600}{12} = 50 \text{ m/s} \dots (ii)$$

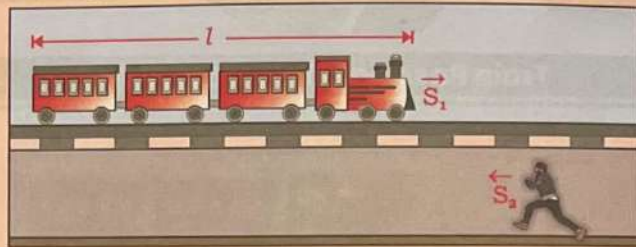
From eqn (i) & (ii)

$$x = 40 \text{ m/s}$$

$$\Rightarrow x = 144 \text{ km/h}$$

When a train crosses 1 or 2 running persons

Case I: A train of length ' l ' m is moving with a speed of ' S_1 ' m/s. A person is moving with a speed of ' S_2 ' m/s in opposite direction of the train.



Time taken by them to pass each other:

$$\text{Time} = \frac{l}{S_1 + S_2}$$

Ex. A 1800 m long train crosses a man walking in opposite direction. The speed of the train and the man is 160 m/s and 20 m/sec, respectively. How much time will the train take to cross the man?

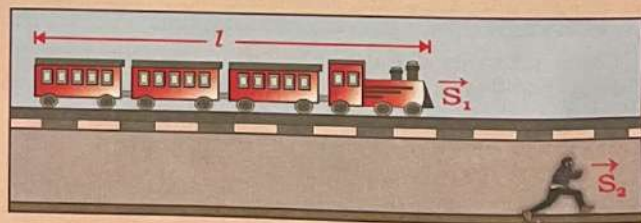
HINTS

$$l = 1800 \text{ m}$$

$$S_1 = 160 \text{ m/s}, S_2 = 20 \text{ m/s}$$

$$\therefore \text{Time} = \frac{l}{S_1 + S_2} = \frac{1800}{160 + 20} = \frac{1800}{180} = 10 \text{ seconds}$$

Case II: A train of length ' l ' m is moving with a speed of ' S_1 ' m/s. A person is moving with a speed of ' S_2 ' m/s in same direction of the train.



Time taken by the train to pass the person:

$$\text{Time} = \frac{l}{S_1 - S_2}$$

Ex. A 228 m long train is running at a speed of 125 km/h. In what time will it pass a man running at 17 km/h in the same direction in which the train is going?

HINTS

$$l = 228 \text{ m}$$

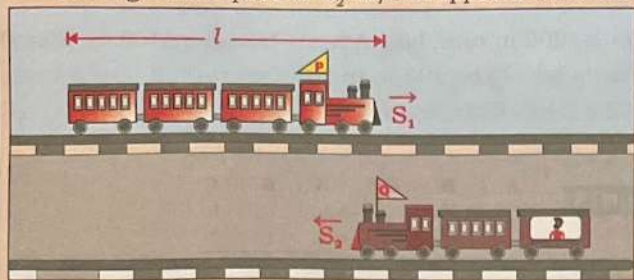
$$S_1 = 125 \text{ km/h}, S_2 = 17 \text{ km/h}$$

$$\therefore \text{Time} = \frac{l}{S_1 - S_2}$$

$$= \frac{228}{(125 - 17) \times \frac{5}{18}} = \frac{228}{108 \times \frac{5}{18}} = \frac{228}{30} = 7.6 \text{ seconds}$$

When a train crosses a person sitting inside another train

Case I: A train P of length l m is moving with a speed of ' S_1 ' m/s. A person is sitting in another train Q which is moving with a speed of ' S_2 ' m/s in opposite direction.



Time taken by train P to pass the person sitting in train Q.

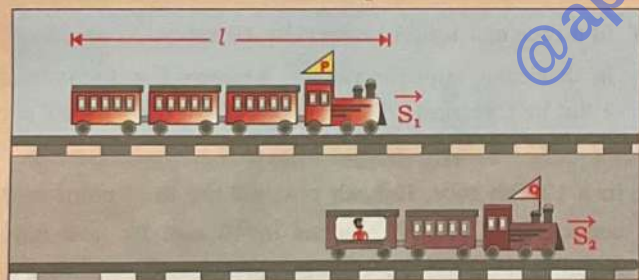
$$\text{Time} = \frac{l}{S_1 + S_2}$$

Ex. Two trains, A and B are moving in opposite direction at the speed of 58 km/h and 50 km/h, respectively. The length of train A is 450 m. How much time will train A take to pass a man sitting in train B?

HINTS $\therefore \text{Time} = \frac{l}{S_1 + S_2}$

$$= \frac{450}{(58 + 50) \times \frac{5}{18}} = \frac{450}{108 \times \frac{5}{18}} = 15 \text{ seconds}$$

Case II: A train P of length l m is moving with a speed of ' S_1 ' m/s. A person is sitting in another train Q which is moving with a speed of ' S_2 ' m/s in same direction:



Time taken by faster train P to pass the person sitting in slower train Q:

$$\text{Time} = \frac{l}{S_1 - S_2}$$

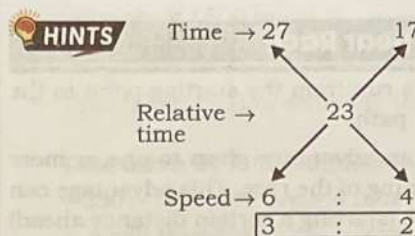
Ex. Two trains are moving in the same direction at the speed of 60 km/h and 42 km/h. The length of the faster train is 180 m. How much time will the faster train take to cross a man sitting in the slower train?

HINTS $\therefore$ Time at which the faster train take to cross a man sitting in the slower train

$$= \frac{l}{S_1 - S_2} = \frac{180}{(60 - 42) \times \frac{5}{18}} = \frac{180}{5} = 36 \text{ seconds}$$

Miscellaneous

Ex. Two trains cross a man standing on a platform in 27 seconds and 17 seconds, respectively. They cross each other in 23 seconds in opposite direction. Find the ratio of the speed of the trains.



$\therefore$ Ratio of speed of trains = 3 : 2

Ex. A 252 m long train is running at a speed of 125 km/h. What is the time (in seconds) in which it will pass a man who starts from the engine running at the speed of 17 km/h in the same direction as that of the train?

HINTS Speed of train = 125 km/h

Speed of man = 17 km/h

Relative speed = $125 - 17 = 108 \text{ km/h}$

$= 108 \times \frac{5}{18} = 30 \text{ m/sec}$

Time = $\frac{252}{30} \text{ sec} = 8.4 \text{ sec}$

Ex. Train 'A' requires 15 seconds to cross train 'B' of length 300 m moving in the opposite direction at a speed of 36 km/h. Further, train 'A' requires 30 seconds to cross a 500 m long stationary train 'C'. Find the length (in m) of train 'A'.

HINTS Let, length of train A = L_A

Time taken by train A = 30 sec.

Length of train C = 500 m

So speed of train A

$$= (S_A) = \frac{L_A + 500}{30}$$

Speed of train B = 36 km/hr

$$= 36 \times \frac{5}{18} = 10 \text{ m/sec}$$

Time = 15 second

Length of B = 300 meter

ATQ,

$$(S_A + 10) = \frac{L_A + 300}{15}$$

$$\Rightarrow (L_A + 300) = \left(\frac{L_A + 500}{30} + 10 \right) \times 15$$

$$\Rightarrow L_A + 300 = \frac{L_A + 500 + 300}{30} \times 15$$

$$\Rightarrow 2L_A + 600 = L_A + 800$$

$$\therefore L_A = 200 \text{ m}$$

LINEAR & CIRCULAR RACE

Linear Race

In a **linear race**, runners run from the starting point to the finish point in a straight path.

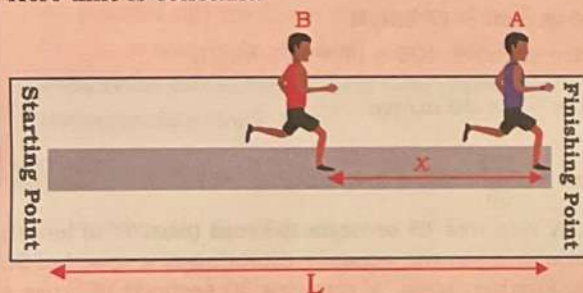
Head start: It refers to an advantage given to one or more competitors at the beginning of the race. This advantage can be in the form of distance (starting a certain distance ahead) or time (starting a certain time earlier).

Dead heat: It occurs in a race when two or more competitors finish at precisely the same time. In other words, there's no single clear winner.

A beats B by 'x' metre:

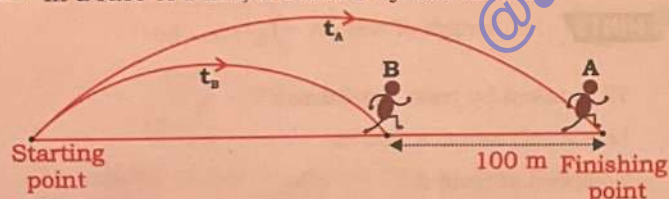
This means that when A crosses the finish point, B is 'x' m behind the finish point.

Here time is constant.



$\therefore \text{Speed} \propto \text{Distance}$ (Time is constant)

In a race of 1 km, A beats B by 100 metre.



In this case, person A covers the distance 'L' of 1 km, but person B covers only $L - x = 1000 - 100 = 900\text{m}$.

Here, time taken by both the persons is same ($t_A = t_B$).

Ex. In a 100-m race, A beats B by 20 m and B beats C by 20 m. By how much distance does A beat C?

HINTS In 100m race, A covers 100m and B covers

$$= 100 - 20 = 80\text{m}$$

In 100m race, B covers 100m and C covers = 80m

Thus,

A	B	C
100	80	80
$\propto 100$		
10000	8000	6400
$\times \frac{1}{100}$		
100	80	64

$\therefore$ In 100m race, A will beat C by $100 - 64 = 36\text{ m}$

Ex. In a 1200 m race, bike A beats bike B by 100 m. Bike B beats bike C by 100 m in a 600 m race. If bike A beats bike C by 30 sec in a 720 m race, then what is the speed of bike C?

HINTS

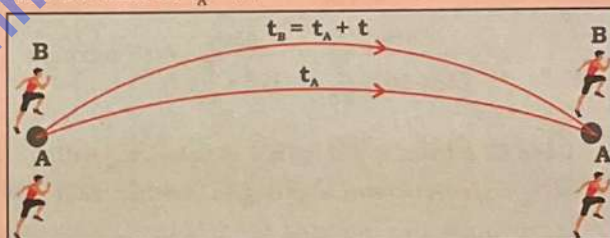
$$\begin{array}{rcl}
 \text{A : B : C} & & \text{A : B : C} \\
 1200 : 1100 & \rightarrow & 12 : 11 : 11 \\
 600 : 500 & \rightarrow & 6 : 5 : 5 \\
 & & \times 10 \quad \times 10 \\
 & & 72 : 55 : 55 \\
 & & \text{Diff.} \\
 & & 170 \text{ m}
 \end{array}$$

$\therefore$ Bike C covers a distance of 170 metre in 30 seconds.

$$\therefore \text{Speed of bike C} = \frac{170}{30} = \frac{17}{3} \text{ m/sec}$$

A beats B by 't' seconds:

A finishes the race 't' seconds earlier than B. If A's time is ' t_A ', then B's time is ' $t_A + t$ '.



In a race of 1 km, A beats B by 10 sec.

In this case, suppose person A covers the distance of 1 km in 't' seconds, then person B will take 't + 10' sec to cover the same distance.

Ex. In a 1200m race, Rakesh reaches the final point in 36 seconds and Rajesh reaches in 40 sec. By how much distance does Rakesh beat Rajesh?

HINTS Distance covered by Rajesh in 36 sec

$$= \frac{1200}{40} \times 36 = 1080 \text{ m}$$

$\therefore$ Rakesh beats Rajesh by $1200 - 1080 = 120 \text{ m}$.

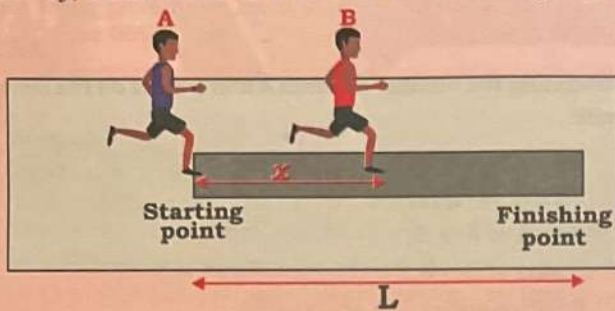
Alternatively

	Rakesh	Rajesh
Time $\rightarrow$	36	40
	9	10
Distance $\rightarrow$	10	9
		Diff.
	1200 m	1 unit $\rightarrow$ 120 m

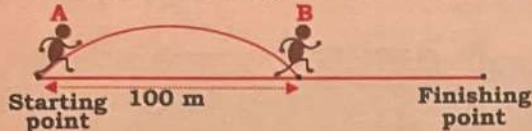
$\therefore$ Rakesh beats Rajesh by a distance of 120 m.

A gives B an 'x' metre start:

This means B starts 'x' metre ahead of the starting line or effectively, B has to run 'L-x' metre if the track length is 'L'.



Ex. In a race of 1 km, A can give B a start of 100 metre.



In this case, person A covers total distance 'L' of 1 km, but person B covers only = $L - x = 1000 - 100 = 900\text{m}$.

Ex. In a 5 km race, S gave a 500 m head start to T. If they finished the race at the same time and the speed of T is 13.5 km/hr, then the speed of S is:

	S	:	T
Distance →	5000	:	4500
Speed →	10	:	9
	$\times 1.5$		$\times 1.5$
	15 km/hr	:	13.5 km/hr

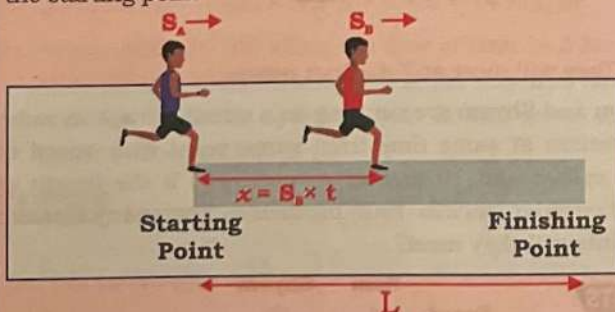
∴ Speed of S = 15 km/hr

Ex. In a 400-metre race, A runs at a speed of 16 m/sec. If A gives B a start of 15 metre and still beats him by 10 sec, then what is the speed of B?

HINTS Time of A = $\frac{\text{Distance}}{\text{Speed}} = \frac{400}{16} = 25\text{sec}$	
A :	B
Distance → 400	: 385
Time → 25	: 35
∴ Speed of B = $\frac{385}{35} = 11\text{ m/s}$	

A gives B a start of 't' minutes:

Person B begins running for 't' minutes before A starts from the starting point.



Ex. In a race of 1 km, A can give B a starting lead of 10 min. In this case, if person A covers total distance 'L' of 1 km in 'T' min and if race becomes dead heat then person B completes the race in 'T+10' min.

Ex. A gives B a head-start of 10 seconds in a 1500 m race and both finish the race at the same time. What is the time taken by A (in minutes) to finish the race if the speed of B is 6 m/s?

HINTS Given,

Distance = 1500 m

Speed of B = 6 m/s

Time taken by B to complete the race = $\frac{1500}{6} = 250\text{ sec}$

Time taken by A to complete the race

= $250 - 10 = 240\text{ sec} = 4\text{ min}$

∴ A will take 4 minutes to complete the race.

Circular Race

A **circular race** is a competition held on a closed, circular track where the starting and the finishing point are the same.

Calculating the time of first meeting at any point on track if A and B are running in same direction:

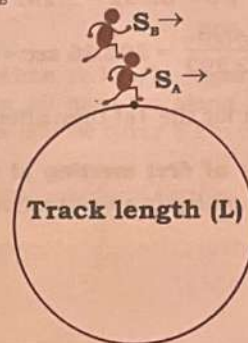
Let,

Length of track = L

Speed of A = S_A

Speed of B = S_B

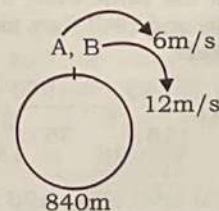
where, $S_A > S_B$



$$\text{Time for first meeting} = \frac{\text{Track length (L)}}{S_A - S_B}$$

Ex. In a circular race of 840 m, A and B start running in the same direction at the same time from the same point at the speed of 6 m/s and 12 m/s, respectively. After how much time will they meet next?

HINTS



Relative speed = $12 - 6 = 6\text{ m/s}$

Time = $\frac{\text{Track length}}{\text{Relative speed}} = \frac{840}{6} = 140\text{ sec}$

∴ They will meet next after 140 sec.

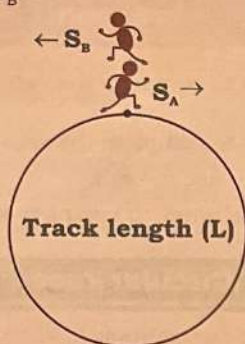
Calculating the time of first meeting at any point on track if A and B are running in opposite direction:

Let,

Length of track = L

Speed of A = S_A

Speed of B = S_B



$$\text{Time for first meeting} = \frac{\text{Track length } (L)}{S_A + S_B}$$

Ex. On a circular path of 693 m, Sujata and Anjali start walking from the same point but in opposite directions at the speed of 2.85 km/hr and 1.5 m/sec, respectively. When will they meet for the first time?

HINTS $2.85 \text{ km/hr} = 2.85 \times \frac{5}{18} \text{ m/s} = 0.792 \text{ m/s}$

Relative speed = $1.5 + 0.792 = 2.292 \text{ m/s}$

Meeting time = $\frac{693}{2.292} = 302.36 \text{ sec} = 5.04 \text{ min}$

$\therefore$ They will meet for the 1st time after 5.04 minutes.

Calculating the time of first meeting at starting point on track if A and B are running in any direction:

Let,

Length of track = L

Speed of A = S_A

Speed of B = S_B

Time of first meeting at starting point = $\text{LCM}\left(\frac{L}{S_A}, \frac{L}{S_B}\right)$

Here for both conditions whether A and B are running in same direction or in opposite direction, formula is same.

Ex. In a circular race of 950 m length, A and B start with speed of 18 km/hr and 36 km/hr, respectively, in the same direction, at the same time, from the same point. After how many seconds will they meet for the first time at the starting point?

HINTS Time = $\text{LCM}\left(\frac{950}{18 \times \frac{5}{18}}, \frac{950}{36 \times \frac{5}{18}}\right)$
 $= \text{LCM}(190, 95) = 190 \text{ sec}$

Ex. Arvind and Chetan started running simultaneously from the same point in the same direction on a circular track of length 210 m. If Arvind takes 180 seconds to complete one round and Chetan takes 420 seconds to complete one round, then after how much time will they meet for the first time at the starting point on the track?

HINTS Time taken by Arvind to complete 1 round = 180 sec
 Time taken by Chetan to complete 1 round = 420 sec
 $\therefore$ Time taken by them to meet at starting point for the first time = $\text{LCM}(180 \text{ sec}, 420 \text{ sec}) = 1260 \text{ sec} = 21 \text{ min}$

Calculating the number of times A and B meet on the circular track:

Let,

Length of track = L

Speed of A = S_A

Speed of B = S_B

Relative speed of A and B = S_{AB}

$$\text{No. of meetings } (n) = \frac{\text{LCM}\left(\frac{L}{S_A}, \frac{L}{S_B}\right)}{\frac{L}{S_{AB}}}$$

Smart tips:

(i) If A and B are running in same direction, then no. of times they meet on circular track

$$= \begin{cases} S_A - S_B; & \text{If } S_A > S_B \\ S_B - S_A; & \text{If } S_B > S_A \end{cases}$$

(ii) If A and B are running in opposite direction, then no. of times they meet on circular track = $S_A + S_B$

? While applying smart tips which are explained above, always calculate simplest form of $S_A : S_B$.
 For example:
 If, $S_A = 8 \text{ m/s}$, $S_B = 6 \text{ m/s}$
 A and B are running in same direction, then:

WRONG SOLUTION

No. of times A & B meet
 $= S_A - S_B = 8 - 6 = 2$

RIGHT SOLUTION

$S_A : S_B = 4 : 3$
 No. of times A & B meet
 $= S_A - S_B = 4 - 3 = 1$

Ex. Anjali and Babita are running on a circular track in opposite directions at same time from same point with speed of 8 m/sec and 6 m/sec, respectively. If the length of the circular track is 960 m, then in how many distinct points will they meet?

HINTS

A	B
Speed $\rightarrow 8$	: 6
4	: 3
Sum	
7	

$\therefore$ They will meet at 7 distinct points.

Ex. Ram and Shyam are running on a circular track in same direction at same time from same point with speed of 14 m/sec and 10 m/sec, respectively. If the length of the circular track is 1000 m, then in how many distinct points will they meet?

HINTS

Ram	Shyam
Speed $\rightarrow 14$	: 10
7	: 5
Diff.	
2	

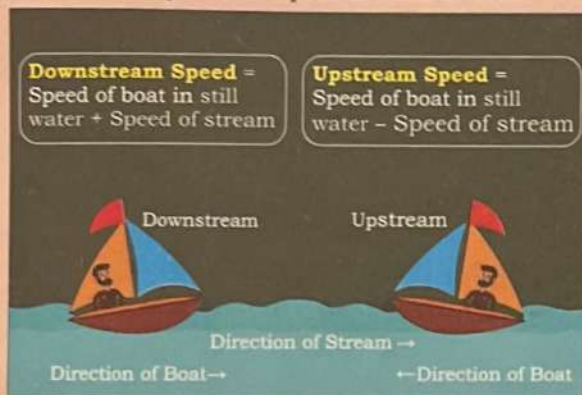
$\therefore$ They will meet at 2 distinct points.

CHAPTER

29

BOAT & STREAM

- **Stream** : The flow of river is called **stream**.
- **Upstream** : If the boat is moving in the opposite direction to the stream, it is called **upstream**. In this case, the net speed of boat is called upstream speed.
- **Downstream** : If the boat is moving along the direction of the stream, it is called **downstream**. In this case, the net speed of boat is called downstream speed.
- **Still water** : Under this circumstance, water is considered to be stationary and the speed of water is zero.



Important Formulae

If speed of boat in still water be 'B' and speed of stream be 'S', then:

- Downstream speed = $B + S$
- Upstream speed = $B - S$
- Speed of boat in still water = $\frac{1}{2} \left(\frac{\text{Downstream}}{\text{Speed}} + \frac{\text{Upstream}}{\text{Speed}} \right)$
- Speed of stream = $\frac{1}{2} \left(\frac{\text{Downstream}}{\text{Speed}} - \frac{\text{Upstream}}{\text{Speed}} \right)$

Ex. If the speed of a swimmer is 9 km/hr in still water and the speed of stream is 5 km/hr then find the speed of swimmer against the stream and along the stream.

HINTS Upstream speed = $B - S = 9 - 5 = 4$ km/hr

Downstream speed = $B + S = 9 + 5 = 14$ km/hr

Ex. A boat rows 30 km along the flow of river in 3 hours 45 minutes and rows 15 km against the flow in 2 hours 30 minutes. Find the speed of boat in still water and the speed of flow (in km/hr).

HINTS Downstream speed = $B + S = \frac{30}{\frac{15}{4}} = 8$ (i)

$$\left(\because 3 \text{ hr } 45 \text{ min} = \frac{15}{4} \text{ hr} \right)$$

Upstream speed = $B - S = \frac{15}{\frac{5}{2}} = 6$ (ii)

$$\left(\because 2 \text{ hr } 30 \text{ min} = \frac{5}{2} \text{ hr} \right)$$

On solving (i) and (ii):

$$B = 7 \text{ km/hr}, S = 1 \text{ km/hr}$$

$$\therefore \text{Speed of boat in still water} = 7 \text{ km/hr}$$

$$\text{Speed of flow} = 1 \text{ km/hr}$$

Ex. A boat takes 2.2 times as long to row a certain distance in upstream as to row the same distance in downstream. If it can row 55 km downstream in 2 hours 30 minutes, what is the speed of the boat in still water?

HINTS Downstream speed = $\frac{55}{2.5} = 22 \text{ km/hr}$

$$\therefore 2.2 = \frac{22}{10} = \frac{11}{5}$$

	Downstream	Upstream
Time →	5	11
Speed →	11	5
	↓ ×2	↓ ×2
	22 km/hr	10 km/hr

$$\therefore \text{Speed of boat in still water} = \frac{1}{2} \times (22 + 10)$$

$$= 16 \text{ km/hr}$$

Ex. A swimmer swims 3 km/hr against the stream and swims 1 km in 10 minutes along the stream, then in how much time will he cover 5 km in still water?

HINTS Upstream speed = 3 km/hr

$$\text{Downstream speed} = \frac{1}{\frac{10}{60}} = 6 \text{ km/hr}$$

$$\text{Speed of swimmer in still water} = \frac{1}{2} (6 + 3) = \frac{9}{2} = 4.5 \text{ km/hr}$$

$$\therefore \text{Required time} = \frac{5}{4.5} = \frac{50}{45} = \frac{10}{9} = 1\frac{1}{9} \text{ hr}$$

Based on : $\frac{B}{S} = \frac{(T_2 + T_1)}{(T_2 - T_1)}$

If the time taken by a boat to row a certain distance in downstream is ' T_1 ' and the same distance in upstream is ' T_2 ', then the ratio of the speed of boat to the speed of stream:

$$\frac{B}{S} = \frac{(T_2 + T_1)}{(T_2 - T_1)}$$

Ex. A motorboat takes 18 hours to go a certain distance downstream and 36 hours to travel the same distance upstream. Find the ratio of the speed of boat in still water to the speed of stream.

HINTS $\frac{B}{S} = \frac{(T_2 + T_1)}{(T_2 - T_1)} = \frac{36+18}{36-18} = \frac{54}{18} = \frac{3}{1} = 3:1$

Boat & Stream

$$\text{Based on : } D = \frac{T(B^2 - S^2)}{2B}$$

Let, distance between two points X and Y be 'D'. If the total time taken by a boat to cover a certain distance from point X to Y and come back from Y to X is 'T', then:

- Distance 'D' between the two points = $\frac{T(B^2 - S^2)}{2B}$
- Average speed of the boat = $\frac{(B^2 - S^2)}{B}$

Ex. A motorboat, whose speed is 15km/hr in still water goes 50km downstream and comes back in a total of 7 hours 30 minutes. The speed of stream (in km/hr) is:

HINTS Let, speed of stream = S km/hr
Speed of boat in still water = 15 km/hr

$$\therefore D = \frac{T(B^2 - S^2)}{2B}$$

$$\Rightarrow 50 = \frac{7.5 \times (15^2 - S^2)}{2 \times 15} \Rightarrow S = 5 \text{ km/h}$$

$\therefore$ Speed of stream = 5km/h

Ex. A boat moves in the upstream from city P to city Q and then it returns in the downstream from city Q to city P. If the speed of boat in still water is 35 km/hr and the speed of stream is 5 km/hr, then what is the average speed of boat in total journey?

HINTS Average speed of boat = $\frac{(B^2 - S^2)}{B}$

$$= \frac{(35)^2 - (5)^2}{35} = \frac{30 \times 40}{35} = \frac{240}{7} = 34.28 \text{ km/hr}$$

$$\text{Based on : } D = \frac{T(B^2 - S^2)}{2S}$$

If a boat takes 'T' hours more to travel a certain distance 'D' upstream than to travel the same distance downstream, then:

$$D = \frac{T(B^2 - S^2)}{2S}$$

Ex. The speed of a boat in still water is 20 km/hr, while the river is flowing at a speed of 8 km/hr and the time taken by the boat to cover a certain distance upstream is 6 hours more than the time taken to cover the same distance downstream. Find the distance covered by the boat in downstream.

HINTS $D = \frac{T(B^2 - S^2)}{2S}$

$$= \frac{6 \times (20^2 - 8^2)}{2 \times 8} = \frac{6 \times 28 \times 12}{2 \times 8} = 126 \text{ km}$$

Alternatively

Let, required distance be 'D' km.

Speed of boat in still water 'x' = 20 km/hr

Speed of stream 'y' = 8 km/hr

ATQ,

$$\frac{D}{x-y} - \frac{D}{x+y} = 6$$

$$\Rightarrow \frac{D}{12} - \frac{D}{28} = 6 \Rightarrow D = 21 \times 6$$

$$\therefore D = 126 \text{ km}$$

UV Concept

Ex. A boat can go 40 km downstream and 25 km upstream in 7 hours 30 minutes. It can go 48 km downstream and 36 km upstream in 10 hours. What is the speed (in km/hr) of the boat in still water?

HINTS Let, speed of boat in still water = x km/hr

Speed of stream = y km/hr

$$\text{Let, } U = \frac{1}{x-y}, D = \frac{1}{x+y}$$

ATQ,

$$\begin{array}{rcl} 40D + 25U & = & 7.5 \text{ hrs} \times 6 \\ 48D + 36U & = & 10 \text{ hrs} \times 5 \\ \hline 240D + 150U & = & 45 \text{ hrs} \\ 240D + 180U & = & 50 \text{ hrs} \\ \hline -30U & = & -5 \end{array}$$

$$U = \frac{1}{6} \Rightarrow x - y = 6 \quad \dots(i)$$

$$\text{Now, } 48D + 36 \times \frac{1}{6} = 10 \Rightarrow 48D = 4$$

$$\Rightarrow D = \frac{1}{12} \Rightarrow x + y = 12 \quad \dots(ii)$$

From equation (i) and (ii):

$$x = 9 \text{ km/hr}$$

$\therefore$ Speed of boat in still water = 9 km/hr

Ex. A boat can go 15 km downstream and 8 km upstream in 2h. It can go 20 km downstream and 12 km upstream in 2h 50 min. What is the speed (in km/h) of the boat while going downstream?

HINTS Let speed of boat in still water = x km/h

Speed of stream = y km/h

$$U = \frac{1}{x-y},$$

$$D = \frac{1}{x+y}$$

$$\begin{array}{rcl} 15D + 8U & = & 2 \text{ hrs} \times 3 \\ 20D + 12U & = & 2 \text{ hrs. } 50 \text{ min} \times 2 \\ \hline 45D + 24U & = & 6 \text{ hrs} \\ 40D + 24U & = & 5 \text{ hrs } 40 \text{ min} \\ \hline 5D & = & 20 \text{ min} \left(\frac{1}{3} \text{ hr} \right) \\ D & = & \frac{1}{15} \Rightarrow x + y = 15 \end{array}$$

$$\therefore \text{Downstream speed} = 15 \text{ km/h}$$

Boat & Stream

Miscellaneous

Ex. A person rowing a boat goes from J to K (upstream) and returns from K to J (downstream) in a total time of 15 hours. The distance between J and K is 300 km. The time taken by a man to row a boat to a distance of 9 km downstream is equal to the time taken by him to row a boat to a distance of 3 km upstream. What is the approximate speed of the boat in still water?

HINTS

	Downstream	Upstream
Distance →	9	3
	3	1
Speed →	3	1

Let, speed of the boat in upstream and downstream be 'x' and '3x', respectively.

So, average speed in the whole journey = $\frac{2 \times 3x \times x}{4x}$

$$\Rightarrow \frac{2 \times 300}{15} = \frac{2 \times 3x \times x}{4x} \Rightarrow x = \frac{80}{3}$$

$$\therefore \text{Speed of boat} = \frac{3x + x}{2} = 2x = 2 \times \frac{80}{3} = 53.33 \text{ km/hr}$$

Ex. X, Y are two points in a river. Point P and Q divide the straight line XY into three equal parts. The river flows along XY and the time taken by a boat to row from X to Q and from Y to Q are in the ratio 4 : 5. The ratio of the speed of boat downstream to that of the river current is:

HINTS



Let,

Travelling time from X to Q = 4 unit

Travelling time from Y to Q = 5 unit

$$\therefore T_{XQ} = \frac{4}{2} \times 3 = 6 \text{ unit}$$

$$T_{YQ} = 5 \times 3 = 15 \text{ unit}$$

	Downstream	Upstream
Time →	6	15
	2	5
Speed →	5	2

$\therefore$ (Speed of boat in downstream) : (Speed of current)

$$= 5 : \frac{5-2}{2} = 5 : \frac{3}{2} = \boxed{10 : 3}$$

CHAPTER

30

PERMUTATION & COMBINATION

Concept of Factorial

The continued product of 'n' natural number is called the "n factorial" and it is denoted by $n!$ or $\underline{n}$.

$$n! = 1 \times 2 \times 3 \times 4 \dots (n-2) (n-1) n$$

OR

$$n (n-1) (n-2) \dots 3 \times 2 \times 1$$

Factorial of fractions or negative integers are not defined. Factorial defined only for whole numbers.

$-3!$ = Not defined

$$\left(\frac{3}{4}\right)! = \text{Not defined}$$

$$3! = 6$$

$$0! = 1$$

$$4! = 24$$

$$1! = 1$$

$$5! = 120$$

$$2! = 2 \times 1 = 2$$

$$6! = 720$$

Ex. Simplify $\frac{70!}{67!}$

HINTS $\frac{70!}{67!} = \frac{70 \times 69 \times 68 \times 67!}{67!} = 70 \times 69 \times 68$

Ex. Find the value of $\frac{11! - 10!}{9!}$

HINTS $\frac{11! - 10!}{9!} = \frac{11 \times 10 \times 9! - 10 \times 9!}{9!} = 110 - 10 = 100$

Ex. Convert it into factorial

$$3 \times 6 \times 9 \times 12 \times 15 \times 18$$

HINTS $= 3 \times 6 \times 9 \times 12 \times 15 \times 18$

$$= 3 (1 \times 2 \times 3 \times 4 \times 5 \times 6)$$

$$= 3 \times 6!$$

Ex. If $(n+3)! = 56 (n+1)!$ Find the value of n .

HINTS $(n+3) (n+2) (n+1)! = 56 (n+1)!$

$$(n+3) (n+2) = 56$$

$$8 \times 7 = 56$$

$$\text{So, } n = 5$$

Ex. If $(n+2)! = 2550 \times n!$ Find the value of n .

HINTS $(n+2) (n+1) n! = 2550 \times n!$

$$(n+2) (n+1) = 2550$$

$$51 \times 50 = 2550$$

$$n = 49$$

Ex. If $(n+1)! = 12 (n-1)!$ Find the value of n .

HINTS $(n+1) (n) (n-1)! = 12 (n-1)!$

$$(n+1) (n) = 12$$

$$4 \times 3 = 12$$

$$n = 3$$

Principle of Addition

If you have n disjoint events, i.e. $A_1, A_2, \dots, A_n$ with outcomes $n_1, n_2, \dots, n_n$ ways respectively, then any of the events can occur with $n_1 + n_2 + \dots + n_n$ ways.

Ex. Shubham has 2 slippers and 3 shoes. How many different pairs of a slippers or a shoes can be choose.



HINTS Only one of the pairs of slippers or shoes can be worn at a time. Hence slippers and shoes are independent of each other. So total number of choosing = $3 + 2 = 5$

Let's give another example for better understanding. Suppose there are 15 NITs and 20 IITs in India and you qualified for the JEE, so there are $(15 + 20) = 35$ ways you can select your college.

Another example, consider you go to a restaurant where you find 8 Chicken items and 5 veg items, now you want to order a dish among these. So, you can choose a dish in $(8 + 5) = 13$ ways.

Principle of Multiplication

Suppose A and B are events with n_1 & n_2 possible outcomes, respectively. Then the total number of outcomes for the sequence of the two events is $n_1 \times n_2$.

Let us explain the principle of multiplication with an example.

Ex. Virat has 3 pants and 4 shirts. How many different pairs of a pant and a shirt can be dressed up with?



HINTS For every choice of a pant, there are four choices of shirt. So this is dependent event so the total ways of dressed up = $4 \times 3 = 12$

Lets take another example on Teacher's day you want to visit your kindergarten, High school, and your college. If there are three ways to connect your kindergarten and High school and five ways to connect your high school and college. Then you can go from your kindergarten to your college in $3 \times 5 = 15$ ways because you can choose any of the three routes for your high school and then again you can choose any of the five routes to your college. Selecting the entire route is dependent on both the routes you choose, hence multiplication.

Another example is, Suppose there are 20 IITs, each with 7 branches. Then the topper can select his branch in $20 \times 7 = 140$ ways.

Permutation

A permutation is a way of arranging a group of items in a particular order. If the order changes, it's considered a different permutation.

OR

'Arrangement where order matters'.

Permutation used when the order is important.

Ex. You have digits: 1, 2, 3. How many 2-digit numbers can you form without repeating digits?

HINTS Possible numbers = 12, 13, 21, 23, 31, 32 = 6 permutations.

Here, 12 is different from 21 because the order of digits matters.

Permutation Formulae

The number of permutations (arrangements) of 'n' different things out of which 'r' things are taken at a time and where the repetition is not allowed, denoted by nP_r or $P(n, r)$:

$${}^nP_r = \frac{n!}{(n-r)!} \quad \text{where, } 0 \leq r \leq n$$

Using the above formula, the total number of ways of arranging n different things (taking all at a time) is n!.

Because in this case, $r = n$

$$\therefore {}^nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = \frac{n!}{1} = n!$$

Some important Result on Permutation:

$${}^nP_0 = \frac{n!}{(n-0)!} = \frac{n!}{n!} = 1$$

$${}^nP_1 = \frac{n!}{(n-1)!} = \frac{n(n-1)!}{(n-1)!} = n$$

$${}^nP_2 = \frac{n!}{(n-2)!} = \frac{n(n-1)(n-2)!}{(n-2)!} = n(n-1)$$

Ex. If the digits are not allowed to repeat in any number formed by using the digits 0, 2, 4, 6, 8 then the number of all numbers greater than 10,000 is equal to-

HINTS 0, 2, 4, 6, 8

2, 4, 6, 8				
4	4	3	2	1

Required no. of numbers = $4 \times 4 \times 3 \times 2 = 96$

Ex. In how many ways can 5 men draw water from 5 taps if no tap can be used more than once?

HINTS 1st man can draw water from any of the 5 taps.

2nd man can draw water from any of the remaining 4 taps.

3rd man can draw water from any of the remaining 3 taps.

4th man can draw water from any of the remaining 2 taps.

5th man can draw water from remaining 1 tap.

5	4	3	2	1
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Hence total number of ways

$$= 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Ex. How many four digit even numbers from 1, 2, 3, 4, 5, 6, 7, 8 are formed if repetition is not allowed?

HINTS 1, 2, 3, 4, 5, 6, 7, 8

			2, 4, 6, 8
7	6	5	$\times 4$

Required total number = $7 \times 6 \times 5 \times 4 = 840$

Ex. How many 100-digit positive number are there?

HINTS 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

1-9				
9	10	10	99 times	

$$\begin{aligned} \text{Required number} &= 9 \times [10 \times 10 \dots\dots 99 \text{ times}] \\ &= 9 \times 10^{99} \end{aligned}$$

Ex. In how many ways can the letters of the word 'QUANTS' be arranged (Repetition is not allowed)?

HINTS Number of ways = ${}^6P_6 = \frac{6!}{(6-6)!}$

$$= \frac{6!}{0} = 6! = 720$$

Ex. How many words with or without meaning, can be formed by using all the letters of the word, 'DELHI' using each letter exactly once?

HINTS The word 'DELHI' has 5 letters and all these letters are different.

Total number of words (with or without meaning) that can be formed using all these 5 letters using each letter exactly once = Number of arrangements of 5 letters taken all at a time

$${}^5P_5 = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Ex. In how many different ways can the letters of the word 'ADITYA' be arranged?

HINTS The word 'ADITYA' has 6 letters.

In these 6 letters, 'A' occurs 2 times,

$$\text{Total Number of ways} = \frac{6!}{2!} = 360$$

Ex. In how many different ways can the letters of the word 'BANKING' be arranged?

HINTS The word 'BANKING' has 7 letters.

In these 7 letters, 'N' occurs 2 times,

$$\text{Total Number of ways} = \frac{7!}{2!} = \frac{5040}{2} = 2520$$

Ex. In how many different ways can the letters of the word 'RUMOUR' be arranged?

HINTS The word 'RUMOUR' has 6 letters.

In these 6 letters, 'R' occurs 2 times, 'U' occurs 2 times and rest of the letters are different.

Hence, number of ways to arrange these letters

$$= \frac{6!}{(2!)(2!)} = 180$$

Ex. How many arrangements can be made out of the letters of the word 'ENGINEERING'?

HINTS The word 'ENGINEERING' has 11 letters.

But in these 11 letters, 'E' occurs 3 times, 'N' occurs 3 times, 'G' occurs 2 times, 'I' occurs 2 times and rest of the letters are different.

Hence, number of ways to arrange these letters

$$= \frac{11!}{(3!)(3!)(2!)(2!)} = 277200$$

Ex. How many words can be formed from the letters of the word 'TRIANGLE' with T always at the beginning and E at the end.

HINTS If we fix up 'T' in the beginning and 'E' at the end, then the remaining 6 letters can be arranged in 6P_6 that is 6! which is 720 ways.

Ex. In how many different ways can the letters of the word 'DESIGN' be arranged so that the vowels are at the two ends?

HINTS Number of ways = $4! \times 2! = 48$

Ex. In how many different ways can the letters of the word 'CORPORATION' be arranged so that the vowels always come together?

HINTS The word 'CORPORATION' has 11 letters. It has the vowels 'O', 'O', 'A', 'I', 'O' in it and these 5 vowels should always come together.

Hence these 5 vowels can be grouped and considered as a single CRPRTN(OOAIO).

Hence we can assume total letters as 7. But in these 7 letters, 'R' occurs 2 times and rest of the letters are different.

Number of ways to arrange these letters

$$= \frac{7!}{2!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 2520$$

In the 5 vowels (OOAIO), 'O' occurs 3 and rest of the vowels are different.

Number of ways to arrange these vowels among themselves = $\frac{5!}{3!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = 20$

Hence, required number of ways = $2520 \times 20 = 50400$

Ex. In how many different ways can the letters of the word 'DETAIL' be arranged such that the vowels must occupy only the odd positions?

HINTS The word 'DETAIL' has 6 letters which has 3 vowels ('E', 'A', 'I') and 3 consonants ('D', 'T', 'L')

The 3 vowels ('E', 'A', 'I') must occupy only the odd positions.

Let's mark the positions as (1) (2) (3) (4) (5) (6). Now, the 3 vowels should only occupy the 3 positions marked as (1), (3) and (5) in any order.

Hence, number of ways to arrange these vowels

$$= {}^3P_3 = 3! = 3 \times 2 \times 1 = 6$$

Total number of ways

Now we have 3 consonants ('D', 'T', 'L') which can be arranged in the remaining 3 positions in any order.

Hence, number of ways to arrange these consonants = ${}^3P_3 = 3! = 3 \times 2 \times 1 = 6$

$\therefore$ Required no. of ways = Number of ways to arrange the vowels $\times$ Number of ways to arrange the consonants = $6 \times 6 = 36$

Ex. Find the number of words that can be formed using letters 'L', 'M', 'N' and 'O' (one letter word is also allowed and repetition is not allowed)?

HINTS Total letters = 4

$$1 \text{ letter word} = 1 \times 4 = 4$$

$$2 \text{ letter word} = 4 \times 3 = 12$$

$$3 \text{ letter word} = 4 \times 3 \times 2 = 24$$

$$4 \text{ letter word} = 4 \times 3 \times 2 \times 1 = 24$$

$$\text{Total number of words} = 4 + 12 + 24 + 24 = 64$$

Ex. How many four-digit numbers can be formed from the digits 1, 2, 3, 4, 5, 6 (Repetition of digits not allowed)?

HINTS Here, $n = 6, r = 4$

$${}^6P_4 = \frac{6!}{(6-4)!} = 360$$

Thus, Total 360 numbers can be formed.

Ex. How many 3 letter-words with or without meaning, can be formed out of the letters of the word, 'LOGARITHMS', if repetition of letters is not allowed?

HINTS The word 'LOGARITHMS' has 10 different letters. Hence, the number of 3 letter-words (with or without meaning formed by using these letters)

$$= {}^{10}P_3 = 10 \times 9 \times 8 = 720$$

Ex. How many words of three letters can be created using only consonants of the English alphabet?

HINTS Number of ways = ${}^{21}P_3 = \frac{21!}{(21-3)!}$

$$= \frac{21 \times 20 \times 19 \times 18!}{18!} = 7980$$

Ex. In how many ways can the letters of the word 'PERMUTATIONS' be arranged if the words start with P and end with S?

HINTS

P	E	R	M	U	T	A	T	I	O	N	S
fix											fix

$$\text{Number of ways} = \frac{10!}{2!} (\because T = 2 \text{ times})$$

Permutations Formula With Repetition

When 'r' things from 'n' things have to be arranged with repetitions then number of permutation = n^r . This is because each of the 'r' things can be Arranged in 'n' different ways, thus giving $n \times n \times n \times \dots \times n$ (r times) = n^r .

Ex. The number of 5 digit numbers which are divisible by 4, with digits from the set [1, 2, 3, 4, 5] and the repetition of digits is allowed?

HINTS Last 2 digit = 12, 24, 32, 44, 52

We can choose last 2 digit from any of the 5 numbers

And the remaining 3 digits can be chosen in 5 ways because repetition is allowed.

5	5	5	5	5

$$\text{Required number} = 5 \times 5 \times 5 \times 5 \times 5 = 625$$

Ex. How many 3 letter words with or without meaning can be formed out of the letters of the word 'SMOKE' when repetition of letters is allowed?

HINTS Number of ways = $n^r = 5^3 = 125$

Ex. How many 3 letter words with or without meaning can be formed out of the letters of the word 'GOVIND' when repetition of the letters is allowed?

HINTS Number of ways = $n^r = 6^3 = 216$

Ex. In how many different ways can the letters of the word 'RECTITUDE' be arranged so that the vowels come together?

HINTS Use the following result to solve such type of questions—

Number of ways that the letters of the given word be arranged so that vowels come together = (No. of vowels)! × (No. of consonants + 1)!

$$\text{Number of ways} = \frac{4! \times 6!}{2! \times 2!} = 4320$$



Number of ways that the letters of the given word be arranged so that consonants come together = (No. of consonants)! × (No. of vowels + 1)!

Ex. Number of ways to declare the result (only 2 possibility win or loss) where 'n' match are played = 2^n

Ex. Five matches are to be played in a tournament with only 2 possibility win or loss. How many ways can results be decided?

HINTS Number of ways = $2^5 = 32$

Ex. There are 20 stations between Hyderabad and Bangalore. How many second class tickets have to be printed, so that a passenger can travel from any station to any other station?

HINTS The total number of stations = 20

From 20 stations we have to choose any two stations and the direction of travel (i.e., Hyderabad to Bangalore is different from Bangalore to Hyderabad) in ${}^{20}P_2$ ways.

$${}^{20}P_2 = 20 \times 19 = 380$$

Ex. There are five stations on a railways line. What is the number of different journey tickets that are required for railway authorities?

HINTS The total number of stations = 5

From 5 stations we have to choose any two stations and the direction of travel (i.e., A to B is different from B to A) in 5P_2 ways.

$${}^5P_2 = 5 \times 4 = 20$$

Ex. In how many ways can 4 boys and 5 girls be seated in a row so that they are alternate?

HINTS

G B G B G B G B G

The diagram shows the possible arrangement of sitting of boys and girls.

Now, 4 boys can be seated in 4 places in ${}^4P_4 = 4!$

and 5 girls in 5 places in $5!$ ways.

$$\therefore \text{Required no. of ways} = 4! \times 5! = 24 \times 120 = 2880$$

Ex. In how many ways can 10 books be arranged on a shelf such that a particular pair of books should always be together?

HINTS We have a total of 10 books.

Given that a particular pair of books should always be together.

Hence, just tie these two books together and consider as a single book.

Hence we can take total number of books as 9.

These 9 books can be arranged in ${}^9P_9 = 9!$ ways.

We had tied two books together.

These books can be arranged among themselves in ${}^2P_2 = 2!$ ways.

Hence, required number of ways = $9! \times 2!$

Ex. 4 boys and 2 girls are to be seated in a row in such a way that the two girls are always together. In how many different ways can they be seated?

HINTS Assume the 2 given girls to be together (i.e. one). Now there are five person.

Possible ways of arranging them are = $5! = 120$

Now, they (two girls) can arrange themselves in $2!$ ways.

Hence total ways = $120 \times 2 = 240$

Circular Permutation

If clockwise and anticlockwise orders are different

The number of circular permutation of n different things taken all at a time is $\frac{{}^nP_n}{n} = (n-1)!$

If clockwise and anticlockwise order are same

The number of circular permutations of 'n' different things taken all at a time is $\frac{1}{2}(n-1)!$

Ex. In how many ways can 7 beads can be arranged to form a necklace?

HINTS In necklace clock wise and anticlock wise arrangement is not different.

Number of arrangements possible

$$= \frac{1}{2}(7-1)! = \frac{1}{2} \times 6! = \frac{1}{2} \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 360$$

Ex. A company has 10 software engineers and 6 civil engineers. In how many ways can they be seated around a round table so that two of the civil engineers will not sit together?

HINTS 10 software engineers can be arranged around a round table in

$$(10-1)! = 9! \text{ ways} \quad \dots(i)$$

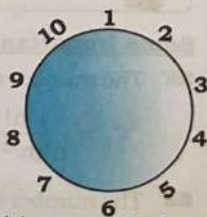
Now we need to arrange civil engineers such that no two civil engineers can be seated together. i.e., we can arrange 6 civil engineers in any of the 10 positions marked as below.

(here 1, 2, 3, 10 represent software engineers)

This can be done in ${}^{10}P_6$ ways

From equation (i) and (ii),

$$\text{Required number of ways} = 9! \times {}^{10}P_6 = 9! \times \frac{10!}{4!}$$



Combination

A combination is a way of selecting items from a group, where the order does not matter.

If you change the order but the selected items are the same, it's still the same combination.

Combination used when the order is NOT important.

Combination Formulae

Combinations are selections made by taking some or all of a number of objects, irrespective of their arrangements. The number of combinations of n different things taken r at a time, denoted by nC_r or $C(n, r)$ and it is given by,

$${}^nC_r = \frac{n!}{r!(n-r)!}, \text{ where } 0 \leq r \leq n$$

When ${}^nC_x = {}^nC_y$, then $x = y$ or $x + y = n$

$${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$${}^nC_r = {}^nC_{n-r}$$

Relation between Permutation and Combination

$${}^nP_r = \frac{n!}{(n-r)!} = \left(\frac{n!}{r!(n-r)!} \right) \times r!$$

$${}^nP_r = {}^nC_r \times r!$$

Ex. Out of 5 students, you want to form a committee of 3.

HINTS Let the students = A, B, C, D, E.

Possible combinations (some examples):

(A, B, C), (A, B, D), (A, B, E), (A, B, D) (...and so on)

$n = 5$ (total students)

$r = 3$ (students to be selected)

Now put the values:

$${}^5C_3 = \frac{5!}{3!(5-3)!} = \frac{5 \times 4 \times 3!}{3! \times 2!} = 10$$

"To write direct value from nC_r formulas, write Numerator by multiplying n by r -times in decreasing order and in Denominator by multiplying r till 1 in decreasing order.



For Example, ${}^5C_2 = \frac{5 \times 4}{2 \times 1} = 10$

Another example ${}^6C_4 = \frac{6 \times 5 \times 4 \times 3}{4 \times 3 \times 2 \times 1} = 15$

Some Important result on Combinations

The number of ways of selecting n objects out of n objects is:

$${}^nC_n = \frac{n!}{n!(n-n)!} = \frac{n!}{n!0!} = 1$$

The number of ways of selecting 0 objects out of n objects is:

$${}^nC_0 = \frac{n!}{0!(n-0)!} = \frac{n!}{0!n!} = 1$$

The number of ways of selecting 1 object out of n objects is:

$${}^nC_1 = \frac{n!}{1!(n-1)!} = \frac{n \times (n-1)!}{(n-1)!} = n$$

The number of Combinations of ' n ' different things taking some or all at a time

$$(i) {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n$$

$$(ii) {}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = 2^{n-1}$$

$$(iii) ({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2 = 2^n {}^nC_n$$

Ex. Amit has 5 blue, 4 red and 3 green socks in his cupboard. Out of this he has to choose 4 socks. In how many ways can he do this?

HINTS Number of ways = ${}^{12}C_4 = \frac{12!}{4!(12-4)!}$
 $= \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} = 495$

Ex. A school team has eight volleyball players. A five member team will be selected out of these eight players. How many different selections can be made?

HINTS Number of ways = ${}^8C_5 = \frac{8 \times 7 \times 6 \times 5!}{5! \times 3 \times 2 \times 1} = 56$

Ex. A group has 3 lawn tennis players S, T and U. A team consisting of 2 players is to be formed. In how many ways can we do so?

HINTS Number of ways = ${}^3C_2 = \frac{3 \times 2!}{2! \times 1} = 3$

Ex. In how many ways can a group of 5 men and 2 women be made out of total of 7 men and 3 women?

HINTS Number of ways = ${}^7C_5 \times {}^3C_2 = 21 \times 3 = 63$

Ex. A question paper has two parts P and Q, each containing 10 questions. If a student needs to choose 8 from part P and 4 from part Q, in how many ways can he do that?

HINTS No. of ways to choose 8 questions from part P = ${}^{10}C_8$
 Number of ways to choose 4 questions from part Q = ${}^{10}C_4$
 Total number of ways = ${}^{10}C_8 \times {}^{10}C_4 = 45 \times 210 = 9450$

Ex. In an examination, a minimum of marks is to be scored in each 6 subjects to pass. In how many ways can a students fail?

HINTS The student will fail if he fails in one or more subjects out of 6 different subjects, i.e.

$${}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6$$

$$= ({}^6C_0 + {}^6C_1 + \dots + {}^6C_6) - {}^6C_0$$

$$= 2^6 - 1 = 64 - 1 = 63 \text{ ways}$$

Alternatively

He fails, if he fails in any of the 6 subjects. With each paper there two possibilities: either fail or pass. This way, for 6 subjects there are $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6 = 64$ possible cases. This also includes the case when he passes in all the 6 subjects. Thus, He can fail in $64 - 1 = 63$ ways.

Ex. A team of 3 members need to be formed out of 5 men and 2 women. In how many ways team can be formed so as to include at least 1 woman?

HINTS Team of 3 can be formed in 2 ways

(a) 2 men and 1 woman - ${}^5C_2 \times {}^2C_1$

(b) 1 man and 2 women - ${}^5C_1 \times {}^2C_2$

So total ways = $({}^5C_2 \times {}^2C_1) + ({}^5C_1 \times {}^2C_2)$

$$= 10 \times 2 + 5 \times 1 = 25 \text{ ways}$$

Ex. Four couples (husband and wife) decide to form a committee of four members. Find the number of different committees that can be formed in which no couple finds a place

HINTS The number of committees of 4 gentlemen = ${}^4C_4 = 1$
 The number of committees of 3 gentlemen, 1 wife = ${}^4C_3 \times {}^1C_1$

(after selecting 3 gentlemen only 1 wife is left who can be included)

The number of committees of 2 gentlemen, 2 wives = ${}^4C_2 \times {}^2C_2$

The number of committees of 1 gentleman, 3 wives = ${}^4C_1 \times {}^3C_3$

The number of committees of 4 wives = 1

$\therefore$ The required number of committees = $1 + 4 + 6 + 4 + 1 = 16$

Ex. An urn contains 5 red marbles, 4 black marbles and 3 white marbles. Then the number of ways in which 4 marbles can be drawn so that at the most three of them are red is _____.

HINTS 0 Red, 1 Red, 2 Red, 3 Red

Number of ways = ${}^7C_4 + {}^5C_1 \times {}^7C_3 + {}^5C_2 \times {}^7C_2 + {}^5C_3 \times {}^7C_1$
 $= 35 + 175 + 210 + 70 = 490$

Important Formulae of Combination

Ex The number of ways of dividing 'm + n' things into two groups containing 'm' and 'n' things respectively

$$= {}^{m+n}C_m \times {}^nC_n = \frac{(m+n)!}{m! \times n!}$$

Ex The number of ways of dividing 'm + n + p' things into three groups containing 'm', 'n' and 'p' things respectively

$$= {}^{m+n+p}C_m \times {}^{n+p}C_n \times {}^pC_p = \frac{(m+n+p)!}{m!n!p!}$$

Ex If there are n non-collinear points in a plane, then

(i) Number of straight lines formed = nC_2

(ii) Number of triangles formed = nC_3

(iii) Number of quadrilaterals formed = nC_4

Ex If there are n points in a plane out of which m are collinear, then

(i) Number of straight lines formed = ${}^nC_2 - {}^mC_2 + 1$

(ii) Number of triangles formed = ${}^nC_3 - {}^mC_3$

Ex Number of diagonals in a polygon of n sides = ${}^nC_2 - n$

Ex. How many triangles can be formed by joining the vertices of an octagon?

HINTS Number of triangles that can be formed by joining the vertices of a polygon of n sides = nC_3

Here, $n = 8$

Hence, number of triangles that can be formed by joining the vertices of an octagon = ${}^8C_3 = 56$

Ex. Find the number of triangles that can be formed using 14 points in a plane such that 4 points are collinear?

HINTS Suppose there are n points in a plane out of which m points are collinear.

Number of triangles that can be formed by joining these n points as vertices = ${}^nC_3 - {}^mC_3$

Here, $n = 14$, $m = 4$

Hence, number of triangles = ${}^nC_3 - {}^mC_3 = {}^{14}C_3 - {}^4C_3 = 360$

Ex. There are 8 points in a plane out of which 3 are collinear. How many straight lines can be formed by joining them?

HINTS Suppose there are n points in a plane out of which m points are collinear. Number of straight lines that can be formed by joining these n points

$$= {}^nC_2 - {}^mC_2 + 1$$

Here, $n = 8$, $m = 3$

Required number of straight lines

$$= {}^nC_2 - {}^mC_2 + 1$$

$$= {}^8C_2 - {}^3C_2 + 1 = 28 - 3 + 1 = 26$$

Ex. There is a polygon of 11 sides. How many triangles can be drawn by only using the vertices of the polygon?

HINTS Number of sides in the polygon = 11

Number of vertices of the polygon = 11

$\therefore$ Number of triangles = ${}^{11}C_3$

$$= \frac{11!}{3!(11-3)!} = \frac{11!}{3!8!} = \frac{11 \times 10 \times 9 \times 8!}{3 \times 2 \times 1 \times 8!}$$

$$= 11 \times 5 \times 3 = 165$$

Ex Let, there are 'n' persons in a hall. If every person shakes his hand with every other person only once, then total number of handshakes—

$${}^nC_2 = \frac{n(n-1)}{2}$$

Ex If in place of handshakes each person gives a gift to another person, then formula changes to = $n(n-1)$

Ex. In a party every person shakes hand with every other person. If there was a total of 210 handshakes in the party, find the no. of persons who were present in the party.

HINTS For each selection of two person there will be one handshake.

So, No. of handshakes in the party = nC_2 ,

Where, n = Number of persons.

Now, ${}^nC_2 = 210$ (given)

$$\frac{n(n-1)}{2} = 210$$

$$n \times (n-1) = 2 \times (2 \times 3 \times 5 \times 7) = 21 \times 20$$

$$\therefore n = 21$$

CHAPTER

31

PROBABILITY

The theory of probability is simply the science of logic quantitatively treated. – C.S. Peirce

Probability

Probability is a measure of the likelihood or chance that a particular event will occur. It quantifies uncertainty by assigning a value between 0 and 1 to an event and it is defined as

Probability of an event E,

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

- If: $P(E) = 0$ means the event is 'Impossible' (can't happen),
- $P(E) = 1$ means the event is 'Certain' (will definitely happen),
- Values between 0 and 1 indicate the varying degrees of likelihood.

Hence, if in an experiment there are 'n' occurrences out of which 'm' occurrences are favourable to a particular event E, then the probability of the event E, denoted by $P(E)$, is defined as

$$P(E) = \frac{m}{n}$$

Also, the non-occurrence of the event E, is denoted by $\bar{E}$, that is (n-m) in this case. So, (n-m) occurrences are not favourable to the event E, and favourable to the event $\bar{E}$.

$$P(\bar{E}) = \frac{n-m}{n}$$

We can observe that, $P(E) + P(\bar{E}) = 1$

Terminologies Involved

The list of the terms related to probability is as follows.

Experiment: An Experiment, also known as a Random Experiment, is the fundamental process or activity conducted in probability theory whose precise outcome can't be predicted in advance with certainty while the set of all possible outcomes is known.

It forms the basis upon which all probability calculations are made.

Trial: A Trial is an execution of a random experiment.

- Each trial yields exactly one outcome.

Event: An event is a specific outcome or a set of outcomes from a well-defined experiment.

- It's a specific result or a collection of results.
- An event is defined by the experiment.

☞ Let's understand experiment, trial & event simultaneously by a coin.

Experiment: Flipping a coin

Trial: Flipping the coin once or two times etc.

Event: Getting "Heads".

Event: Getting "Tails".

☞ Let's understand experiment, trial & event simultaneously by a dice.

Experiment: Rolling a standard six-sided dice

Trial: Rolling the dice two times.

Event: Getting a "4".

Event: Getting an "even number"

Event: Getting a number greater than 5.

Outcome: This is the result of a trial. In the process of a sportsperson hitting a ball towards the goal post, there are two clear outcomes. He may either make the goal or miss the goal.

Equally likely Outcomes: An experiment in which each of the outcomes has an equal probability, such outcomes are referred to as equally likely outcomes. In the process of rolling a six-faced dice, the probability of getting any particular number is equal.

$$P(\text{Getting any particular number}) = \frac{1}{6}$$

Sample Space: It is the set of outcomes of all the trials in an experiment. On rolling a dice, the possible outcomes are 1, 2, 3, 4, 5 and 6. These outcomes make up the sample space. $S = \{1, 2, 3, 4, 5, 6\}$

Probable Event: An event that can be predicted is called a probable event. We can calculate the probability of such events. The probability of a particular child being promoted to the next class can be calculate, hence, we can refer to this as a probable event.

Impossible Event: An event that is not a part of the experiment, or which does not belong to the sample space of the outcomes of the experiment can be referred to as an impossible event. There is no snowfall in a temperate climatic region. Here, the snowfall can be referred to as an impossible event because the probability of occurrence of such an event is zero.

Complementary Events: Complementary events occur when there are just two events, and one event is exactly opposite to another event. For an event with probability $P(A)$, its compliment is $P(\bar{A})$ or $P(A')$.

$$P(A) + P(\bar{A}) = 1$$

In an examination, the event of success and the event of failure are complementary events.

$$P(\text{Success}) + P(\text{Failure}) = 1$$

Ex. If the probability of winning a game is 0.3, what is the probability of losing it?

💡 **HINTS** $P(W) + P(\bar{W}) = 1$

$$0.3 + P(\bar{W}) = 1$$

$$P(\bar{W}) = 0.7$$

At least

Mathematically, if 'n' is the total number of occurrences, and 'x' is the specified number, then "At least x" includes: x, x + 1, x + 2, upto n.

Therefore, the probability of "At least x" can be calculated as:

$$P(\text{At least } x) = 1 - P(\text{Less than } x)$$

$$\therefore P(\text{At least one}) = 1 - P(\text{No one})$$

Consider an Experiment where a fair coin is flipped 4 times then getting "at least 3 Heads" means that we are interested in getting exactly 3 Heads or exactly 4 Heads.

Ex. Out of 17 applicants 8 boys and 9 girls. Two persons are to be selected for the job. Find the probability that at least one of the selected persons will be a girl.

HINTS $P(\text{At least one girls}) = 1 - P(\text{No girls})$

$$= 1 - \frac{{}^8C_2}{{}^{17}C_2} = 1 - \frac{8 \times 7}{17 \times 16} = \frac{27}{34}$$

At most

Mathematically, if 'x' is the specified number, then "At most x" includes:

0, 1, x - 3, x - 2, x - 1, x.

Therefore, the probability of "At most x" can be calculated as:

$$P(\text{At most } x) = 1 - P(\text{Greater than } x)$$

Consider an Experiment where a fair coin is flipped 4 times then getting "At most 3 Heads" means that we are interested in getting exactly 1 Heads or 2 Heads or 3 Heads or zero Heads (all tail).

Ex. A class has 8 boys and 6 girls. A group of 3 students is to be formed. Find the probability that the group has at most 2 boys.

HINTS $P(\text{At most 2 boys}) = 1 - P(3 \text{ boys})$

$$= 1 - \frac{{}^8C_3}{{}^{14}C_3} = 1 - \frac{8 \times 7 \times 6}{14 \times 13 \times 12} = \frac{11}{13}$$

Based on Coins

1 Coin
Tossed



2 Coin
Tossed



When n coins are tossed or single coin tossed n times,

Possible outcomes = 2^n

When an unbiased coin is tossed once:

Possible outcomes are

$$2^n = 2^1 = 2$$

Sample space $\rightarrow \{H, T\}$

When two unbiased coins are tossed simultaneously.

Possible outcomes are $2^2 = 4$

Sample space $\rightarrow \{(H, H) (H, T) (T, H) (T, T)\}$

When three unbiased coins are tossed simultaneously.

Possible outcomes are $2^3 = 8$

Sample Space $\rightarrow \{(HHH) (HHT) (HTH) (THH) (HTT) (THT) (TTH) (TTT)\}$

Ex. Two unbiased coins are tossed up simultaneously. Find the probability of getting

- (i) Two tails
- (ii) One head
- (iii) At least one tail
- (iv) At most one tail
- (v) No tail

HINTS Sample space = $\{(H, H) (H, T) (T, H) (T, T)\}$

$$(i) P(\text{Two tails}) = \frac{1}{4}$$

$$(ii) P(\text{One head}) = \frac{2}{4} = \frac{1}{2}$$

$$(iii) P(\text{At least one tail}) = \frac{3}{4}$$

$$(iv) P(\text{At most one tail}) = \frac{3}{4}$$

$$(v) P(\text{No tail}) = \frac{1}{4}$$

Ex. If two coins are tossed simultaneously, what is the probability that one coin will get head and the other coin will get tail?

HINTS Sample space = $\{HT, TH, TT, HH\}$

$$\text{Probability} = \frac{2}{4} = \frac{1}{2}$$

Alternatively

$$\text{Probability} = {}^2C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^1 = \frac{2}{4} = \frac{1}{2}$$

Ex. Three unbiased coins are tossed up together. Find the probability of getting

- (i) All heads
- (ii) At least one tail
- (iii) Exactly 2 heads

HINTS Sample Space

= $\{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

$$(i) P(\text{All heads}) = \frac{1}{8}$$

$$(ii) P(\text{At least one tail}) = \frac{7}{8}$$

$$(iii) P(\text{Exactly 2 heads}) = \frac{3}{8}$$

Ex. Three coins are tossed, what is the probability that neither 3 heads nor 3 tails are obtained?

HINTS Sample Space

= $\{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

$$\text{Probability} = \frac{6}{8} = \frac{3}{4}$$

Alternatively

$$P(3 \text{ Heads}) = {}^3C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^0 = \frac{1}{8}$$

$$P(3 \text{ Tails}) = {}^3C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$P(\text{Neither 3H nor 3T}) = 1 - \frac{1}{8} - \frac{1}{8} = \frac{3}{4}$$

Ex. Four coins are tossed simultaneously, what is the probability that at least three coins get heads or tails?

HINTS Sample Space = $\{HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT\}$

$$\text{Probability} = \frac{10}{16} = \frac{5}{8}$$

Alternatively

Total cases = $2^4 = 16$ No. of unfavourable cases = ${}^4C_2 = \frac{4!}{2! \times 2!} = \frac{4 \times 3 \times 2!}{2! \times 2!} = 6$ Favourable case = $16 - 6 = 10$ $\therefore$ Required Probability = $\frac{10}{16} = \frac{5}{8}$ **Ex.** A coin is tossed 5 times, what is a probability of getting exactly 2 heads?**HINTS** Total cases = $2^5 = 32$ No. of favourable cases = ${}^5C_2 = \frac{5!}{2! \times 3!} = \frac{5 \times 4 \times 3!}{2! \times 3!} = 10$ Probability = $\frac{10}{32} = \frac{5}{16}$

Alternatively

Probability = ${}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = \frac{10}{2^5} = \frac{5}{16}$ **Ex.** A coin is tossed 8 times, what is a probability of getting exactly 6 tails?**HINTS** Total cases = $2^8 = 256$ No. of favourable cases = ${}^8C_6 = \frac{8!}{2! \times 6!} = \frac{8 \times 7 \times 6!}{2! \times 6!} = 28$ Probability = $\frac{28}{256} = \frac{7}{64}$

Alternatively

P(Exactly 6 tails) = ${}^8C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^2 = \frac{28}{2^8} = \frac{28}{256} = \frac{7}{64}$ **Ex.** A coin is tossed 9 times. Find the probability of at most 2 tails.**HINTS** Total cases = $2^9 = 512$ 0 Tail $\rightarrow {}^9C_0 = \frac{9!}{9! \times 0!} = 1$, 1 Tail $\rightarrow {}^9C_1 = \frac{9!}{8! \times 1!} = 9$ 2 Tails $\rightarrow {}^9C_2 = \frac{9!}{7! \times 2!} = 36$ $\therefore$ Total no. of favourable cases = $1 + 9 + 36 = 46$ Probability = $\frac{46}{512} = \frac{23}{256}$

Alternatively

P(1 Tail) = ${}^9C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^8 = \frac{9}{2^9}$ P(2 Tails) = ${}^9C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^7 = \frac{36}{2^9}$ P(Zero tail) = ${}^9C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^9 = \frac{1}{2^9}$ $\therefore$ P(At most two tails) = $\frac{9}{2^9} + \frac{36}{2^9} + \frac{1}{2^9} = \frac{46}{2^9} = \frac{23}{256}$ **Ex.** A coin is tossed 7 times. Find the probability of at least 2 tails.**HINTS** Total cases = $2^7 = 128$ 0T $\rightarrow {}^7C_0 = 1$, 1T $\rightarrow {}^7C_1 = 7$

2T, 3T, 4T, 5T, 6T, 7T ✓

Favourable cases = $128 - (1 + 7) = 120$ Probability = $\frac{120}{128} = \frac{15}{16}$

Alternatively

P(1 Tail) = ${}^7C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^6 = \frac{7}{2^7}$ P(0 Tail) = ${}^7C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^7 = \frac{1}{2^7}$ P(At least two tails) = $1 - P(1 \text{ Tail}) - P(0 \text{ Tail})$ $= 1 - \frac{7}{2^7} - \frac{1}{2^7} = 1 - \frac{8}{128} = \frac{120}{128} = \frac{15}{16}$ **Ex.** In a biased coin the probability of getting a head is 0.4, if we toss a coin 10 times, what is the probability of getting exactly three heads?**HINTS** Direct shortcut, Probability = ${}^nC_r (P)^r (1-P)^{n-r}$ P(Head) = $\frac{2}{5}$ P(Not head) = $\frac{3}{5}$ Probability = ${}^{10}C_3 \left(\frac{2}{5}\right)^3 \left(\frac{3}{5}\right)^7 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} \times \frac{2^3 \times 3^7}{5^{10}} = \frac{2^6 \times 3^8}{5^9}$

Based on Dice

When a dice is thrown once

Sample space

 $= \{1, 2, 3, 4, 5, 6\} = 6$ **Ex.** A dice is thrown, find the probability of:

- (i) A prime number
- (ii) A multiple of 2 or 3
- (iii) A number greater than 4

HINTS Sample space = $\{1, 2, 3, 4, 5, 6\} = 6$

- (i) Prime number = 2, 3, 5

Probability = $\frac{3}{6} = \frac{1}{2}$

- (ii) Multiple of 2 or 3 = 2, 3, 4, 6

Probability = $\frac{4}{6} = \frac{2}{3}$

- (iii) Numbers greater than 4 = 5, 6

Probability = $\frac{2}{6} = \frac{1}{3}$ **Ex.** A dice is thrown, find the probability of the following outcomes-

- (i) Getting an even number
- (ii) Getting an odd number
- (iii) Getting a natural number less than 7
- (iv) Getting a natural number greater than 6

HINTS Sample space = $\{1, 2, 3, 4, 5, 6\} = 6$

- (i) Number of favourable outcomes = 3 (2, 4, 6)

 $\therefore P(E_1) = \frac{3}{6} = \frac{1}{2}$

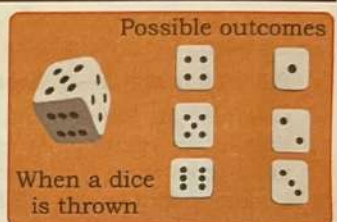
- (ii) $P(E_2) = P(\bar{E}_1) = 1 - \frac{1}{2} = \frac{1}{2}$ or else, we can say

 $P(E_2) = \frac{3}{6} = \frac{1}{2}$

- (iii) Here, number of favourable outcomes = 6

 $\therefore P(E) = 1 \rightarrow$ This is called a certain event as there is no chance of failure.

- (iv) Here number of favourable outcomes = 0

 $\therefore P(E) = 0 \rightarrow$ This is called impossible event, i.e., there is no chance of success.

Probability

When two dice are thrown simultaneously

Total number of possible cases = $6^2 = 36$

Sample space

		→ Second Throw					
		1	2	3	4	5	6
First Throw ↓	1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
	2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
	3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
	4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
	5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
	6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

If $2 \leq \text{Sum} \leq 6$, Number of cases = $\text{Sum} - 1$
 $7 \leq \text{Sum} \leq 12$, Number of cases = $13 - \text{Sum}$

Sum	Cases	No. of cases	Probability
2	(1,1)	1	$\frac{1}{36}$
3	(1,2), (2,1)	2	$\frac{2}{36}$
4	(1,3), (2,2), (3,1)	3	$\frac{3}{36}$
5	(1,4), (2,3), (3,2), (4,1)	4	$\frac{4}{36}$
6	(1,5), (2,4), (3,3), (4,2), (5,1)	5	$\frac{5}{36}$
7	(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)	6	$\frac{6}{36}$
8	(2,6), (3,5), (4,4), (5,3), (6,2)	5	$\frac{5}{36}$
9	(3,6), (4,5), (5,4), (6,3)	4	$\frac{4}{36}$
10	(4,6), (5,5), (6,4)	3	$\frac{3}{36}$
11	(5,6), (6,5)	2	$\frac{2}{36}$
12	(6,6)	1	$\frac{1}{36}$
Total		36	

Ex. A dice is thrown twice, what is the probability of getting a sum of 7?

HINTS Sum = 7

$$\text{Probability} = \frac{(13-7)}{36} = \frac{1}{6}$$

Ex. Two Dice are thrown simultaneously. Find the probability of-

- Sum of 8
- At least a sum of 9
- A doublet i.e., same number on both the dice
- A multiple of 2 on one dice and a multiple of 3 on the other dice

HINTS Total number of possible cases = $6^2 = 36$

- Sum = 8
 $\text{Probability} = \frac{(13-8)}{36} = \frac{5}{36}$
- At least a sum of 9 means sum can be 9, 10, 11 or 12.
 Favourable cases for 9 as sum = $13 - 9 = 4$

Favourable cases for 10 as sum = $13 - 10 = 3$

Favourable cases for 11 as sum = $13 - 11 = 2$

Favourable cases for 12 as sum = $13 - 12 = 1$

$$\text{Probability} = \frac{(4+3+2+1)}{36} = \frac{10}{36} = \frac{5}{18}$$

(iii) Favourable cases = (1,1) (2,2) (3,3) (4,4) (5,5) (6,6) = 6

$$\text{Probability} = \frac{6}{36} = \frac{1}{6}$$

(iv) Favourable cases = (2, 3) (2, 6) (4, 3) (4, 6) (6, 3) (6, 6)
 (3, 2) (3, 4) (3, 6) (6, 2) (6, 4) = 11

$$\text{Probability} = \frac{11}{36}$$

Ex. Two Dice are thrown Simultaneously. Find the probability that the number on the first dice is more than the number on the second dice?

HINTS No. of favourable cases

- (2, 1)
 (3, 1) (3, 2)
 (4, 1) (4, 2) (4, 3)
 (5, 1) (5, 2) (5, 3) (5, 4)
 (6, 1) (6, 2) (6, 3) (6, 4) (6, 5) = 15

$$\text{Probability} = \frac{15}{36} = \frac{5}{12}$$

Total cases = 36

There will be 6 such outcomes in which the same number will be obtained on both the dice. Out of the remaining 30 pairs, there will be 15 pairs in which the first number will be larger and 15 will be such pairs in which the second number will be larger. In this way, one can get number of favourable cases without writing down all the cases.

Ex. Two fair dice are thrown. Find the probability of getting a number whose sum is divisible by 2 or 4?

HINTS Sum is divisible by 2 or 4 means, sum can be 2, 4, 6, 8, 10, 12.

Favourable cases = (1 + 1, 1 + 3, 3 + 1, 2 + 2, 1 + 5, 5 + 1, 2 + 4, 4 + 2, 3 + 3, 2 + 6, 6 + 2, 3 + 5, 5 + 3, 4 + 4, 5 + 5, 6 + 4, 4 + 6, 6 + 6) = 18 cases

$$\text{Probability} = \frac{18}{36} = \frac{1}{2}$$

Ex. Two fair dice are thrown. Find the probability of getting a number whose sum is divisible by 2 and 4?

HINTS Sum is divisible by 2 and 4 means, sum can be 4, 8, 12.

Favourable cases = (1 + 3, 3 + 1, 2 + 2, 2 + 6, 6 + 2, 3 + 5, 5 + 3, 4 + 4, 6 + 6) = 9 cases

$$\text{Probability} = \frac{9}{36} = \frac{1}{4}$$

Based on Playing Cards

Deck of Cards

52

Red Suit (26)

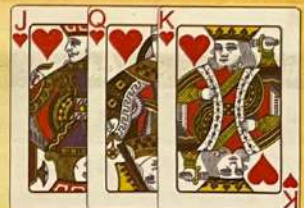
Black Suit (26)



Aces (4)

Number Cards (36)

Face Cards (12)



Ex. One card is drawn from a pack of 52 cards. What is the probability that the card drawn is—

- (i) An ace (ii) "10" of a red suit
(iii) A face card (iv) Either red or king
(v) Black and a queen

HINTS No. of possible cases = 52

- (i) Probability = $\frac{4}{52} = \frac{1}{13}$
(ii) Probability = $\frac{2}{52} = \frac{1}{26}$
(iii) Probability = $\frac{12}{52} = \frac{3}{13}$
(iv) Probability = $\frac{(26+4-2)}{52} = \frac{28}{52} = \frac{7}{13}$
(v) Probability = $\frac{2}{52} = \frac{1}{26}$

Ex. One card is drawn from a pack of 52 cards. What is the probability that the card drawn is—

- (i) A jack, queen or a King (ii) Neither a heart nor a King
(iii) Other than Ace (iv) Red King

HINTS No. of possible cases = 52

- (i) Probability = $\frac{(4+4+4)}{52} = \frac{3}{13}$
(ii) Heart = 13 & King = 4
Heart or King = $13 + 4 - 1 = 16$
Probability = $\frac{(52-16)}{52} = \frac{9}{13}$
(iii) Probability = $\frac{48}{52} = \frac{12}{13}$
(iv) Probability = $\frac{2}{52} = \frac{1}{26}$

Ex. Jacks, Queens, Kings and Aces of red color are removed. From the remaining cards a card is drawn at random. Find the probability, that the card drawn is a black queen?

HINTS Total no. of cards = 52

$$\text{No. of cards removed} = 4 \times 2 = 8$$

$$\therefore \text{Remaining Cards} = 52 - 8 = 44$$

$$\text{Probability} = \frac{2}{44} = \frac{1}{22}$$

Ex. A card is drawn at random from a deck of 52 cards. then find the probability that this card drawn is a king or a queen?

HINTS Probability = $\frac{{}^4C_1 + {}^4C_1}{{}^{52}C_1} = \frac{4+4}{52} = \frac{2}{13}$

Ex. Three cards are drawn at random from 52 cards. Find the probability of these cards being a king, a queen and jack.

HINTS

$$P = \frac{{}^4C_1 \times {}^4C_1 \times {}^4C_1}{{}^{52}C_3} = \frac{4 \times 4 \times 4}{\frac{52 \times 51 \times 50}{3 \times 2 \times 1}} = \frac{64}{22100} = \frac{16}{5525}$$

Ex. Find the probability that when a hand of 3 cards is drawn from a well shuffled deck. What is the probability that it has:

- (i) All face cards (ii) At least one king

HINTS

$$(i) \text{ Probability} = \frac{{}^{12}C_3}{{}^{52}C_3} = \frac{\frac{12 \times 11 \times 10}{3 \times 2 \times 1}}{\frac{52 \times 51 \times 50}{3 \times 2 \times 1}} = \frac{11}{1105}$$

$$(ii) P(\text{At least one king}) = 1 - P(\text{No king}) \\ = 1 - \frac{{}^{48}C_3}{{}^{52}C_3} = 1 - \frac{48 \times 47 \times 46}{52 \times 51 \times 50} \\ = 1 - \frac{4324}{5525} = \frac{1201}{5525}$$

Ex. A person draws 3 cards from a pack of 52 cards. Find the probability of getting exactly two red cards?

HINTS

$$\text{Probability} = \frac{{}^{26}C_2 \times {}^{26}C_1}{{}^{52}C_3} = \frac{\frac{26 \times 25}{2 \times 1} \times 26}{\frac{52 \times 51 \times 50}{3 \times 2 \times 1}} = \frac{13}{34}$$

Ex. A person draws 3 cards from a pack of 52 cards (with replacement). Find the probability of getting following outcomes:

- (i) All three red cards
(ii) That none of the cards is an ace
(iii) First card a Jack, second a black card and third card an even numbered card?

HINTS

$$(i) \text{ Probability} = \frac{26}{52} \times \frac{26}{52} \times \frac{26}{52} = \frac{1}{8}$$

$$(ii) \text{ Probability} = \frac{48}{52} \times \frac{48}{52} \times \frac{48}{52} = \frac{1728}{2197}$$

$$(iii) \text{ Required probability} = \frac{4}{52} \times \frac{26}{52} \times \frac{20}{52} = \frac{5}{338}$$

Based on Balls/Marbles

- We have all the marbles or balls collected in the jar or bag.

- The first thing we need to do is get the total number of colored marbles or balls in the jar or bag. We will use this number in all of our probability calculations.



Ex. There are 3 white and 4 black balls in an urn. A ball is selected randomly. What is the probability to be white?

HINTS Total number of balls = 3 + 4 = 7

$$\text{No. of white balls} = 3$$

$$\text{Probability} = \frac{3}{7}$$

Ex. A glass jar contains 6 white, 8 black, 4 red and 3 blue marbles. If a single marble is chosen at random from the jar, what is the probability that it is black or blue?

HINTS Probability = $\frac{3+8}{21} = \frac{11}{21}$

Ex. A jar contains 54 marbles each of which is either blue or green or white. The probability of selecting a blue marble at random from the jar is $\frac{1}{3}$, and the probability of selecting a green marble at random is $\frac{4}{9}$. How many white marbles does the jar contain?

HINTS Let, B = Blue marbles in jar,

G = Green marbles in jar, and

W = White marbles in jar.

$$\text{ATQ, } P(\text{Getting blue}) = \frac{1}{3}$$

$$\Rightarrow \frac{B}{54} = \frac{1}{3} \Rightarrow B = 18$$

$$\text{Again, } P(\text{Getting green}) = \frac{4}{9} \Rightarrow \frac{G}{54} = \frac{4}{9} \Rightarrow G = 24$$

$$\text{Total number of white marbles} = 54 - (18 + 24) = 12$$

Ex. There are 2 bags. First Bag contains 3 Red & 4 Black balls and the second bag contains 5 Red & 2 Black balls. One ball is picked from each of the bags. Find the probability that both balls are of :

- (i) Same color (ii) Different color

HINTS

(i) Bag - I

3 Red, 4 Black

Bag - II

5 Red, 2 Black

$$P(\text{Both Red}) = \frac{3}{7} \times \frac{5}{7} = \frac{15}{49}$$

$$P(\text{Both Black}) = \frac{4}{7} \times \frac{2}{7} = \frac{8}{49}$$

$$P(\text{Same Colour}) = \frac{15}{49} + \frac{8}{49} = \frac{23}{49}$$

$$(ii) P(\text{Different Colour}) = 1 - P(\text{Same colour})$$

$$= 1 - \frac{23}{49} = \frac{26}{49}$$

Ex. A box contains 5 green, 4 yellow and 3 white marbles. Three marbles are drawn at random. What is the probability that they are of the same color?

HINTS $P = \frac{{}^5C_3 + {}^4C_3 + {}^3C_3}{{}^{12}C_3} = \frac{10+4+1}{220} = \frac{15}{220} = \frac{3}{44}$

Probability

Ex. A bag contains 6 red balls, 5 yellow balls and 3 pink balls. Two balls are drawn at random. What is the probability that none of the balls are red?

HINTS Probability = $\frac{{}^8C_2}{{}^{14}C_2} = \frac{4}{13}$

Ex. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?

HINTS Probability = $\frac{{}^5C_2}{{}^7C_2} = \frac{5 \times 4}{7 \times 6} = \frac{10}{21}$

Ex. A bag contains 3 red balls, 5 yellow balls and 7 pink balls. If 1 ball is drawn at random from the bag, what is the Probability that it is pink or red in colour?

HINTS Probability = $\frac{{}^7C_1 + {}^3C_1}{{}^{15}C_1} = \frac{10}{15} = \frac{2}{3}$

Ex. A bag contains 6 red balls, 4 green balls and 8 yellow balls. Three balls are drawn at random from the bag. What is the probability that two balls are red and one ball is green?

HINTS Probability = $\frac{{}^6C_2 \times {}^4C_1}{{}^{18}C_3} = \frac{5}{68}$

Ex. A bag contains 4 white, 4 red and 2 green balls. Two balls are drawn at random. What is the probability that at least one balls is green?

HINTS Probability = $1 - \frac{{}^8C_2}{{}^{10}C_2} = 1 - \frac{28}{45} = \frac{17}{45}$

Ex. A bag contains 3 red, 6 white and 7 black balls. Three balls are drawn at random, then find the probability that all the three balls are black?

HINTS Probability = $\frac{{}^7C_3}{{}^{16}C_3} = \frac{1}{16}$

Ex. A basket contains three blue and four red balls. If three balls are drawn at random from the basket, what is the probability that all three are either blue or red?

HINTS Probability = $\frac{{}^3C_3 + {}^4C_3}{{}^7C_3} = \frac{1}{7}$

Based on Set Theory & Venn diagram

Mutually Exclusive Events

If two events are such that they can't occur simultaneously for any random experiment are said to be mutually exclusive events.

If X and Y are two mutually exclusive events,

$$X \cap Y = \phi \Rightarrow P(X \cap Y) = 0$$

For example, events in rolling of a dice are "Even face" and "Odd face" which are known as mutually exclusive events. But "Odd-face" and "Multiple of 3" are not mutually exclusive, because when "Face-3" occurs both the events "Odd face" and "Multiply of 3" are said to be occurred simultaneously.

Mutually Non-Exclusive Events

Two events A and B are said to be mutually non-exclusive events if both the events A and B have at least one common outcome between them.

The events A and B can't prevent the occurrence of one another so from here we can say that the events A and B have something common in them. When a dice rolled,

Let 'X' is denoted as event of getting an 'Odd-face' and 'Y' is denoted as event of getting 'Less than 4'

$$X = \{1, 3, 5\}$$

$$Y = \{1, 2, 3\}$$

Between the events X and Y the common outcomes are 1 and 3. Therefore, the events X and Y are compatible events or mutually non-exclusive.

Addition Theorem Based on Mutually Exclusive Events

If X and Y are two mutually exclusive events, then the probability of 'X union Y' is as below-

$$P(X \cup Y) = P(X) + P(Y)$$

Addition Theorem Based on Mutually Non-Exclusive Events

If X and Y are two mutually Non- Exclusive Events, then the probability of 'X union Y' is as below-

$$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$$

Independent Events

Two events are said to be independent if the occurrence of one event does not affect the probability of the occurrence of the other event.

In other words, events A and B are independent if:

$$P(A \cap B) = P(A) \times P(B)$$

Independent Events	Mutually Exclusive Events
Occurrence of one does not affect the other.	Occurrence of one prevents the other.
Can happen at the same time.	Can not happen at the same time.
$P(A \cap B) = P(A) \times P(B)$	$P(A \cap B) = 0$
No dependency.	Highly dependent (if one happens, the other can't).
If two coins are tossed, flipping a head on the first coin is independent of flipping a head on the second coin i.e head may appear on both coins at the same time.	If flip a coin one time, the head and tail cannot appear simultaneously. So here head and tail are mutually exclusive.

Important Result

$P(A) \rightarrow$ Probability of event A

$P(B) \rightarrow$ Probability of event B

$P(A \cup B) \rightarrow$ Probability of event A or B

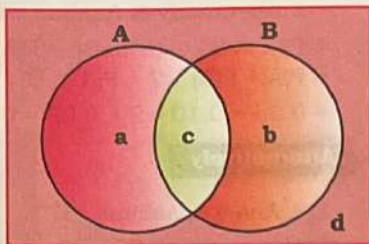
$P(A \cap B) \rightarrow$ Probability of event A and B

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$
- $P(A') = 1 - P(A)$
- $P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B)$
- $P(A' \cup B') = P(A \cap B)' = 1 - P(A \cap B)$

Ex. A and B are two events such that $P(A) = 0.54$, $P(B) = 0.69$ and $P(A \cap B) = 0.35$, then find $P(A \cap B')$

HINTS

$$\begin{aligned} P(A) &= a + c = 0.54 \\ P(A \cap B) &= c = 0.35 \\ P(A \cap B') &= a \\ &= P(A) - P(A \cap B) \\ &= 0.54 - 0.35 = 0.19 \end{aligned}$$



Ex. Tossing a fair coin and rolling a fair six-sided dice at the same time. Find the probability of getting Head on coin and 4 on dice?

HINTS

Let: A = Event that the coin shows Head,
B = Event that the dice shows 4.
The outcome of the coin toss does not affect the dice roll, and vice versa.
Probability of A = $\frac{1}{2}$, Probability of B = $\frac{1}{6}$
Probability of both A and B happening together:
 $P(A \cap B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

Ex. If A and B are two events such that, $P(A \cup B) = \frac{5}{6}$, $P(A) = \frac{1}{2}$ & $P(B) = \frac{1}{3}$ then the events A and B are ____.

HINTS

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $\frac{5}{6} = \frac{1}{2} + \frac{1}{3} - P(A \cap B) \Rightarrow P(A \cap B) = 0$
Since $P(A \cap B)$ is zero so, A and B are mutually exclusive events.

Ex. If A and B are two events such that $P(A \cup B) = \frac{3}{4}$, $P(A \cap B) = \frac{1}{4}$ & $P(B) = \frac{1}{2}$, then the events A and B are ____.

HINTS

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $\frac{3}{4} = P(A) + \frac{1}{2} - \frac{1}{4}$
 $P(A) = \frac{2}{4} = \frac{1}{2}$
If A and B are independent event they must satisfy the following:
 $P(A \cap B) = P(A) \times P(B)$
 $\frac{1}{4} = \frac{1}{2} \times \frac{1}{2}$
Since, it's satisfy this condition
Hence, it is independent.

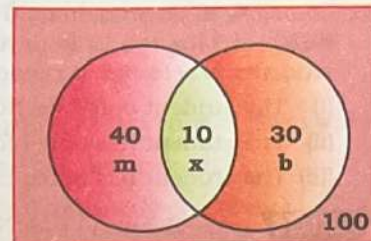
Ex. In Class XI of a school, 40% of the students study Mathematics and 30% study Biology. 10% of the class study both Mathematics and Biology. If a student is selected at random from the class, find the probability that he will be studying Mathematics or Biology.

HINTS

$$\begin{aligned} P(M) &= 40\% = \frac{2}{5} \text{ \& } P(B) = 30\% = \frac{3}{10} \\ P(M \text{ and } B) &= P(M \cap B) = 10\% = \frac{1}{10} \\ P(M \text{ or } B) &= P(M \cup B) = P(M) + P(B) - P(M \cap B) \\ &= \frac{2}{5} + \frac{3}{10} - \frac{1}{10} = \frac{4+3-1}{10} = \frac{6}{10} = 0.6 \end{aligned}$$

Alternatively

$$\begin{aligned} P(M \cup B) &= \frac{m+b-x}{100} \\ &= \frac{40+30-10}{100} = 0.6 \end{aligned}$$



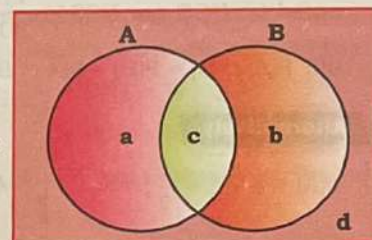
Ex. In an entrance test that is graded on the basis of two examinations, the probability of a randomly chosen student passing the first examination is 0.8, and the probability of passing the second examination is 0.7. The probability of passing at least one of them is 0.95. What is the probability of passing both?

HINTS

Let, Test - I = A, Test - II = B
Here, $P(A) = 0.8$, $P(B) = 0.7$ & $P(A \cup B) = 0.95$
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $\Rightarrow 0.95 = 0.8 + 0.7 - P(A \cap B)$
 $\Rightarrow P(A \cap B) = 1.5 - 0.95 = 0.55$

Alternatively

$P(A) = a + c = 0.8$
 $P(B) = b + c = 0.7$
 $P(\text{At least one}) = a + b + c = 0.95$
 $P(\text{Both passed}) = c$
 $\Rightarrow P(\text{Both passed}) = c$



$$\begin{aligned} P(A \cap B) &= P(A) + P(B) - P(\text{At least one}) \\ \Rightarrow P(\text{Both passed}) &= 0.8 + 0.7 - 0.95 = 0.55 \end{aligned}$$

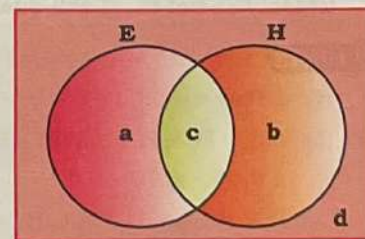
Ex. The probability that a student will pass the final examination in both English and Hindi is 0.5, and the probability of passing neither is 0.1. If the probability of passing the English examination is 0.75, what is the probability of passing the Hindi examination?

HINTS

$P(E) = 0.75$, $P(E \cap H) = 0.5$ &
 $P(\text{Neither E nor H}) = P(E' \cap H') = 0.1$
We know that, $E' \cap H' = (E \cup H)'$
 $P(E \cup H) = 0.1$
 $P(E \cup H) = 1 - P(E \cup H)' = 1 - 0.1 = 0.9$
 $P(E \cup H) = P(E) + P(H) - P(E \cap H)$
 $P(H) = 0.9 - 0.75 + 0.5 = 0.65$
 $\therefore$ Probability of passing the Hindi examination = 0.65

Alternatively

$P(E) = a + c = 0.75$
 $P(H) = b + c$
 $P(E \text{ and } H \text{ both}) = c = 0.5$
 $P(\text{None of them passed}) = d = 0.1$
 $\Rightarrow d = 1 - (a + b + c)$
 $\Rightarrow 0.1 = 1 - (0.75 + b) \Rightarrow b = 0.15$
 $\therefore P(H) = b + c = 0.15 + 0.5 = 0.65$



Ex. In a class of 60 students, 30 opted for NCC, 32 opted for NSS and 24 opted for both NCC and NSS. If one of these students is selected at random, find the probability that

- The student opted for NCC or NSS
- The student has opted for neither NCC nor NSS
- The student has opted for NSS but not NCC

HINTS Let, Event A $\rightarrow$ for NCC, Event B $\rightarrow$ for NSS

$$P(A) = \frac{30}{60} = \frac{1}{2} \text{ \& } P(B) = \frac{32}{60} = \frac{8}{15} \text{ \& } P(A \cap B) = \frac{24}{60} = \frac{2}{5}$$

$$(i) P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ = \frac{1}{2} + \frac{8}{15} - \frac{2}{5} = \frac{19}{30}$$

$\therefore$ The probability that the selected students has opted for NCC or NSS is $\frac{19}{30}$.

$$(ii) P(\text{Neither A nor B}) = P(A' \cap B') = P(A' \cap B')$$

$$\& A' \cap B' = (A \cup B)' \Rightarrow P(A' \cap B') = P(A \cup B)'$$

$$\Rightarrow P(A' \cap B') = 1 - P(A \cup B) = 1 - \frac{19}{30} = \frac{11}{30}$$

$\therefore$ The probability that the selected students has neither opted for NCC nor NSS is $\frac{11}{30}$

$$(iii) P(A' \cap B) = P(B) - P(A \cap B) = \frac{8}{15} - \frac{2}{5} = \frac{2}{15}$$

Alternatively

$$P(A) = a + c = \frac{1}{2},$$

$$P(B) = b + c = \frac{8}{15},$$

$$P(A \cap B) = c = \frac{2}{5}$$

$$(i) P(A \text{ or } B) = a + b + c \\ = P(A) + P(B) - c = \frac{1}{2} + \frac{8}{15} - \frac{2}{5} = \frac{19}{30}$$

$$(ii) P(\text{Neither A nor B}) = d = 1 - (a + b + c) \\ = 1 - \frac{19}{30} = \frac{11}{30}$$

$$(iii) P(A' \cap B) = b = P(B) - c = \frac{8}{15} - \frac{2}{5} = \frac{2}{15}$$

Ex. Two students Anil and Ashima appeared in an examination. The probability that Anil will qualify the examination is 0.05 and that Ashima will qualify the examination is 0.10. The probability that both will qualify the examination is 0.02. Find the probability that-

- Both Anil and Ashima will not qualify the examination
- At least one of them will not qualify the examination
- Only one of them will qualify the examination.

HINTS Let, Anil = A, Ashima = B

Here, $P(A) = 0.05$, $P(B) = 0.10$ & $P(A \cap B) = 0.02$

$$(a) P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ = 0.05 + 0.10 - 0.02 = 0.13$$

$$P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.13 = 0.87$$

$$(b) P(\text{At least one of them will not qualify the examination}) \\ = 1 - P(\text{Both of them will qualify}) \\ = 1 - P(A \cap B) = 1 - 0.02 = 0.98$$

(c) Only one of them qualify

= (Anil will qualify & Ashima will not) 'OR' (Anil will not qualify & Ashima will)

$$= P\{(A \cap B') \cup (A' \cap B)\} = P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$= P(A) + P(B) - 2 \times P(A \cap B)$$

$$= 0.05 + 0.10 - 2 \times 0.02 = 0.15 - 0.04 = 0.11$$

Alternatively

Let, Anil = A, Ashima = B

$$P(A) = a + c = 0.05$$

$$P(B) = b + c = 0.10$$

$$P(A \cap B) = c = 0.02$$

$$(a) P(\text{Both disqualified}) = d$$

$$\Rightarrow d = 1 - (a + b + c) \Rightarrow d = 1 - (P(A) + P(B) - c)$$

$$\Rightarrow d = 1 - (0.05 + 0.10 - 0.02) \Rightarrow d = 0.87$$

Hence, The probability of both disqualify is 0.87.

$$(b) P(\text{At least one disqualified}) = a + b + d$$

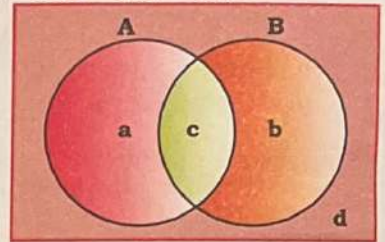
$$= P(A) + P(B) - 2c + d = 0.05 + 0.10 - 0.04 + 0.87 = 0.98$$

Hence, the probability of at least one disqualify is 0.98.

$$(c) P(\text{Only one qualified})$$

$$= a + b = P(A) + P(B) - 2c = 0.05 + 0.10 - 2 \times 0.02$$

$$= 0.15 - 0.04 = 0.11$$



Odds (In Favour & Against)

Odds in probability of a particular event, means the ratio between the number of favourable outcomes to the number of unfavourable outcomes.

Odds in favour of event A

Odds in favour of a particular event are given by Number of favourable outcomes to Number of unfavourable outcomes.

$$\frac{n(A)}{n(S) - n(A)} = \frac{P(A)}{1 - P(A)} \text{ OR } \frac{P(A)}{P(A')}$$

If, Odds in favour = 0 $\Rightarrow$ Impossible Event

Odds in favour = $\infty \Rightarrow$ Certain Event

Odds against event A

Odds against is given by Number of unfavourable outcomes to number of favourable outcomes.

$$\frac{n(S) - n(A)}{n(A)} = \frac{1 - P(A)}{P(A)} \text{ OR } \frac{P(A')}{P(A)}$$

If, Odds against = 0 $\Rightarrow$ Certain Event

Odds against = $\infty \Rightarrow$ Impossible Event

Ex. A fair dice is thrown once, what are the odds against the event that number 3 or 4 turns up?

HINTS When a fair dice is thrown once, the sample space S is given by

$$S = \{1, 2, 3, 4, 5, 6\} \Rightarrow n(S) = 6$$

Let, A = the event that number 3 or 4 turns up

$$\therefore A = \{3, 4\} \Rightarrow n(A) = 2$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

$$\therefore P(A') = 1 - P(A) = \frac{2}{3}$$

$$\therefore \text{Odds against event A} = P(A') : P(A)$$

$$= \frac{2}{3} : \frac{1}{3} = 2 : 1$$

Probability

Ex. If $P(A \cup B) = \frac{5}{9}$, $P(\bar{A} \cap \bar{B}) = \frac{13}{27}$, $P(A) = \frac{11}{18}$, then the odds against the event B are:

HINTS $P(A' \cup B') = P(A \cap B)$
 $\Rightarrow \frac{13}{27} = 1 - P(A \cap B)$
 $\Rightarrow P(A \cap B) = \frac{14}{27}$
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $\Rightarrow \frac{5}{9} = \frac{11}{18} + P(B) - \frac{14}{27}$
 $\Rightarrow P(B) = \frac{5}{9} - \frac{11}{18} + \frac{14}{27} = \frac{30 - 33 + 28}{54} = \frac{25}{54}$
Odds against the event B = $\frac{P(B')}{P(B)} = \frac{1 - \frac{25}{54}}{\frac{25}{54}}$
 $= \frac{29}{54} \times \frac{54}{25} = \frac{29}{25}$

Ex. The odds against a certain event is 5 : 2 and the odds in favour of another event, independent to the former is 6 : 5. Find the probability that

- (i) At least one of the events will occur.
(ii) None of the events will occur.

HINTS $\frac{P(A')}{P(A)} = \frac{5}{2} \Rightarrow P(A) = \frac{2}{7}$ & $P(A') = \frac{5}{7}$

$\frac{P(B)}{P(B')} = \frac{6}{5} \Rightarrow P(B) = \frac{6}{11}$ & $P(B') = \frac{5}{11}$

- (i) $P(\text{At least one event occurs}) = P(A \cup B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Since both the events are independent therefore

$P(A \cap B) = P(A) \times P(B)$

$\therefore P(A \cup B) = P(A) + P(B) - P(A) \times P(B)$

$P(A \cup B) = \frac{2}{7} + \frac{6}{11} - \frac{2}{7} \times \frac{6}{11}$
 $= \frac{22 + 42 - 12}{77} = \frac{52}{77}$

- (ii) $P(\text{None of the event occurs}) = 1 - P(A \cup B)$

$= 1 - \frac{52}{77} = \frac{25}{77}$

Ex. The odds against student X solving a statistics problem are 8 : 6 and odds in favour of student Y solving the same problem are 14 : 16. Find is the chance that neither of them solves the problem.

HINTS $\frac{P(X')}{P(X)} = \frac{8}{6} \Rightarrow P(X') = \frac{8}{14} = \frac{4}{7}$ & $P(X) = \frac{6}{14} = \frac{3}{7}$

$\frac{P(Y)}{P(Y')} = \frac{14}{16} = \frac{7}{8} \Rightarrow P(Y) = \frac{7}{15}$ & $P(Y') = \frac{8}{15}$

Required probability = $P(X' \cap Y')$

Since, X & Y are independent events. X' & Y' are also independent events.

Required probability $P(X' \cap Y') = P(X') \times P(Y')$

$= \frac{4}{7} \times \frac{8}{15} = \frac{32}{105}$

Ex. The odds against a husband who is 60 years old, living till he is 85 are 7 : 5. The odds against his wife who is now 56, living till she is 81 are 5 : 3.

- (i) Find the probability that at least one of them will be alive 25 years more from current age.
(ii) Find the probability that exactly one of them will be alive 25 years hence.

HINTS $\frac{P(H')}{P(H)} = \frac{7}{5} \Rightarrow P(H') = \frac{7}{12}$ & $P(H) = \frac{5}{12}$

$\frac{P(W')}{P(W)} = \frac{5}{3} \Rightarrow P(W') = \frac{5}{8}$ & $P(W) = \frac{3}{8}$

- (i) Required probability = $P(H \cup W)$

$= P(H) + P(W) - P(H) \times P(W)$

$= \frac{5}{12} + \frac{3}{8} - \frac{5}{12} \times \frac{3}{8}$

$= \frac{40 + 36 - 15}{96} = \frac{61}{96}$

- (ii) The required event will happen if husband is alive and wife is dead or husband is dead and wife is alive.

$\therefore$ The required probability = $P((H \cap W') \cup (H' \cap W))$

$= P(H \cap W') + P(H' \cap W)$

$\therefore$ Events $H \cap W'$ and $H' \cap W$ are mutually exclusive)

$= P(H) \times P(W') + P(H') \times P(W)$

$\therefore$ Events are independent)

$= \frac{5}{12} \times \frac{5}{8} + \frac{7}{12} \times \frac{3}{8} = \frac{46}{96} = \frac{23}{48}$

Conditional Probability

If X and Y are two events belonging to the same sample space of a random experiment, then the conditional probability of event X when event Y has already occurred is denoted by $P\left(\frac{X}{Y}\right)$ and calculated using the following formula.

$P\left(\frac{X}{Y}\right) = \frac{P(X \cap Y)}{P(Y)}, P(Y) \neq 0$

Similarly, Probability of Y When X has already occurred.

$P\left(\frac{Y}{X}\right) = \frac{P(X \cap Y)}{P(X)}, P(X) \neq 0$

Graphical representation of conditional probability:

If B has already occurred then our sample space must be within B

$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$



Now A can occur only within sample space B

Multiplication Theorem of Probability

If X and Y are independent, then probabilities of both X and Y is given by

$$P(X \cap Y) = P(X) \times P(Y).$$

General form:

$$P(A_1 \cap A_2 \cap A_3 \dots \dots \cap A_n) \\ = P(A_1) \times P\left(\frac{A_2}{A_1}\right) \times P\left(\frac{A_3}{A_1 \cap A_2}\right) \dots \dots \dots P\left(\frac{A_n}{A_1 \cap A_2 \dots \dots A_{n-1}}\right)$$

If A, B and C are three events of same sample space then,

$$P(A \cap B \cap C) = P(A) \times P\left(\frac{B}{A}\right) \times P\left(\frac{C}{A \cap B}\right)$$

Ex. If $P(A) = \frac{7}{13}$, $P(B) = \frac{9}{13}$ and $P(A \cap B) = \frac{4}{13}$, evaluate $P\left(\frac{A}{B}\right)$.

HINTS $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{4}{13} \times \frac{13}{9} = \frac{4}{9}$

Ex. If $P\left(\frac{A}{B}\right) = \frac{1}{2}$, $P\left(\frac{B}{A}\right) = \frac{1}{3}$ & $P(A \cap B) = \frac{1}{6}$ then find the value of $P(A \cup B)$.

HINTS $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

$$\therefore \frac{1}{2} = \frac{1}{6} \times \frac{1}{P(B)} \Rightarrow P(B) = \frac{1}{3}$$

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$$

$$\therefore \frac{1}{3} = \frac{1}{6} \times \frac{1}{P(A)} \Rightarrow P(A) = \frac{1}{2}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ = \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{5}{6} - \frac{1}{6} = \frac{2}{3}$$

Ex. A family has two children. What is the probability that both the children are boys given that at least one of them is a boy?

HINTS $S = \{(g, g), (g, b), (b, g), (b, b)\}$

E : Both the children are boys

$$E = \{(b, b)\} \Rightarrow P(E) = \frac{1}{4}$$

& F : At least one of the children is a boy.

$$F = \{(g, b), (b, g), (b, b)\} \Rightarrow P(F) = \frac{3}{4}$$

$$\& (E \cap F) = \{(b, b)\} \Rightarrow P(E \cap F) = \frac{1}{4}$$

$$\text{Now, } P\left(\frac{E}{F}\right) = \frac{P(E \cap F)}{P(F)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Alternatively

Let, S = At least one of them is a boy

$$\Rightarrow S = \{(g, b), (b, g), (b, b)\} = 3$$

Favourable outcome = (b, b) = 1

$$\text{Probability} = \frac{1}{3}$$

Ex. Ten cards numbered 1 to 10 are placed in a box, mixed up thoroughly and then one card is drawn randomly. If it is known that the number on the drawn card is more than 3, what is the probability that number is an even number?

HINTS A: The number on the card drawn is even

B : The number on the card drawn is greater than 3

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{4, 5, 6, 7, 8, 9, 10\}$$

$$A \cap B = \{4, 6, 8, 10\}$$

$$\text{Also, } P(A) = \frac{5}{10}, P(B) = \frac{7}{10}$$

$$P(A \cap B) = \frac{4}{10}$$

$$\therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{4}{10} \times \frac{10}{7} = \frac{4}{7}$$

Alternatively

Let, S = Number more than 3

$$\Rightarrow S = \{4, 5, 6, 7, 8, 9, 10\} = \text{Total 7 number}$$

Favourable outcomes = (4, 6, 8, 10) = Total 4 outcomes

$$\text{Probability} = \frac{4}{7}$$

Ex. Two dice are thrown. Find the probability that sum of the numbers coming up on them is 9, if it is known that the number 5 always occurs on the first dice.

HINTS Total no. of cases = $6^2 = 36$

A : The sum of the numbers coming up is 9.

B: Occurrence of 5 on the first dice.

$$A : \{(3, 6), (6, 3), (4, 5), (5, 4)\}$$

$$B : \{(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)\}$$

$$(A \cap B) = \{(5, 4)\}$$

$$P(A) = \frac{4}{36}, P(B) = \frac{6}{36} \& P(A \cap B) = \frac{1}{36}$$

$$\therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{1}{36} \times \frac{36}{6} = \frac{1}{6}$$

Alternatively

Let, S = Total cases with 5 on first dice

$$\Rightarrow S = \{(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)\}$$

Favourable outcome = Sum is 9 = (5, 4)

$$\text{Probability} = \frac{1}{6}$$

Ex. If $P(\bar{A}) = 0.3$, $P(B) = 0.4$, $P(A \cap \bar{B}) = 0.5$, find the $P\left(\frac{B}{A \cup \bar{B}}\right)$.

HINTS $P\left(\frac{B}{A \cup \bar{B}}\right) = \frac{P(B \cap (A \cup \bar{B}))}{P(A \cup \bar{B})} = \frac{P(A \cap B)}{P(A) + P(\bar{B}) - P(A \cap \bar{B})}$

$$= \frac{P(A) - P(A \cap \bar{B})}{P(A) + P(\bar{B}) - P(A \cap \bar{B})}$$

$$= \frac{0.7 - 0.5}{0.8} = \frac{1}{4}$$

Ex. If $P(A) = \frac{1}{2}P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$ find $P\left(\frac{A}{B}\right)$

HINTS $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{P(A \cup B)}{P(B)} = \frac{1 - P(A \cup B)}{1 - P(B)} = \frac{1 - \frac{7}{12}}{1 - \frac{1}{3}} = \frac{5}{8}$$

Ex. A committee of 4 students is selected at random from a group consisting 8 boys and 4 girls. Given that there is at least one girl in the committee, calculate the probability that there are exactly 2 girls on the committee.

HINTS A : At least one girl will be chosen

B : Exactly 2 girls will be chosen

A' : No girl is chosen $\Rightarrow$ 4 Boys are chosen

$$P(A') = \frac{{}^8C_4}{{}^{12}C_4} = \frac{70}{495} = \frac{14}{99} \Rightarrow P(A) = 1 - \frac{14}{99} = \frac{85}{99}$$

Now, $P(A \cap B) = P(2 \text{ boys \& } 2 \text{ girls})$

$$= \frac{{}^8C_2 \times {}^4C_2}{{}^{12}C_4} = \frac{6 \times 28}{495} = \frac{56}{165}$$

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{56}{165} \times \frac{99}{85} = \frac{168}{425}$$

Ex. There are 10 cards, 5 of these have 'I' & other 5 have 'T' printed on them. 3 cards are drawn one by one without replacement & are kept in same order, then probability of making the word IIT is.

HINTS A : Getting a card with mark I in first draw.

B : Getting a card with mark I in second draw.

C : Getting a card with mark T in third draw.

$$\therefore P(A \cap B \cap C) = P(A) \times P\left(\frac{B}{A}\right) \times P\left(\frac{C}{A \cap B}\right) = \frac{5}{10} \times \frac{4}{9} \times \frac{5}{8} = \frac{5}{36}$$

Baye's Theorem

We know that,

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \quad \dots(i)$$

$$\& P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} \quad \dots(ii)$$

From (i) & (ii), we get-

$$P\left(\frac{A}{B}\right) = \frac{P(A) \times P\left(\frac{B}{A}\right)}{P(B)}$$

This is called Baye's theorem. It is a relation between $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$. Here probability of event is not asked but it is given.

GENERAL FORM

$$P\left(\frac{E_i}{A}\right) = \frac{P(E_i) \times P\left(\frac{A}{E_i}\right)}{\sum_{r=1}^n P(E_r) \times P\left(\frac{A}{E_r}\right)}$$

Whenever the event of an experiment is given & probability of it being occurring through a particular path is asked, then Baye's theorem is applied. Paths are denoted by E's and event is denoted by A.



Ex. Amy has two bags. Bag I has 7 red and 4 blue balls and bag II has 5 red and 9 blue balls. Amy draws a ball at random and it turn out be red. Determine the probability that the ball was from the bag I.

HINTS A : The event of drawing a red ball

E_1 : Choosing the bag I, E_2 : Choosing the bag II

We know that the probability of choosing a bag is $\frac{1}{2}$, i.e.,

$$P(E_1) = P(E_2) = \frac{1}{2}$$

$$P(\text{Drawing a red ball from bag I}) = P\left(\frac{A}{E_1}\right) = \frac{7}{11}$$

$$P(\text{Drawing a red ball from bag II}) = P\left(\frac{A}{E_2}\right) = \frac{5}{14}$$

We need to determine the value of P(the ball drawn from

the bag I given that it is a red ball), i.e., $P\left(\frac{E_1}{A}\right)$.

by Baye's Theorem

$$P\left(\frac{E_1}{A}\right) = \frac{P\left(\frac{A}{E_1}\right) \times P(E_1)}{P\left(\frac{A}{E_1}\right) \times P(E_1) + P\left(\frac{A}{E_2}\right) \times P(E_2)} = \frac{\frac{7}{11} \times \frac{1}{2}}{\frac{7}{11} \times \frac{1}{2} + \frac{5}{14} \times \frac{1}{2}} = \frac{98}{153} = 0.64$$

Ex. Box I contains 2 white and 3 blue balls, box II contains 4 white and 2 blue balls and box III contains 1 white and 3 blue balls. Probabilities of box I, II & III being selected is 0.2, 0.3 and 0.5 respectively. A ball is drawn at random and is found to be white. Find the probability that it came from box I.

HINTS E_1 : Box I is selected, E_2 : Box II is selected

E_3 : Box III is selected, A : White ball is drawn

$$P(E_1) = 0.2, P(E_2) = 0.3 \& P(E_3) = 0.5$$

$$P\left(\frac{A}{E_1}\right) = \text{Probability of white ball from box I} = \frac{2}{5}$$

$$P\left(\frac{A}{E_2}\right) = \text{Probability of white ball from box II} = \frac{4}{6}$$

$$P\left(\frac{A}{E_3}\right) = \text{Probability of white ball from box III} = \frac{1}{4}$$

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1) \times P\left(\frac{A}{E_1}\right)}{P(E_1) \times P\left(\frac{A}{E_1}\right) + P(E_2) \times P\left(\frac{A}{E_2}\right) + P(E_3) \times P\left(\frac{A}{E_3}\right)} = \frac{0.2 \times \left(\frac{2}{5}\right)}{(0.2) \times \left(\frac{2}{5}\right) + (0.3) \times \left(\frac{4}{6}\right) + (0.5) \times \left(\frac{1}{4}\right)} = \frac{\frac{4}{50}}{\frac{243}{600}} = \frac{16}{81}$$

Probability

Ex. A man is known to speak truth 3 out of 4 times. He throws a dice and reports that it is a six. Find the probability that it is actually a six.

HINTS A : The event that the man reports a six.

E_1 : The event that 6 occurs

E_2 : The event that 6 doesn't occur

$$P(E_1) = \frac{1}{6} \text{ \& } P(E_2) = 1 - \frac{1}{6} = \frac{5}{6}$$

$P\left(\frac{A}{E_1}\right)$ = Probability when man reports that actually 6 has occurred and when it is true = $\frac{3}{4}$

$P\left(\frac{A}{E_2}\right)$ = Man reports that 6 has occurred and that has not occurred = $1 - \frac{3}{4} = \frac{1}{4}$

By Baye's theorem

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1) \times P\left(\frac{A}{E_1}\right)}{P(E_1) \times P\left(\frac{A}{E_1}\right) + P(E_2) \times P\left(\frac{A}{E_2}\right)}$$

$$= \frac{\frac{1}{6} \times \frac{3}{4}}{\frac{1}{6} \times \frac{3}{4} + \frac{5}{6} \times \frac{1}{4}} = \frac{3}{8}$$

Ex. A man is known to speak truth $\frac{3}{4}$ times. He draws a card and reports it is king. Find the probability that it is actually a king.

HINTS A : Event that the man reports that king is drawn from the pack of 52 cards

E_1 : King is drawn,

E_2 : King is not drawn

$$P(E_1) = \frac{1}{13} \text{ and } P(E_2) = 1 - \frac{1}{13} = \frac{12}{13}$$

$P\left(\frac{A}{E_1}\right)$ = Probability that the man says the truth that king is drawn when actually king is drawn = $\frac{3}{4}$

$P\left(\frac{A}{E_2}\right)$ = Probability that the man says that king is drawn when actually king is not drawn = $1 - \frac{3}{4} = \frac{1}{4}$

By baye's theorem-

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1) \times P\left(\frac{A}{E_1}\right)}{P(E_1) \times P\left(\frac{A}{E_1}\right) + P(E_2) \times P\left(\frac{A}{E_2}\right)}$$

$$= \frac{\frac{1}{13} \times \frac{3}{4}}{\frac{1}{13} \times \frac{3}{4} + \frac{12}{13} \times \frac{1}{4}} = \frac{3}{15} = \frac{1}{5} = 0.20$$

Miscellaneous

Ex. What is the probability of a prime number, if we select a number from first 20 natural numbers?

HINTS Natural numbers = 1, 2, 3 20

Prime numbers = 2, 3, 5, 7, 11, 13, 17, 19

$$\text{Probability} = \frac{8}{20} = \frac{2}{5}$$

Ex. A bag contains cards which are numbered from 2 to 90. A card is drawn at random from the bag. Find the probability that the card number is a perfect square.

HINTS Perfect square numbers from 2 to 90

= 4, 9, 16, 25, 36, 49, 64, 81

$$\text{Probability} = \frac{8}{90}$$

Ex. Find the probability that a number selected at random from first hundred natural numbers is a multiple of 3 or 5?

HINTS Multiple of 3 = $\frac{99}{3} = 33$

$$\text{Multiple of 5} = \frac{100}{5} = 20$$

$$\text{Multiple of both 3 \& 5} = \frac{90}{15} = 6$$

$$\therefore \text{Total no. of favourable conditions} = 33 + 20 - 6 = 47$$

$$\text{Probability} = \frac{47}{100}$$

Ex. A basket contains 10 apples and 20 oranges out of which 3 apples and 5 oranges are defective. If we choose two fruits at random, what is the probability that either both are oranges or both are non-defective?

HINTS $P = \frac{{}^{20}C_2 + {}^{22}C_2 - {}^{15}C_2}{{}^{30}C_2}$

$$= \frac{190 + 231 - 105}{435} = \frac{316}{435}$$

	A	O
D	3	5
N	7	15

Ex. A problem is given to three students whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved?

HINTS $P(\text{Solved}) = 1 - P(\text{Not solved})$

$$= 1 - \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = 1 - \frac{1}{4} = \frac{3}{4}$$

Ex. The probability that three students A, B and C solve a problem are $\frac{3}{7}$, $\frac{5}{9}$ and $\frac{1}{5}$ respectively. What is the probability that the problem is solved?

HINTS $P(\text{Solved}) = 1 - P(\text{Not solved})$

$$= 1 - \frac{4}{7} \times \frac{4}{9} \times \frac{4}{5} = 1 - \frac{64}{315} = \frac{251}{315}$$

Ex. A man and his wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $\left(\frac{1}{7}\right)$ and the probability of wife's selection is $\left(\frac{1}{5}\right)$. What is the probability that only one of them is selected?

HINTS $\text{Probability} = \frac{1}{7} \times \frac{4}{5} + \frac{6}{7} \times \frac{1}{5} = \frac{10}{35} = \frac{2}{7}$

Ex. A brother and a sister appear for an interview against two vacant posts in an office. The probability of the brother's selection is $\frac{1}{5}$ and that of the sister's selection is $\frac{1}{3}$. What is the probability that only one of them is selected?

HINTS Probability = $\frac{1}{5} \times \frac{2}{3} + \frac{4}{5} \times \frac{1}{3}$
 $= \frac{2}{15} + \frac{4}{15} = \frac{6}{15} = \frac{2}{5}$

Ex. A speaks truth in 80% of the cases and B speaks truth in 60% of the cases. Find the probability that they contradict each other in narrating an incident.

HINTS Let event A : A speaks the truth, event B : B speaks the truth.
 $\therefore P(A) = \frac{80}{100} = \frac{4}{5}$ and $P(B) = \frac{60}{100} = \frac{3}{5}$
 $\therefore P(A') = 1 - P(A) = 1 - \frac{4}{5} = \frac{1}{5}$
 and, $P(B') = 1 - P(B) = 1 - \frac{3}{5} = \frac{2}{5}$
 $\therefore P(\text{A and B contradict each other})$
 $= P(\text{A speaks the truth and B lies}) + P(\text{A lies and B speaks the truth})$
 $= P(A \cap B') + P(A' \cap B)$
 $= P(A) \times P(B') + P(A') \times P(B)$
 $= \left(\frac{4}{5} \times \frac{2}{5}\right) + \left(\frac{1}{5} \times \frac{3}{5}\right) = \frac{8}{25} + \frac{3}{25} = \frac{11}{25}$

Ex. A speaks truth in 75% of cases and B in 80% of cases. What is the probability that they likely to contradict each other, narrating the same incident?

HINTS Probability = $\frac{3}{4} \times \frac{1}{5} + \frac{1}{4} \times \frac{4}{5} = \frac{7}{20}$

Ex. A speaks the truth 5 out of 7 times and B speaks truth 8 out of 9 times. What is the probability that they contradict each other in stating the same fact?

HINTS Probability = $\frac{5}{7} \times \frac{1}{9} + \frac{2}{7} \times \frac{8}{9} = \frac{21}{63} = \frac{1}{3}$

Ex. In a cricket world cup the probability that India will win the cup is $\frac{1}{4}$. The probability of Pakistan winning the cup is $\frac{1}{5}$ and of Australia winning the cup is $\frac{1}{6}$. What is the probability that either of these 3 teams India, Pakistan and Australia would win the world cup? Only one team can win the cup.

HINTS Let, W = win, L = loss

$I_w = \text{Ind - win}$	$I_L = \text{Ind - loss}$
$P_w = \text{Pak - win}$	$P_L = \text{Pak - loss}$
$A_w = \text{Aust - win}$	$A_L = \text{Aus - loss}$

$P(I_w) = \frac{1}{4}$; $P(P_w) = \frac{1}{5}$; $P(A_w) = \frac{1}{6}$

World cup will end with a team as a winner if:

$(I_w \text{ and } P_L \text{ and } A_L) \text{ or } (I_L \text{ and } P_w \text{ and } A_L) \text{ or } (I_L \text{ and } P_L \text{ and } A_w)$

Probability = $\left(\frac{1}{4} \times \frac{4}{5} \times \frac{5}{6}\right) + \left(\frac{3}{4} \times \frac{1}{5} \times \frac{5}{6}\right) + \left(\frac{3}{4} \times \frac{4}{5} \times \frac{1}{6}\right)$
 $= \frac{20+15+12}{120} = \frac{47}{120}$

Ex. Two letters are chosen from English Alphabet. What is the probability that both are vowels?

HINTS Probability = $\frac{{}^5C_2}{{}^{26}C_2} = \frac{5 \times 4}{26 \times 25} = \frac{2}{65}$

Ex. The probability that a man who is 45 years old will be alive till he becomes 70 is $\frac{5}{12}$. The probability that his wife who is 40 years old will be alive till she becomes 65 is $\frac{3}{8}$. What is the probability that, after 25 years -

- (i) The couple will be alive
- (ii) Exactly one of them will be alive

HINTS Let, A = Event that man who is 45 will be alive till age 70
 B = Event that wife who is 40 will be alive till age 65

It is given that, $P(A) = \frac{5}{12}$, $P(B) = \frac{3}{8}$

$P(A') = 1 - P(A) = 1 - \frac{5}{12} = \frac{7}{12}$

$P(B') = 1 - P(B) = 1 - \frac{3}{8} = \frac{5}{8}$

Since A and B are independent events,

A' and B' are also independent events.

(i) Let Event C: Both man and his wife will be alive.

$P(C) = P(A \cap B) = P(A) \times P(B) = \frac{5}{12} \times \frac{3}{8} = \frac{5}{32}$

(ii) Let Event D: Exactly one of them will be alive.

$P(D) = P(A' \cap B) + P(A \cap B')$
 $= P(A') \times P(B) + P(A) \times P(B')$
 $= \left(\frac{7}{12} \times \frac{3}{8}\right) + \left(\frac{5}{12} \times \frac{5}{8}\right) = \frac{21}{96} + \frac{25}{96} = \frac{46}{96} = \frac{23}{48}$

Ex. A letter is taken out at random from 'ASSISTANT' and another is taken out from 'STATISTICS'. What is the probability that they are the same letter?

HINTS

AA	A
SSS	SSS
I	II
TT	TTT
N	C

Probability = $\frac{2}{9} \times \frac{1}{10} + \frac{3}{9} \times \frac{3}{10} + \frac{1}{9} \times \frac{2}{10} + \frac{2}{9} \times \frac{3}{10} = \frac{19}{90}$

Ex. A person can hit a target 5 times out of 8 shots. If he fires 10 shots, what is the probability that he will hit the target twice?

HINTS Direct shortcut, Probability = ${}^nC_r (P)^r (1 - P)^{n-r}$

$P(\text{Hit}) = \frac{5}{8}$, $P(\text{not hit}) = \frac{3}{8}$

Probability = ${}^{10}C_2 \left(\frac{5}{8}\right)^2 \left(\frac{3}{8}\right)^8$

$= \frac{10 \times 9}{2 \times 1} \times \frac{5^2}{8^2} \times \frac{3^8}{8^8} = \frac{1125 \times 3^8}{8^{10}}$

Probability

Ex. A word consists of 9 letters; 5 consonants and 4 vowels. Three letters are chosen at random. What is the probability that more than one vowel will be selected?

HINTS Probability that more than one vowel

$$= \frac{{}^4C_2 \times {}^5C_1 + {}^4C_3}{{}^9C_3} = \frac{17}{42}$$

Alternatively

$$\begin{aligned} P(\text{More than 1 vowel}) &= 1 - P(\text{No vowel}) - P(1 \text{ Vowel}) \\ &= 1 - \frac{{}^5C_3}{{}^9C_3} - \frac{{}^5C_2 \times {}^4C_1}{{}^9C_3} = 1 - \frac{5}{42} - \frac{10}{21} = \frac{17}{42} \end{aligned}$$

Ex. Find the probability that a leap year will have:

- (i) At least 52 Sundays, (ii) Exactly 52 Sundays
- (iii) Exactly 53 Sundays

HINTS Leap year = 366 days = 52 weeks + 2 days

Remaining 2 days can be = MT, TW, WTh, ThF, FS, SS, SM = 7 (Total ways)

- (i) Since in a leap year 52 weeks are there, hence

$$P(\text{At least 52 Sundays}) = 1$$

- (ii) For exactly 52 Sundays, remaining 2 days can be = MT, TW, WTh, ThF, FS = 5

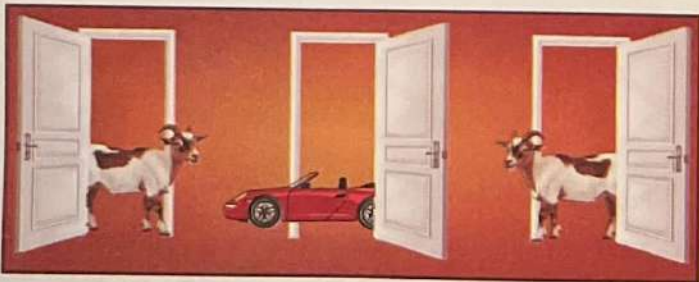
$$P(\text{Exactly 52 Sundays}) = \frac{5}{7}$$

- (iii) For exactly 53 Sundays, remaining 2 days can be = SS, SM = 2

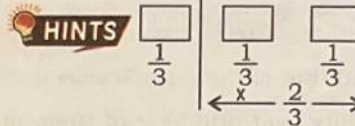
$$P(\text{Exactly 53 Sundays}) = \frac{2}{7}$$

Monty Hall Problems

The Monty Hall problem is a probability puzzle that is loosely based on the American game show "Let's Make a Deal", hosted by Monty Hall. While the problem itself was originally posed by Steve Selvin in a letter to the American Statistician in 1975, it gained widespread attention when it was presented in Marilyn vos Savant's "Ask Marilyn" column in Parade magazine in 1990. The problem is named after the show's host, Monty Hall, due to its connection to the show's format.

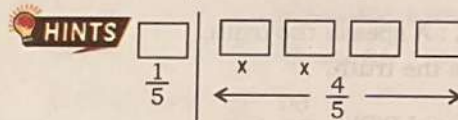


Ex. In the Monty Hall game, host Chunnilal presents you with 3 doors. One door has a car behind it and two other doors have goats behind them. After you choose one door, host Chunnilal (who knows what is behind which door) opens another door with a goat. So what is the probability of winning a car if you switch?



Winning chance is $\frac{2}{3}$

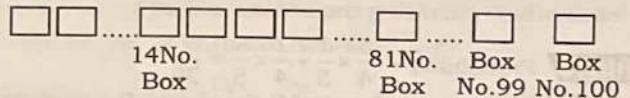
Ex. You have 5 doors, one has a car behind it, the rest have goats behind them. You choose one door, Mishra ji who knows what is behind each door opens the door with two goats. When you change your door the probability of winning is $\frac{x}{y}$. Find the value of $x \times y$.



Probability, if you switch your door = $\frac{4}{5} \times \frac{1}{2} = \frac{2}{5} = \frac{x}{y}$
 $\therefore x \times y = 10$

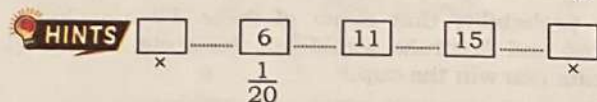
Ex. There are 100 boxes in the Monty Hall game. One box contains SSC Maths book and the rest are empty. Aditya Sir asked one of his students to select the box containing the book and the student selected box number 14. After Aditya Sir opened 98 boxes, now only box number 14 and box number 81 are closed. The book is now in one of those two boxes. So, if you change the box then what is a possibility of winning the book.

HINTS



if you select box number 81 then possibility of winning the book is 99%.

Ex. You are given 20 boxes one has gold coin. You pick box 6. The host opens 1 box at a time, all empty, until only 3 boxes remain box 6, box 11 and box 15. Now you are allowed to switch to either of the other two boxes. If you switch, what's your odds in favour of winning.



Probability, if you switch, $P(E) = \frac{19}{20} \times \frac{1}{2} = \frac{19}{40}$

$$P(E') = 1 - \frac{19}{40} = \frac{21}{40}$$

Odds in favour of winning = $\frac{\text{Favourable Outcomes}}{\text{Unfavourable Outcomes}}$

$$= \frac{P(E)}{P(E')} = \frac{\frac{19}{40}}{\frac{21}{40}} = \frac{19}{21}$$

CHAPTER

32

STATISTICS

Statistics:- The study of data gathering, analysis, interpretation, presentation and organization is known as statistics.

Frequency:- No. of observation falling in a particular class is called frequency of that class.

Frequency Distribution:- The distribution of frequency in various classes is known as frequency distribution.

Cumulative frequency:- The cumulative frequency of a particular class is the sum of frequency of that class and of those prior to it.

MEAN

The mean of a given data is the sum of all observations divided by the number of observations.

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

If $x_1, x_2, x_3, \dots, x_n$ are n values of a variable x , then the arithmetic mean or simply mean of these values is denoted by $\bar{x}$ (pronounced as x bar) and is defined as:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \quad \text{or} \quad \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Ex. A cricketer's scores in five ODI matches are as follows: 12, 34, 45, 50, 24. Find the mean.

HINTS

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

$$= \frac{12 + 34 + 45 + 50 + 24}{5} = \frac{165}{5} = 33$$

Ex. The mean of the following data is _____
23, 17, 20, 19, 21

HINTS

$$\bar{x} = \frac{23 + 17 + 20 + 19 + 21}{5} = \frac{100}{5} = 20$$

Alternatively

Let the mean of given data = 20

23, 17, 20, 19, 21

↓ ↓ ↓ ↓ ↓

Deviation → +3, -3, 0, -1, +1

Here, Net deviation = 3 - 3 + 0 - 1 + 1 = 0

∴ Mean = 20

Ex. If the mean of six observations 5, 7, 9, α , 11 and 12 is 9 then the value of α is :

HINTS

$$\bar{x} = \frac{5 + 7 + 9 + \alpha + 11 + 12}{6} = \frac{44 + \alpha}{6} = 9$$

$$\Rightarrow 44 + \alpha = 54 \Rightarrow \alpha = 10$$

Alternatively

Let the mean of given data = 9

5, 7, 9, α , 11, 12

↓ ↓ ↓ ↓ ↓

Deviation → -4, -2, 0, x , +2, +3

$$\Rightarrow x - 1 = 0 \Rightarrow x = 1$$

$$\therefore \alpha = 9 + x = 9 + 1 = 10$$

For Discrete Frequency Distribution

If x takes values $x_1, x_2, \dots, x_n$ with corresponding frequencies $f_1, f_2, \dots, f_n$ respectively, then arithmetic mean of these

$$\text{values is: } \bar{x} = \frac{f_1 x_1 + f_2 x_2 + f_3 x_3 + \dots + f_n x_n}{f_1 + f_2 + f_3 + \dots + f_n}$$

$$\text{or } \bar{x} = \frac{\sum_{i=1}^n f_i x_i}{N} \quad \text{where } N = \sum_{i=1}^n f_i = f_1 + f_2 + f_3 + \dots + f_n$$

Ex. Find the mean of the following distribution:

(x)	4	6	9	10	15
(f)	5	10	10	7	8

HINTS

f	x	fx
5	4	20
10	6	60
10	9	90
7	10	70
8	15	120
$\Sigma f = 40$		$\Sigma fx = 360$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{360}{40} = 9$$

Alternatively

Let the mean of given data = 9

f	x	Deviation d = x - A	f × d
5	4	-5	-25
10	6	-3	-30
10	9 = A	0	0
7	10	+1	+7
8	15	+6	+48
$\Sigma f = 40$			$\Sigma fd = 0$

∴ Mean = 9

Ex. If the mean is 6, find the value of p .

(x)	2	4	6	10	$p+5$
(f)	3	2	3	1	2

HINTS

f	x	fx
3	2	6
2	4	8
3	6	18
1	10	10
2	$p+5$	$2p+10$
$\Sigma f = 11$		$\Sigma fx = 52+2p$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} \Rightarrow \frac{52+2p}{11} = 6 \Rightarrow 52+2p = 66 \Rightarrow 2p = 14 \Rightarrow p = 7$$

Statistics

Ex. If the mean of the following data is 15, then find the value of k.

(x)	5	10	15	20	25
(f)	6	k	6	10	5

HINTS

f	x	fx
6	5	30
k	10	10k
6	15	90
10	20	200
5	25	125
$\Sigma f = 27+k$		$\Sigma fx = 445+10k$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} \Rightarrow \frac{445+10k}{27+k} = 15$$

$$\Rightarrow 445 + 10k = 405 + 15k$$

$$\Rightarrow 5k = 40 \Rightarrow k = 8$$

Mean of a Continuous Frequency Distribution

Ex. Find the mean of the following frequency distribution:

Class interval	0-10	10-20	20-30	30-40	40-50
No. of workers (f)	7	10	15	8	10

HINTS

No. of Workers (f)	Class Int.	x	fx
7	0-10	5	35
10	10-20	15	150
15	20-30	25	375
8	30-40	35	280
10	40-50	45	450
$f = 50$			$\Sigma fx = 1290$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{1290}{50} = 25.8$$

Alternatively

Let, Mean = 25

No. of Workers (f)	Class Int.	x	Deviation $d = x - A$	$f \times d$
7	0-10	5	-20	-140
10	10-20	15	-10	-100
15	20-30	25 = A	0	0
8	30-40	35	+10	80
10	40-50	45	+20	200
$\Sigma f = 50$				$\Sigma fd = +40$

$$\text{Net Deviation} = \frac{+40}{50} = +0.8$$

$$\therefore \text{Mean} = 25 + 0.8 = 25.8$$

Ex. The mean of the following distribution is 26, then what is the value of k?

Class	0-10	10-20	20-30	30-40	40-50
Frequency	8	10	k	6	12

HINTS

Frequency (f)	Class Int.	x	fx
8	0-10	5	40
10	10-20	15	150
k	20-30	25	25k
6	30-40	35	210
12	40-50	45	540
$\Sigma f = 36 + k$			$\Sigma fx = 940 + 25k$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} \Rightarrow \frac{940+25k}{36+k} = 26$$

$$\Rightarrow 940 + 25k = 936 + 26k \Rightarrow k = 4$$

Alternatively

$\therefore$ Mean = 26

Frequency (f)	Class Int.	x	Deviation $d = x - A$	$f \times d$
8	0-10	5	-21	-168
10	10-20	15	-11	-110
k	20-30	25	-1	-k
6	30-40	35	+9	54
12	40-50	45	+19	228
$\Sigma f = 36 + k$				$\Sigma fd = 4 - k$

$$4 - k = 0 \Rightarrow k = 4$$

MEDIAN

The value of the middle most observation, obtained after arranging the data in ascending or descending order, is called the median of the data.

Ex. Consider the data: 4, 4, 6, 3, 2. Find the median

HINTS

Let's arrange this data in ascending order:

2, 3, 4, 4, 6. There are 5 observations.

Thus, median = middle value i.e. 4.

How to find median

Step I: Arrange the observations $x_1, x_2, \dots, x_n$ in ascending or descending order of magnitude.

Step II: Determine the total number of observations, say, n

Step III: If n is odd, then median is the value of $\left(\frac{n+1}{2}\right)^{\text{th}}$ observation.

- If n is even, then median is the arithmetic mean of the values of $\left(\frac{n}{2}\right)^{\text{th}}$ and $\left(\frac{n}{2}+1\right)^{\text{th}}$ observations.

Ex. The median of the following data will be _____.
32, 25, 33, 27, 35, 29 and 30

HINTS Arrange the given data in ascending order

25, 27, 29, 30, 32, 33, 35

Here, $n = 7$ which is odd

$$\therefore \text{Median} = \left(\frac{7+1}{2} \right)^{\text{th}} = 4^{\text{th}} \text{ Observation} = 30$$

Ex. The median of a set of observations 15, 16, 18, 22, $x+2$, $x+3$, 26, 27, 30 arranged in ascending order is 24, then find the value of x .

HINTS Given set of observations is -

15, 16, 18, 22, $x+2$, $x+3$, 26, 27, 30

Here, $n = 9$

$$\therefore \text{Median} = 5^{\text{th}} \text{ observation} = x+2$$

ATQ,

$$x+2 = 24 \Rightarrow x = 22$$

Ex. The median of observations $k - \frac{3}{2}$, $k+2$, $k-1$, $k+4$,

$k + \frac{1}{2}$, $k-3$, $k + 4\frac{1}{2}$ is ____.

HINTS $k - \frac{3}{2}$, $k+2$, $k-1$, $k+4$, $k + \frac{1}{2}$, $k-3$, $k + 4\frac{1}{2}$

Put $k = 0$

-1.5, 2, -1, 4, 0.5, -3, 4.5

Ascending order = -3, -1.5, -1, 0.5, 2, 4, 4.5 $\Rightarrow n = 7$

$$\therefore \text{Median} = 4^{\text{th}} \text{ observation} = 0.5 = \frac{1}{2} + k$$

Median of Discrete Frequency Distribution

Step I: Find the cumulative frequencies (c.f.)

Step II: Find $\frac{N}{2}$, where $N = \sum_{i=1}^n f_i$

Step III: See the cumulative frequency (c.f.) just greater than $\frac{N}{2}$ and determine the corresponding value of the variable.

Step IV: The value obtained in step III is the median.

Ex. Find the median for the following frequency distribution:

x	1	2	3	4	5	6	7	8	9
f	8	10	11	16	20	25	15	9	6

HINTS

x	f	cf
1	8	8
2	10	8 + 10 = 18
3	11	18 + 11 = 29
4	16	29 + 16 = 45
5	20	45 + 20 = 65
6	25	65 + 25 = 90
7	15	90 + 15 = 105
8	9	105 + 9 = 114
9	6	114 + 6 = 120
Total	$\Sigma f = 120 = N$	

$$\frac{N}{2} = \frac{120}{2} = 60$$

c.f. just more than $\frac{N}{2}$ is 65,

$\therefore$ Median = 5

"Arrange the values of the variable in ascending or descending order."

Ex. Consider the following frequency distribution:

x	8	5	6	10	9	4	7
f	6	4	5	8	9	6	4

What is the median for the distribution?

HINTS To find median we need to arrange the values of ' x ' in ascending order.

x	Frequency (f)	Cumulative Frequency (c.f.)
4	6	6
5	4	6 + 4 = 10
6	5	10 + 5 = 15
7	4	15 + 4 = 19
8	6	19 + 6 = 25
9	9	25 + 9 = 34
10	8	34 + 8 = 42
Total	$\Sigma f = 42 = N$	

$$\text{Here, } \frac{N}{2} = \frac{42}{2} = 21$$

c.f. just more than $\frac{N}{2}$ is 25, $\Rightarrow \therefore$ Median = 8

Median of a continuous frequency Distribution

Step I: Prepare the cumulative frequency table

Step II: Find the median class, i.e. the class whose cumulative frequency is equal or just greater than $\left(\frac{N}{2} \right)$

Step III: The median value is given by the formula

$$\text{Median} = l + \left(\frac{\left(\frac{N}{2} \right) - cf^-}{f} \right) \times h$$

Where,

l = Lower limit of the median class

N = Total frequency

f = Frequency of the median class

h = Width of the median class

cf^- = Cumulative frequency of the class preceding the median class

Let's take an example on finding median of continuous data, that is the only case where finding median is sometimes challenging

Statistics

Ex. Median, for the following distribution is

Wages (Rs.)	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
No. of labour	22	38	46	35	20

HINTS

Class	f	Cf
0-10	22	22
10-20	38	60
20-30	46	106
30-40	35	141
40-50	20	161
	N = 161	

Here, $\frac{N}{2} = \frac{161}{2} = 80.5 \Rightarrow$ Median class is 20 - 30.

$$\text{Median} = l + \left(\frac{\left(\frac{N}{2} \right) - cf^-}{f} \right) \times h$$

$$= 20 + \left(\frac{80.5 - 60}{46} \right) \times 10 = 24.46$$

Ex. Calculate the median from the following distribution:

Class	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45
Frequency	5	6	15	10	5	4	2	2

HINTS

Class	f	cf
5-10	5	5
10-15	6	11
15-20	15	26
20-25	10	36
25-30	5	41
30-35	4	45
35-40	2	47
40-45	2	49
	N = 49	

Here, $\frac{N}{2} = \frac{49}{2} = 24.5$

$\Rightarrow$ Median class = 15 - 20

$$\text{Median} = l + \left(\frac{\left(\frac{N}{2} \right) - cf^-}{f} \right) \times h$$

$$= 15 + \left(\frac{24.5 - 11}{15} \right) \times 5 = 19.5$$

Mode

The mode or modal value of a distribution is that value of the variable for which the frequency is maximum.

Ex. What will be the mode of the following data?

13, 15, 31, 12, 27, 13, 27, 30, 27, 28 and 16

HINTS Mode = 27 (Because 27 comes maximum no. of times)

Ex. What is the mode of the given data?

5, 7, 9, 7, 3, 7, 5, 7, 8, 6, 7

HINTS Mode = 7 (Because 7 comes maximum no. of times)

Computation of Mode for a continuous frequency distribution

In continuous frequency distribution the computation of mode is done by the following formula

$$\text{Mode } (M_0) = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

l = Lower limit of class

f_1 = Frequency of modal class

f_0 = Frequency of the class just preceding the modal class

f_2 = Frequency of the class just following the modal class

i = Class width of modal class

Ex. A set of data presented in the form of a frequency distribution table with class intervals and their respective frequencies had a mode of 48.5. The lower boundary of the modal class was 46.5; the frequency of the modal class was 34; the frequency of the class interval just preceding the modal class was 32 and the frequency of the class interval just succeeding the modal class was 25. What was the width of the modal class?

HINTS Here we have, $M_0 = 48.5$

$l = 46.5, f_1 = 34, f_0 = 32, f_2 = 25$

$$\text{Mode } (M_0) = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

$$\Rightarrow 48.5 = 46.5 + \frac{34 - 32}{2 \times 34 - 32 - 25} \times i$$

$$\Rightarrow i = \frac{2 \times 11}{2} = 11$$

Ex. Given below is the data of the age of the various children. What is the difference between the mean and mode of the ages?

Age (years) (X)	9	12	5	8	7	6
No. of Children (F)	1	6	9	7	4	3

HINTS

Age (Years) (X)	No. of Children (F)	X × F
9	1	9
12	6	72
5	9	45
8	7	56
7	4	28
6	3	18
	$\Sigma F = 30$	$\Sigma FX = 228$

$$\text{Mean} = \frac{228}{30} = 7.6$$

& Mode = 5 ($\because$ frequency of 5 is maximum)

Difference between mean & mode = $7.6 - 5 = 2.6$

Ex. For the following grouped frequency distribution, find the mode:

Class	3-6	6-9	9-12	12-15	15-18	18-21	21-24
Frequency	2	5	10	23	21	12	3

HINTS Class of maximum frequency (23) = 12 - 15

$$f_1 = 23, f_0 = 10 \text{ \& } f_2 = 21$$

$$l = 12 \text{ \& } i = 3$$

$$\text{Mode } (M_0) = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

$$= 12 + \left(\frac{23 - 10}{46 - 10 - 21} \right) \times 3 = 12 + \frac{13 \times 3}{15} = 14.6$$

Relation between mean, Median & Mode

Emperical formula for mode:-

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$$

Ex. Mean of a sample data = 60 and median = 48. Find the mode of this distribution.

HINTS Mode = $3 \times 48 - 2 \times 60 = 144 - 120 = 24$

Range

Difference between highest and lowest numbers, is called Range.

How to find the Range:

- Arrange the numbers in ascending order.
- Subtract the lowest value from the largest.

$$\text{Range} = \text{largest value} - \text{smallest value}$$

Ex. Find the range of 12, 22, 7, 1, 5, 27, 30, 43.

HINTS Range = $43 - 1 = 42$

Ex Coefficient of Range = $\frac{\text{Highest} - \text{Lowest}}{\text{Highest} + \text{Lowest}}$

Variance

Variance is a measure of variability in statistics. It assesses the average squared difference between data values and the mean. It is denoted by sigma (σ^2).

How to compute variance and standard deviation?

Step I - Compute the simple mean $\bar{x}$.

Step II - Calculate the difference of $(x_i - \bar{x})$ for each value in the data set.

Step III - Calculate the squared difference $(x_i - \bar{x})^2$ for each value in the data set.

Step IV - Sum of squared difference $\sum_{i=1}^n (x_i - \bar{x})^2$

Step V - Divide the sum of squared differences with n,

$$\text{variance } (\sigma^2) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

Ex. The variance of the seven observations 6, 7, 10, 12, 13, 8, 14 is:

HINTS

$$\bar{x} = \frac{6+7+10+12+13+8+14}{7} = \frac{70}{7} = 10$$

$$\text{Variance} = \frac{(-4)^2 + (-3)^2 + 0^2 + (2)^2 + (3)^2 + (-2)^2 + (4)^2}{7}$$

$$= \frac{16+9+0+4+9+4+16}{7} = \frac{58}{7} = 8.29$$



If we add/subtract from the given numbers there will be no change in variance & if we multiply by z to the given numbers then variance will become z^2 times.

Ex. If the variance of 10 observations is 6 & each observation is multiplied by 3 then new variance is:

HINTS $\sigma^2 = 6$

$$\text{New variance} = 3^2 \times 6 = 54$$

Ex. The mean and variance of 10 observations are found to be 10 and 4 respectively. On rechecking it was found that an observation 8 was incorrect. If it is replaced by 18, then the correct variance is.

HINTS

$$\sum x_{(old)} = 10 \Rightarrow \frac{\sum x_i}{10} = 10 \Rightarrow \sum x_i (old) = 100$$

$$\sum x_{i(new)} = 100 - 8 + 18 = 110$$

$$\bar{x}_{(new)} = \frac{110}{10} = 11$$

$$\text{Var}_{(old)} = 4 \Rightarrow \frac{\sum x_{i(old)}^2}{10} - (\bar{x}_{(old)})^2 = 4 \Rightarrow \sum x_{i(old)}^2 = 1040$$

$$\sum x_{i(new)}^2 = 1040 - 64 + 324 = 1300$$

$$\text{Var}_{(new)} = \frac{1300}{10} - (11)^2 = 130 - 121 = 9$$



Did You Know?

- Variance of first n natural number = $\frac{(n^2 - 1)}{12}$
- Variance of first n even/odd natural number = $\frac{n^2 - 1}{3}$
- If observations are in AP then variance = $\left(\frac{n^2 - 1}{12} \right) \times d^2$
Where n = number of observations, d = difference

Ex. The variance of first 50 even number is

HINTS Variance = $\frac{50^2 - 1}{3} = 833$

Ex. If the variance of the first n natural number is 10 and the variance of the first 'm' even natural number is 16, then m + n is equal to _____.

HINTS $\frac{n^2 - 1}{12} = 10 \Rightarrow n = 11$

$$\frac{m^2 - 1}{3} = 16 \Rightarrow m = 7$$

$$\therefore m + n = 7 + 11 = 18$$

Standard Deviation

If σ^2 is the variance, then σ is called the standard deviation.

$$\text{Standard Deviation } (\sigma) = \sqrt{\text{variance}}$$

Ex. Calculate the standard deviation for the following data.

3, 4, 5, 6, 7

HINTS Mean $(\bar{x}) = \frac{3+4+5+6+7}{5} = 5$

$$\text{Variance} = \frac{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}{5} = \frac{10}{5} = 2$$

$$\text{Standard Deviation} = \sqrt{2}$$



If we multiply by z to the given numbers then standard deviation will become z times.

• Mean deviation = $\frac{4}{5}$ standard deviation

Ex. The standard deviation of 12 values is 3. If each value is increased by 4, then find the variance of the new set of values.

HINTS S.D. = 3
Variance = $3^2 = 9$
New Variance = 9

Ex. The mean of 100 observations is 50 and their standard deviation is 5. The sum of squares of all the observations is.

HINTS Given, $\sigma = 5$, $\bar{x} = 50$, $n = 100$

We know that,

$$\text{Standard Deviation } (\sigma) = \sqrt{\text{variance}}$$

$$\Rightarrow (\text{standard deviation})^2 = \text{variance}$$

$$\Rightarrow \sigma^2 = \frac{\sum x^2}{n} - (\bar{x})^2 \Rightarrow 5^2 = \frac{\sum x^2}{100} - (50)^2$$

$$\Rightarrow 25 = \frac{\sum x^2}{100} - (50)^2$$

$$\Rightarrow \sum x^2 = 252500$$

Thus sum of squares of all the observations is 252500.

Ex. A student scores 45, 54, 41, 57, 43 respectively in five tests. His score is not known for the sixth test. If the mean score is 48 in the six tests, then the standard deviation of the marks in six tests is:

HINTS Mean score = 48

Let unknown score be x .

$$\therefore \bar{x} = \frac{41+45+54+57+43+x}{6} \Rightarrow 48 = \frac{x+240}{6}$$

$$\Rightarrow x+240 = 288 \Rightarrow x = 48$$

$$\text{Now, } \sigma^2 = \frac{1}{6} [(48-41)^2 + (48-45)^2 + (48-54)^2 + (48-57)^2 + (48-43)^2 + (48-48)^2]$$

$$\Rightarrow \sigma^2 = \frac{1}{6} (49 + 9 + 36 + 81 + 25) = \frac{200}{6} = \frac{100}{3}$$

$$\Rightarrow \sigma = \sqrt{\frac{100}{3}} = \frac{10}{\sqrt{3}}$$

What is Mean Deviation

The mean deviation of a given standard distribution is the average of the deviation from the central tendency. Central Tendency can be computed using the Arithmetic Mean, Median or Mode of the data. It is used to show how far the observations are situated from the central point of the data (the central point can be either mean, median or mode).

Mean Deviation about Mean

The mean deviation about mean is defined as a statistical measure that is used to calculate the average deviation from the mean value of the given data set.

$$\text{Ungrouped data M.D.} = \frac{\sum |X_i - \text{Mean}|}{n}$$

$$\text{Coefficient of M. D.} = \frac{\text{M.D.}}{\bar{X}} ; \text{ where } \bar{X} \text{ is Mean}$$

The mean deviation of the data values can be easily calculated using the below procedure.

Step I: Find the mean value for the given data values

Step II: Now, subtract the mean value from each of the data values given (Note: Ignore the minus symbol)

Step III: Now, find the mean of those values obtained in step II.

Ex. Determine the mean deviation for the data values 5, 3, 7, 8, 4, 9.

HINTS $\therefore \text{Mean } (\bar{x}) = \frac{5+3+7+8+4+9}{6} = 6$

$$\begin{aligned} \text{Mean Deviation} &= \frac{|-1| + |-3| + 1 + 2 + |-2| + 3}{6} \\ &= \frac{1+3+1+2+2+3}{6} = \frac{12}{6} = 2 \end{aligned}$$

Ex. What is the mean deviation of first 10 even natural numbers?

HINTS Numbers = 2, 4, 6, 8, 10, 12, 14, 16, 18, 20

$$(\bar{x}) = \frac{2+20}{2} = 11$$

$$\text{M.D.} = \frac{9+7+5+3+1+1+3+5+7+9}{10} = \frac{50}{10} = 5$$

Ex. Find the coefficient of mean deviation for the following data:

3, 8, 7, 6, 5, 1

HINTS $(\bar{x}) = \frac{3+8+7+6+5+1}{6} = 5$

$$\text{M.D.} = \frac{2+3+2+1+0+4}{6} = \frac{12}{6} = 2$$

$$\text{Coefficient of M.D.} = \frac{2}{5} = 0.4$$

Ex. Calculate the mean deviation from mean for the following frequency distribution:

Variable (x)	10	30	50	70	90
Frequency (f)	4	24	28	16	8

HINTS Let us make the table of the given data and append other columns after calculations

x_i	f_i	$x_i f_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
10	4	40	40	160
30	24	720	20	480
50	28	1400	0	0
70	16	1120	20	320
90	8	720	40	320
Sum	80	4000		1280

The sum of calculated data $N = \sum_{i=1}^5 f_i = 80$,

$$\sum_{i=1}^5 f_i x_i = 4000$$

$$\Rightarrow \bar{x} = \frac{1}{N} \sum_{i=1}^5 f_i x_i = \frac{1}{80} \times 4000 = 50$$

The absolute values of the deviations from the mean, i.e., $|x_i - \bar{x}|$, as shown in the table

From the table, $\sum_{i=1}^5 f_i |x_i - \bar{x}| = 1280$

$$\therefore \text{M.D.} (\bar{x}) = \frac{1}{N} \sum_{i=1}^5 f_i |x_i - \bar{x}| = \left(\frac{1}{80}\right) \times 1280 = 16$$

Mean Deviation about Median

The mean deviation about median is defined as a statistical measure that is used to calculate the average deviation from the median value of the given data set.

$$\text{Ungrouped data M.D.} = \frac{\sum_{i=1}^n |X_i - \text{Median}|}{n}$$

$$\text{Coefficient of M.D.} = \frac{\text{M.D.}}{M}; \text{ Where } M \text{ is Median}$$

The mean deviation of the data values can be easily calculated using the below procedure.

Step I: Find the Median for the given data values

Step II: Now, subtract the median from each of the data values given (Note: Ignore the minus symbol)

Step III: Now, find the mean of those values obtained in step I.

Ex. The mean deviation about median for the given data: 52, 56, 66, 70, 75, 80, 82

HINTS Here $n = 7$, which is odd.

$$\Rightarrow \text{Median} = 4^{\text{th}} \text{ observation} = 70$$

$$\text{M.D.} = \frac{18+14+4+0+5+10+12}{7} = \frac{63}{7} = 9$$

Ex. Compute Coefficient of M.D. from median for following series.

3, 7, 12, 14, 15, 18, 22

HINTS Here $n = 7$, which is odd.

$$\Rightarrow \text{Median} = 4^{\text{th}} \text{ observation} = 14$$

$$\text{M.D.} = \frac{11+7+2+0+1+4+8}{7} = \frac{33}{7}$$

$$\text{Coefficient} = \frac{\text{MD}}{\text{Median}} = \frac{\frac{33}{7}}{14} = \frac{33}{98} = 0.336$$

"Mean deviation is minimum about Median".

Ex. Find the minimum possible number of the mean deviations from the following data

Class interval	0-6	6-12	12-18	18-24	24-30
Frequency	4	5	3	6	2

HINTS

Class int.	f_i	x_i	cf	$d_i = x_i - \bar{m}_d $	$f_i d_i$
0-6	4	3	4	11	44
6-12	5	9	9	5	25
12-18	3	15	12	1	3
18-24	6	21	18	7	42
24-30	2	27	20	13	26
Total	N=20				$\sum f_i d_i = 140$

$$\therefore \frac{N}{2} = \frac{20}{2} = 10$$

So, the median class is (12-18)

$$\text{Median} = l + \left(\frac{\frac{N}{2} - C.f'}{f} \right) \times h = 12 + \left(\frac{10-9}{3} \right) \times 6$$

$$= 12 + \frac{6}{3} (10-9) = 14$$

$$\therefore \text{MD} = \frac{\sum f_i d_i}{\sum f_i} = \frac{140}{20} = 7$$

Mean Deviation about Mode

The mean deviation about mode is defined as a statistical measure that is used to calculate the average deviation from the mode value of the given data set.

$$\text{Ungrouped data M.D.} = \frac{\sum_{i=1}^n |X_i - \text{Mode}|}{n}$$

$$\text{Coefficient of M.D.} = \frac{\text{M.D.}}{Z}; \text{ Where } Z \text{ is Mode.}$$

The mean deviation of the data values can be easily calculated using the below procedure.

Step I: Find the mode for the given data values

Step II: Now, subtract the mode from each of the data values given (Note: Ignore the minus symbol)

Step III: Now, find the mean of those values obtained in step 2.

Statistics

Ex. The mean deviation about mode for the given data:

7, 4, 10, 9, 15, 12, 7, 9, 7

HINTS Given data,

7, 4, 10, 9, 15, 12, 7, 9, 7

Mode (M) = 7

x_i	7	4	10	9	15	12	7	9	7	
$\Sigma x_i - M $	0	3	3	2	8	5	0	2	0	23

$$\text{Mean deviation} = \frac{23}{9} = 2.56$$

Coefficient of variation

It describes variability by expressing the standard deviation as a proportion of mean, usually a percentage.

The formula for it as a percentage is

$$\text{Coefficient of variation} = \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

Ex. If the mean is 25 and the standard deviation is 5 then the coefficient of variation is:

HINTS Coefficient of variation

$$= \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

$$= \frac{5}{25} \times 100 = 20\%$$

Ex. If the standard deviation of a set of numbers is 3 and the arithmetic mean of these numbers is 6, what is the coefficient of variation of these numbers?

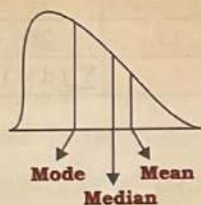
HINTS

$$\text{Coefficient of variation} = \frac{\text{S.D.}}{\text{Mean}} \times 100$$

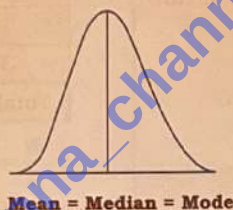
$$= \frac{3}{6} \times 100 = 50\%$$

Karl Pearson's coefficient of skewness

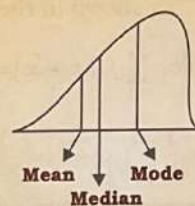
Karl Pearson's coefficient of skewness provides a standardized way to quantifies how much a distribution deviates from being perfectly symmetrical. It is a widely used measure in statistics for understanding the shape of data distributions, particularly when dealing with continuous variables.



- Positive Skewness (Right Skewed):
- Mean > Median > Mode
- Long tail on the right.
- Skewness is positive



- Zero Skewness (Symmetrical):
- Mean = Median = Mode
- Looks like a bell curve.
- Skewness is close to 0



- Negative Skewness (Left Skewed):
- Mean < Median < Mode
- Long tail on the left.
- Skewness is negative

- In essence, Karl Pearson's coefficient of skewness measures the asymmetry of a distribution. It is calculated as three times the difference between the mean and the median, divided by the standard deviation.
- The value of this coefficient typically ranges between -3 and +3, indicating the extent and direction of the skew.

$$\text{SKP} = \frac{(\text{Mean} - \text{Mode})}{\text{Standard deviation}}$$

(When mode is given)

$$\text{SKP} = \frac{3(\text{Mean} - \text{Median})}{\text{Standard deviation}}$$

(When mode is not given)

Ex. For a distribution, Mean = 100, mode = 40 and Standard Deviation = 80. Find the pearson coefficient of skewness.

HINTS Given, Mean = 100, Mode = 40, SD = 80

$$\text{SKP} = \frac{(\text{Mean} - \text{Mode})}{\text{Standard deviation}} = \frac{100 - 40}{80} = \frac{60}{80} = 0.75$$

Ex. For a frequency distribution the mean is 200, the coefficient of variation is 8% and Karl Pearson's coefficient of skewness is 0.3. Find the mode and median of the distribution.

HINTS Coefficient of variation = $\frac{\text{S.D.}}{\text{Mean}} \times 100$

$$\Rightarrow 8\% = \frac{\text{S.D.}}{200} \times 100$$

$$\Rightarrow \text{SD} = 16$$

Now,

$$\text{SKP} = \frac{(\text{Mean} - \text{Mode})}{\text{Standard deviation}}$$

$$\Rightarrow 0.3 = \frac{200 - \text{mode}}{16} \Rightarrow 4.8 = 200 - \text{Mode}$$

$$\Rightarrow \text{Mode} = 195.2$$

Using empirical formula,

$$\text{Mode} = 3 \text{ median} - 2 \text{ mean}$$

$$\Rightarrow 195.2 = 3 \text{ median} - 2 \times 200$$

$$\Rightarrow 595.2 = 3 \text{ median} \Rightarrow \text{median} = 198.4$$

Ex. If mean = 75, median = 70 and Karl Pearson's coefficient of skewness = 0.6 then find standard deviation?

HINTS Given, Mean = 75, Median = 70, Skewness = 0.6

$$SKP = \frac{3(\text{Mean} - \text{Median})}{\text{Standard deviation}}$$

$$\Rightarrow 0.6 = \frac{3(75 - 70)}{SD} \Rightarrow SD = 25$$

Ex. Find Karl Pearson's coefficient of skewness for the following data?

12, 6, 2, 6, 3, 4, 4, 4, 5, 15

HINTS

Arrange data in ascending order

→ 2, 3, 4, 4, 4, 5, 6, 6, 12, 15

Mode = 4

$$\text{Mean } (\bar{x}) = \frac{2+3+4+4+4+5+6+6+12+15}{10} = \frac{61}{10} = 6.1$$

$$\sum x^2 = 2^2 + 3^2 + 4^2 + 4^2 + 4^2 + 5^2 + 6^2 + 6^2 + 12^2 + 15^2$$

$$= 527$$

$$\text{Standard Deviation} = \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2}$$

$$= \sqrt{\frac{527}{10} - (6.1)^2} = \sqrt{52.7 - 37.21} = \sqrt{15.49} = 3.9$$

$$SKP = \frac{(\text{Mean} - \text{Mode})}{\text{Standard deviation}} = \frac{6.1 - 4}{3.9} = \frac{2.1}{3.9} = \frac{7}{13} = 0.53$$

$$\therefore SKP > 0$$

Then this shows that distribution is positively skewed or right skewed.

(Sigma) Σ = Summation

$\bar{x}$ = Mean

σ^2 = Variance.

(Sigma) σ = Standard Deviation

Z = Mode.

R = Range of data

$\bar{R}$ = Average Range of data

CHAPTER

33

LOGARITHM

Logarithm

A logarithm is the exponent or power to which a base must be raised to yield a given number. Expressed mathematically, x is the logarithm of n to the base b .

If $b^x = n$ then $x = \log_b n$, $b \neq 1$ and $n, b > 0$

Ex. $16 = 2^4 \Rightarrow \log_2 16 = 4$

Common logarithm:

Logarithms to the base 10 are called common logarithms. There are two notation to represent common logarithm i.e., $\log b$ (read as 'log b') and $\log_{10} b$.

Mathematically, $\log x = \log_{10} x$

Natural logarithm:

Logarithms to the base "e" are called natural logarithms. There are also two notations to represent natural logarithms i.e., $\ln b$ and $\log_e b$.

These both represent the logarithm of b to the base e .

Mathematically, $\ln x = \log_e x$.

Log of zero or negative numbers are not defined and the base of logarithm can never be equal to 1. i.e.,

$\log 0 \rightarrow$ Not defined.

$\log_e(-5) \rightarrow$ Not defined

$\log_1 a \rightarrow$ Not defined



Properties of Logarithm

Let m and n be positive number such that $a, m, n, b, c > 0$ and $a, b, c \neq 1$

(i) $\log_a a = 1$

(ii) $\log_a 1 = 0$

(iii) $\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$

(iv) $\log_a (m \times n) = \log_a m + \log_a n$

(v) $\log_b a = \frac{1}{\log_a b}$

(vi) $\log_c a = \log_b a \times \log_c b = \frac{\log_b a}{\log_b c}$

(vii) $\log_a (m^n) = n \log_a m$

(viii) $\log_{a^q} (m) = \frac{1}{q} \log_a m$

(ix) $\log_{a^q} (m^p) = \frac{p}{q} \log_a m$

(x) $a^{\log_a n} = n$

(xi) $a^{\log_e x} = x^{\log_e a}$, x & $a > 0$ and $x, a \neq 1$

(xii) $\log_a b = -\log_{\frac{1}{a}} b$

Remember:

$$\log 2 = 0.3010$$

$$\log 3 = 0.4771$$

$$\log 5 = 0.6990$$

Ex. If $\log_{10} 5 + \log_{10} (5x + 1) = \log_{10} (x + 5) + 1$, and then x is equal to:

HINTS We have

$$\Rightarrow \log_{10} 5 + \log_{10} (5x + 1) = \log_{10} (x + 5) + 1$$

$$\Rightarrow \log_{10} 5 + \log_{10} (5x + 1) = \log_{10} (x + 5) + \log_{10} 10$$

We know that,

$$\log_a (m \times n) = \log_a m + \log_a n$$

$$\Rightarrow \log_{10} [5 (5x + 1)] = \log_{10} [(x + 5) \times 10]$$

$$\Rightarrow 5x + 1 = 2x + 10$$

$$\Rightarrow 3x = 9 \Rightarrow x = 3$$

Ex. What is the value of $\frac{1}{10} \log_5 1024 - \log_5 10 + \frac{1}{5} \log_5 3125$

HINTS $\frac{1}{10} \log_5 1024 - \log_5 10 + \frac{1}{5} \log_5 3125$

We know that,

$$\log_a (m^n) = n \log_a m$$

$$= \frac{1}{10} \log_5 (2)^{10} - \log_5 10 + \frac{1}{5} \log_5 (5)^5$$

$$= \frac{10}{10} \log_5 2 - \log_5 10 + \frac{5}{5} \log_5 5$$

$$= \log_5 2 - \log_5 (2 \times 5) + \log_5 5$$

$$= \log_5 2 - \log_5 2 - \log_5 5 + \log_5 5 = 0$$

Ex. If the roots of the quadratic equation $x^2 - 4x - \log_{10} N = 0$ are real, then what is the minimum value of N ?

HINTS We have

$$x^2 - 4x - \log_{10} N = 0$$

Compare with $ax^2 + bx + c = 0$

$$a = 1, b = -4 \text{ and } c = -\log_{10} N$$

Roots of quadratic equation are real

then $D \geq 0$ i.e. $b^2 - 4ac \geq 0$

$$\Rightarrow (-4)^2 - 4 \times 1 \times (-\log_{10} N) \geq 0$$

$$\Rightarrow 4 \log_{10} N \geq -16$$

$$\Rightarrow N \geq 10^{-4}$$

$$N \geq 0.0001$$



Logarithm

Ex. What is the value of $\log_{\sqrt{b}} a \times \log_{\frac{1}{c^3}} b \times \log_{\frac{1}{a^4}} c$?

HINTS $(\log_{\sqrt{b}} a) \times \left(\log_{\frac{1}{c^3}} b\right) \times \left(\log_{\frac{1}{a^4}} c\right)$

$$= \left(\frac{1}{\frac{1}{2}} \log_b a\right) \left(\frac{1}{\frac{1}{3}} \log_c b\right) \left(\frac{1}{\frac{1}{4}} \log_a c\right)$$

$$= 2 \frac{\log a}{\log b} \times 3 \frac{\log b}{\log c} \times 4 \frac{\log c}{\log a}$$

$$= 2 \times 3 \times 4 = 24$$

Ex. If, $\log_2 x = 6$, Find x ?

HINTS $x = 2^6 = 64$

Ex. Simplify: $\log_5 125 - \log_5 5$

HINTS $= \log_5 (5^3) - \log_5 5$

$$= 3 \log_5 5 - \log_5 5$$

$$= 2 \log_5 5 = 2$$

Ex. If, $\log_{10} 2 = 0.3010$, find $\log_{10} 8$?

HINTS $\log_{10} 8 = \log_{10} (2^3) = 3 \log_{10} 2$

$$= 3 \times 0.3010$$

$$= 0.9030$$

Ex. Solve it, $\log_{10} 100 - \log_{10} 10$

HINTS $\log_{10} (10^2) - \log_{10} 10$

$$= 2 \log_{10} (10) - \log_{10} 10$$

$$= \log_{10} 10 = 1$$

Ex. If $\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b}$, then the value of $a^b b^c c^a$ is:

HINTS Let, $\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b} = k$

$$\log a = k(b-c)$$

$$\log b = k(c-a)$$

$$\log c = k(a-b)$$

$$\text{Let } x = a^b b^c c^a$$

Taking log both sides

$$\log x = \log a^b b^c c^a$$

$$\log x = a \log a + b \log b + c \log c$$

$$= ka(b-c) + kb(c-a) + kc(a-b)$$

$$= k(ab - ac + bc - ba + ca - cb)$$

$$\log x = 0 \Rightarrow x = 1 \Rightarrow a^b b^c c^a = 1$$

Ex. The value of $\left[\frac{1}{\log_{\frac{q}{p}} x} + \frac{1}{\log_{\frac{p}{r}} x} + \frac{1}{\log_{\frac{r}{q}} x} \right]$ is:

HINTS $\frac{1}{\log_{\frac{q}{p}} x} + \frac{1}{\log_{\frac{p}{r}} x} + \frac{1}{\log_{\frac{r}{q}} x}$

$$= \log_x \left(\frac{q}{p} \right) + \log_x \left(\frac{p}{r} \right) + \log_x \left(\frac{r}{q} \right)$$

$$= \log_x \left(\frac{q}{p} \times \frac{p}{r} \times \frac{r}{q} \right) = \log_x 1 = 0$$

CHAPTER

34

CALCULATION

Despite having sufficient knowledge, in various competitive exams many students are not able to pass the examination due to lack of good basic calculation speed. If you practice for 20-30 minutes every day using the techniques given in this chapter, your calculation speed can increase up to 5-times within 2 months.

How to enhance Speed in Quantitative aptitude?

Learn some values

- Learn tables from 1 to 20
(Refer Table-05 of this chapter)
- Learn squares from 1 to 30
(Refer Table-01 of this chapter)
- Learn cubes from 1 to 20
(Refer Table-02 of this chapter)
- Basic exponent values
(Refer Table-04 of this chapter)
- Some percentage values
(Refer percentage chapter)

Work on Basic Mathematical operations

Addition Subtraction Multiplication Division

Addition

Sum of two or more number is known as addition.

CONCEPT-01

Ex. $(69 + 37)$

Step-1: Break the number (69) as $(60 + 9)$ and (37) as $(30 + 7)$

Step-2: Add base number first $\rightarrow (60 + 30) = 90$

Step-3: Add unit digits of number $\rightarrow (9 + 7) = 16$

Step-4: Now add both the results $\rightarrow (90 + 16) = 106$

DO IT LIKE PRO

N_1	N_2	$N_1 + N_2$	N_1	N_2	$N_1 + N_2$
69	49		67	48	
65	29		89	37	
87	48		74	51	
69	52		73	37	
79	42		79	28	

Ex. $(187 + 287)$

Step-1: Break the number (187) as $(100 + 80 + 7)$ and (287) as $(200 + 80 + 7)$

Step-2: Add base number first $\rightarrow$

$(100 + 200) = 300, (80 + 80) = 160$

Step-3: Add unit digits $\rightarrow (7 + 7) = 14$

Step-4: Now add the results $\rightarrow$

$(300 + 160 + 14) = 474$

Ex. $4639 + 1235$

Step-1: Keep largest number in mind

4639

Step-2: Break the second number 1235 as

$1000 + 200 + 30 + 5$

Step-3: Add the numbers one after another

$4639 + 1000 = 5639$

$5639 + 200 = 5839$

$5839 + 30 = 5869$

$5869 + 5 = 5874$

DO IT LIKE PRO

N_1	N_2	$N_1 + N_2$	N_1	N_2	$N_1 + N_2$
169	249		637	448	
1265	629		269	137	
563	438		264	511	
639	252		2273	1237	
3479	1128		4576	1445	

CONCEPT-02

In this approach, numbers which are very close to each other are added.

Ex. $(45 + 52)$

Step-1: Double the smaller number, i.e. $\rightarrow (45 \times 2) = 90$

Step-2: Find the gap between the numbers i.e. $(52 - 45) = 7$

Step-3: Now add both the results i.e. $\rightarrow (90 + 7) = 97$

Ex. $(333 + 338)$

Step-1: Double the smaller number, i.e. $333 \times 2 = 666$

Step-2: Find the gap between the numbers i.e. $338 - 333 = 5$

Step-3: Now add both the results i.e. $666 + 5 = 671$

DO IT LIKE PRO

N_1	N_2	$N_1 + N_2$	N_1	N_2	$N_1 + N_2$
69	74		87	92	
137	146		269	277	
363	358		1264	1277	
1639	1645		2273	2282	
4576	4584		3489	3492	

CONCEPT-03

In this approach, number is added with the number which is close to 100, 200, and 300 and so on.

Ex. $(57 + 94)$

Step-1: Assume 94 as 100 (6 is extra here)

Step-2: Add 57 with 100 i.e. 157

Step-3: Then subtract the extra number i.e.

$157 - 6 = 151$

Calculation

Ex. $(314 + 399)$ **Step-1:** Assume 399 as 400 $\rightarrow$ (1 is extra here)**Step-2:** Add 314 with 400 i.e. 714**Step-3:** Then subtract the extra number you added i.e. $714 - 1 = 713$

DO IT LIKE PRO

N_1	N_2	$N_1 + N_2$	N_1	N_2	$N_1 + N_2$
189	84		181	92	
153	283		363	358	
1196	146		1264	1277	
1721	1342		2585	2289	
3489	3492		4576	4584	

CONCEPT-04

In this approach, during addition you should be very alert to find some pair of number which are getting added and forming base numbers like 100, 200, 300 etc.

Ex. $(45 + 53 + 75)$ **Step-1:** Add those numbers first which will give you base number like 100, 200, 300 etc. i.e. $\rightarrow 45 + 75 = 120$ **Step-2:** Now add Result with remaining number i.e. $120 + 53 = 173$ Ex. $(157 + 186 + 143)$ **Step-1:** Add those numbers first which will give you base number like 100, 200, 300 etc. like $157 + 143 = 300$ **Step-2:** Now add Result with remaining number i.e. $300 + 186 = 486$

DO IT LIKE PRO

N_1	N_2	N_3	SUM	N_1	N_2	N_3	SUM
34	43	66		79	52	22	
46	93	56		57	78	43	
67	88	47		167	196	148	
163	177	284		227	258	298	

CONCEPT-05

(A) In this approach first we count how many times the number one appears most. Then we right the numbers from 1 upto that count.

Ex. $1 + 11 + 111$ **Step-1:** Count the maximum times of number 1. $111 = 3$ times**Step-2:** Now write the number from 1 upto the 3
123**Step-3:** So $1 + 11 + 111 = 123$ Ex. $1 + 11 + 111 + 1111$ **Step-1:** Count the maximum times of number 1. $1111 = 4$ times**Step-2:** Now write the number from 1 upto the 4
1234**Step-3:** So $1 + 11 + 111 + 1111 = 1234$

(B) In this approach first we count how many times the number one appears most. Then we right the numbers from 1 upto that count and in last multiply the resultant value by number which is on repetition.

Ex. $2 + 22 + 222$ **Step-1:** Count the max. times of numbers occurred $222 = 3$ times**Step-2:** Now write the numbers from 1 upto 3.
123**Step-3:** Multiply by 2 $\Rightarrow 123 \times 2 = 246$ **Step-4:** $2 + 22 + 222 = 246$ Ex. $2 + 22 + 222 + 2222 + 22222$ **Step-1:** Count the max. times of numbers occurred $22222 = 5$ times**Step-2:** Now write the numbers from 1 upto 5.
12345**Step-3:** Multiply by 2 $\Rightarrow 12345 \times 2 = 24690$ **Step-4:** $2 + 22 + 222 + 2222 + 22222 = 24690$ Ex. $3 + 33 + 333$ **Step-1:** $333 = 3$ times**Step-2:** 123**Step-3:** $123 \times 3 = 369$ Ex. $3 + 33 + 333 + 3333$ **Step-1:** $3333 = 4$ times**Step-2:** 1234**Step-3:** $1234 \times 3 = 3702$

Subtraction

Subtraction is also used to find the difference between two or more numbers.

CONCEPT-01

(A) In this type of approach, double digit number is getting subtracted from its reversed number and we need to multiply the difference of digits with 9 always.

Ex. $(73 - 37)$ **Step-1:** Difference between digits $(7 - 3 = 4)$ **Step-2:** Multiply the result always by 9 i.e. $\rightarrow 4 \times 9 = 36$

DO IT LIKE PRO

N_1	N_2	$N_1 - N_2$	N_1	N_2	$N_1 - N_2$
61	16		87	78	
54	45		72	27	
63	36		93	39	
75	57		74	47	
84	48		91	19	

CONCEPT-02

In this approach, a number which is closer to a base number like 100, 200, 300... is subtracted from number.

Ex. $(267 - 94)$

(Consider 94 as 100, 6 is extra here)

Step-1: First Subtract 100 from 267 i.e. $\rightarrow 167$ **Step-2:** Then add extra number (6) i.e. $\rightarrow 167 + 6 = 173$ Ex. $(567 - 393)$

(Take 393 as 400, 7 is extra here)

Step-1: First Subtract 400 from 567 i.e. $\rightarrow 167$ **Step-2:** Then add extra number (7) i.e. $\rightarrow 167 + 7 = 174$

Calculation

DO IT LIKE PRO

N_1	N_2	$N_1 - N_2$	N_1	N_2	$N_1 - N_2$
169	94		137	92	
146	89		178	98	
278	86		264	197	
247	192		227	192	
576	184		348	192	

CONCEPT-03

In this approach, two numbers are getting subtracted which are closer to a base number (100, 200, 300,). So add the margins of both the numbers from the base number.

Ex. (815-777)

Find Base Which is closer to 777 i.e. $\rightarrow 800$

Step-1: 815 is 15 above than 800

Step-2: 777 is 23 less than 800

Add the margins, $15 + 23 = 38$

Ex. (1305-878)

Find Base Which is closer to 878 like 900

Step-1: 1305 is 405 above than 900

$1305 - 900 = 405$

Step-2: 878 is 22 less than 900

Add the margins $405 + 22 = 427$

DO IT LIKE PRO

N_1	N_2	$N_1 - N_2$	N_1	N_2	$N_1 - N_2$
169	124		387	292	
237	196		277	269	
363	158		1264	277	
1639	1345		2273	2182	
4576	4184		3492	2892	

CONCEPT-04

It is not necessary to always add first. Sometimes we can subtract also if it is easier. $35 + 73 - 55$

Ex. $(35 + 73 - 55)$

Step-1: Since 73 and 55 are closer to each other, so First $73 - 55 = 18$

Step-2: Add result with 35, $35 + 18 = 53$

Ex. $(-267 + 196 + 253)$

Step-1: Since 253 and 267 are closer to each other, so First $253 - 267 = -14$

Step-2: Add result with 196, $196 - 14 = 182$

DO IT LIKE PRO

Expression	Solve	Expression	Solve
$28 - 65 - 66$		$47 - 54 + 57$	
$76 + 62 - 72$		$57 + 78 - 43$	
$63 - 58 + 34$		$64 + 77 - 103$	
$-267 + 164 + 245$		$573 + 382 - 234$	
$576 + 684 - 272$		$489 - 249 + 125$	

CONCEPT-05

In this approach number is subtracted from 100 or multiple of 100 like 200, 300, etc. So check how much you should add to smaller number to reach the bigger number.

Ex. $(100 - 46)$

Try to make 46 reach 100

Step-1: Add (4) to 46, it becomes 50

Step-2: Add (50) to 50, it becomes 100.

We have added 4 and 50 so answer is 54

OR

46 is $(40 + 6)$

Step-1: Subtract 40 from 100, $100 - 40 = 60$

Step-2: Subtract 6 from 60 now, $\rightarrow 60 - 6 = 54$

Ex. $(1000 - 357)$

357 is $300 + 50 + 7$

Step-1: Subtract 300 from 1000, $\rightarrow (1000 - 300) = 700$

Step-2: Now subtract 57 from 700, $\rightarrow (700 - 57) = 643$

DO IT LIKE PRO

N_1	N_2	$N_1 + N_2$	N_1	N_2	$N_1 + N_2$
50	26		60	32	
67	46		69	27	
263	135		661	277	
1639	645		1273	682	
2376	1184		2489	1492	

Square

For calculating squares of any number very fast, you should know the squares from 1 to 30.

How to calculate square of numbers that ends on 5.

$$(10x + 5)^2 = 100x^2 + 100x + 25$$

$$= 100x(x + 1) + 25$$

$$= x(x + 1)/25$$

Ex. (i) $(65)^2 = 6 \times 7/25 = 4225$

(ii) $(75)^2 = 7 \times 8/25 = 5625$

(iii) $(95)^2 = 9025$ (iv) $(35)^2 = 1225$

(v) $(45)^2 = 2025$ (vi) $(85)^2 = 7225$

$$(x5)^2 = x(x + 1)/25$$

This can be understood as

$$(x5)^2 = (x^2 + x)/25$$

Ex. (i) $(105)^2 = 10^2 + 10/25 = 11025$

(ii) $(125)^2 = 12^2 + 12/25 = 15625$

(iii) $(145)^2 = 21025$ (iv) $(195)^2 = 38025$

(v) $(165)^2 = 27225$ (vi) $(205)^2 = 42025$

Application of Base method in calculating squares from 31 to 130

Case I : 101 - 130

Here, Base = 100

Let $N = 100 + x$

Then $N^2 = (N + x)/x^2$

"Here, x^2 is to be written in 2 digits."

Calculation

Ex. (i) $(108)^2 = (108 + 8)/8^2 = 11664$

(ii) $(116)^2 = (116 + 16)/16^2 = 132/256 = 13456$

(iii) $(123)^2 = (123 + 23)/23^2 = 146/529 = 15129$

Case II : 71 - 100

Here, Base = 100

Let $N = 100 - x$

Then $N^2 = (N - x)/x^2$

"Here, x^2 is to be written in 2 digit."

(i) $(94)^2 = (94 - 6)/6^2 = 8836$

(ii) $(89)^2 = (89 - 11)/11^2 = 78/121 = 7921$

(iii) $(78)^2 = (78 - 22)/22^2 = 56/484 = 6084$

Case III : 31 - 70

Here, Base = 50

Let $N = 50 \pm x$

Then $N^2 = (25 \pm x)/x^2$

"Here, x^2 is to be written in 2 digits."

(i) $(57)^2 = (25 + 7)/7^2 = 3249$

(ii) $(42)^2 = (25 - 8)/8^2 = 1764$

Duplex Method of calculating Squares

Duplex of a single digit number (a) is its square.

D(a) = a²

Ex. D(4) = 16

D(7) = 49

D(2) = 4

D(8) = 64

Duplex of a two digit number (ab) is twice of their product.

D(ab) = 2ab

Ex. D(10) = $2 \times 1 \times 0 = 0$

D(12) = $2 \times 1 \times 2 = 4$

D(93) = 54

D(51) = 10

Duplex of a three digit number (abc) is twice the product of the outer pair + the square of the middle digit

D(abc) = 2ac + b²

Ex. D(123) = 10

D(512) = 21

D(108) = 16

D(287) = 92

Square of Two-Digit Numbers

$(ab)^2 = D(a) / D(ab) / D(b)$

Ex. (i) $(23)^2 = D(2) / D(23) / D(3) = 4/12/9 = 529$

(ii) $(73)^2 = D(7) / D(73) / D(3) = 49/42/9 = 5329$

(iii) $(43)^2 = D(4) / D(43) / D(3) = 16/24/9 = 1849$

Square of Three-Digit Numbers

$(abc)^2 = D(a)/D(ab)/D(abc)/D(bc)/D(c)$

Ex. (i) $(123)^2 = D(1)/D(12)/D(123)/D(23)/D(3)$
 $= 1/4/10/12/9 = 15129$

(ii) $(271)^2 = D(2)/D(27)/D(271)/D(71)/D(1)$
 $= 4/28/53/14/1 = 73441$

Special Case - 01

Square of a number which is nearer to 10ⁿ

$(98)^2 = (98 + 2)(98 - 2) + 2^2$

$= 100 \times 96 + 4 = 9604$

$(103)^2 = (103 - 3)(103 + 3) + 3^2$

$= 100 \times 106 + 9 = 10609$

$(993)^2 = (993 + 7)(993 - 7) + 7^2$

$= 1000 \times 986 + 49 = 986049$

$(1008)^2 = (1008 + 8)(1008 - 8) + 8^2$

$= 1016 \times 1000 + 64$

$= 1016064$

Special Case - 02

Let's observe

$3^2 = 09$

$33^2 = 1089$

$333^2 = 110889$

$3333^2 = 11108889$

$6^2 = 36$

$66^2 = 4356$

$666^2 = 443556$

$6666^2 = 44435556$

$9^2 = 81$

$99^2 = 9801$

$999^2 = 998001$

$9999^2 = 99980001$

Special Case - 03

Let's observe

$1^2 = 1$

$11^2 = 121$

$111^2 = 12321$

$1111^2 = 1234321$

Square Root**Square Root of perfect square**

Unit digit of N	Unit digit of N ²
1, 9	1
2, 8	4
3, 7	9
4, 6	6
5	5

Ex. Calculate the square root of 5184

Step-1: The unit digit of 5184 is 4, so its square root will be end on 2 or 8.

Step-2: Now leave two digits from end and check the remaining number. In this, case it is 51.

$7^2 < 51 < 8^2$

So, the tens digit of square root is 7.

Step-3: Now the square root of 5184 can be 72 or 78. For this, we have to check whether 51 is closer to 7^2 or 8^2 . As 51 is closer to 7^2 .

Square root of 5184 is 72.

DO IT LIKE PRO

$\sqrt{N}$	N	$\sqrt{N}$	N
$\sqrt{7744}$		$\sqrt{1764}$	
$\sqrt{6561}$		$\sqrt{7396}$	
$\sqrt{2916}$		$\sqrt{7569}$	
$\sqrt{4489}$		$\sqrt{4225}$	
$\sqrt{3249}$		$\sqrt{1764}$	
$\sqrt{4624}$		$\sqrt{9216}$	

Ex. Calculate the square root of 15129

Step-1: The unit digit of 15129 is 9, so its square root will be end on 3 or 7.

Step-2: Now leave two digits from end and check the remaining number. In this case, it is 151.

$12^2 < 151 < 13^2$

So, the first two digits of square root of 15129 is 12.

Calculation

Step-3: Now the square root of 15129 can be 123 or 127. For this, we have to check whether 151 is closer to 12^2 or 13^2

As 151 is closer to 12^2 .

Square root of 15129 is 123.

DO IT LIKE PRO

$\sqrt{N}$	N	$\sqrt{N}$	N
$\sqrt{18769}$		$\sqrt{63001}$	
$\sqrt{24964}$		$\sqrt{70225}$	
$\sqrt{54756}$		$\sqrt{33489}$	
$\sqrt{69169}$		$\sqrt{75076}$	

Square Root of Non-perfect square

$$\sqrt{A \pm B} = \sqrt{A} \pm \frac{B}{2\sqrt{A}}$$

Ex. $\sqrt{10} = \sqrt{9+1} = \sqrt{9} + \frac{1}{2\sqrt{9}} = 3 + \frac{1}{2 \times 3} = \frac{19}{6} = 3.17$

Ex. $\sqrt{14} = \sqrt{16-2} = \sqrt{16} - \frac{2}{2\sqrt{16}} = 4 - \frac{2}{2 \times 4} = \frac{30}{8} = 3.75$

DO IT LIKE PRO

$\sqrt{N}$	N	$\sqrt{N}$	N
$\sqrt{57}$		$\sqrt{265}$	
$\sqrt{374}$		$\sqrt{445}$	

Cube

Cube of two Digit Number

+ Carry+ Carry+ Carry

$$(ab)^3 = a^3 / 3a^2b / 3ab^2 / b^3$$

$\uparrow$ $\uparrow$ $\uparrow$
 1 digit 1 digit 1 digit

Ex. Find the cube of 14

Here, a = 1 & b = 4

$$14^3 = 1^3 / 3 \cdot 1^2 \cdot 4 / 3 \cdot 1 \cdot 4^2 / 4^3 = 1 / 12 / 48 / 64 = 2744$$

Ex. Find the cube of 79

Here, a = 7 & b = 9

$$79^3 = 7^3 / 3 \cdot 7^2 \cdot 9 / 3 \cdot 7 \cdot 9^2 / 9^3 = 343 / 1323 / 1701 / 729 = 493039$$

DO IT LIKE PRO

16^3		34^3	
84^3		87^3	
69^3		43^3	
51^3		59^3	

Cube of three Digit Number

+ Carry + Carry + Carry

$$(abc)^3 = a^3 / 3a^2b / 3ab^2 / b^3$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 2 digit 2 digit 2 digit 2 digit

Ex. Find the cube of 104

Here, a = 1 & b = 4

$$104^3 = 1^3 / 3 \cdot 1^2 \cdot 4 / 3 \cdot 1 \cdot 4^2 / 4^3 = 1 / 12 / 48 / 64 = 1124864$$

DO IT LIKE PRO

806^3		506^3	
109^3		408^3	
711^3		912^3	

Cube Root

Cube Root of perfect cube

A perfect cube is a number that is the cube root of an integer.

Unit digit of N	Unit digit of N^3
1, 4, 5, 6, 9	1, 4, 5, 6, 9
2	8
8	2
3	7
7	3

Ex. Calculate the cube root of 157464

First make a group of 3-digits starting from the unit digit.

157464

Unit digit of 157464 is 4.

So the unit digit of cube root will be 4.

Now, calculate the tens digit of cube root of 157464, we should find a natural number whose cube root is just less than 157

In this case, it is 5.

Cube root of 157464 is 54.

DO IT LIKE PRO

$\sqrt[3]{21952}$		$\sqrt[3]{148877}$	
$\sqrt[3]{32768}$		$\sqrt[3]{91125}$	
$\sqrt[3]{24389}$		$\sqrt[3]{50653}$	
$\sqrt[3]{68921}$		$\sqrt[3]{753571}$	
$\sqrt[3]{438976}$		$\sqrt[3]{35937}$	
$\sqrt[3]{46656}$		$\sqrt[3]{85184}$	

Cube Root of Non-perfect cube

$$\sqrt[3]{N} = A \pm \frac{B}{3A^2}$$

First find a number which is a perfect cube near to the number N, let it be A and the difference between N and A^3 is B.

"+" sign is used if $N > A^3$

- sign is used if $N < A^3$

Ex. $\sqrt[3]{60} = 4 - \frac{4}{3 \times 4^2} = \frac{188}{48} = 3.92$

Ex. $\sqrt[3]{72} = 4 + \frac{8}{3 \times 4^2} = \frac{200}{48} = 4.17$

Calculation

DO IT LIKE PRO

$\sqrt[3]{10}$		$\sqrt[3]{66}$	
$\sqrt[3]{118}$		$\sqrt[3]{226}$	
$\sqrt[3]{525}$		$\sqrt[3]{706}$	
$\sqrt[3]{953}$		$\sqrt[3]{1365}$	
$\sqrt[3]{1500}$		$\sqrt[3]{255}$	

Multiplication

Multiplication means adding the same number again and again.

CONCEPT-01

In this approach, we multiply numbers by 5^n .

- (A) If the number is even, we divide it by 2 and then add a 0 to the end of the result.

Ex. 74×5

Step-1: 74 is a even number, so we will divide it by 2.

$$\frac{74}{2} = 37$$

Step-2: Add 0 at the end of resulting number.

370

Step-3: $74 \times 5 = 370$

Ex. 6274×5

Step-1: Divide the number by 2.

$$\frac{6274}{2} = 3137$$

Step-2: Add 0 in the resulting value.

31370

Step-3: $6274 \times 5 = 31370$

- (B) In this approach if the number is odd then add a 0 to the number then divide the resulting number by 2

Ex. 43×5

Step-1: Add 0 at the end of the number = 430

Step-2: Divide the resulting number by 2.

$$\frac{430}{2} = 215$$

Step-3: $43 \times 5 = 215$

Ex. 7235×5

Step-1: Add 0 at the end of the number = 72350

Step-2: Divide the resulting number by 2.

$$\frac{72350}{2} = 36175$$

Step-3: $7235 \times 5 = 36175$

DO IT LIKE PRO

N_1	N_2	$N_1 \times N_2$	N_1	N_2	$N_1 \times N_2$
272	5		437	5	
836	5		295	5	
4278	5		921	5	

CONCEPT-02

Multiply by 25: In this approach first add '00' to the number because of $5^2 = 25$. Then divide the resulting number by 4.

Ex. 27×25

Step-1: Add 00 at the end of the number. $\rightarrow 2700$

Step-2: Then divide the resulting number by 4. $\rightarrow$

$$\frac{2700}{4} = 675$$

Step-3: $27 \times 25 = 675$

Multiply by 125: In this approach, we first add '000' to the number because $5^3 = 125$. Then, we divide the resulting number by 8.

Ex. 672×125

Step-1: Add 000 at the end of the number. $\rightarrow 672000$

Step-2: Then divide the resulting number by 8. $\rightarrow$

$$\frac{672000}{8} = 84000$$

Step-3: $672 \times 125 = 84000$

DO IT LIKE PRO

N_1	N_2	$N_1 \times N_2$	N_1	N_2	$N_1 \times N_2$
29	25		421	125	
724	25		529	125	
321	25		7294	125	
920	25		8125	125	

CONCEPT-03

Multiply by 9, 99, 999

In this method, first subtract 1 from the given number. Then, subtract the result from 99, 999, 9999, etc. (depending on the number of digits). Finally, place both results side by side to get the final answer.

Ex. 40×99

Step-1: Subtract 1 from the number to be multiplied.

$$40 - 1 = 39$$

Step-2: Subtract the resulting value from 99.

$$99 - 39 = 60$$

Step-3: Placed both result together $\rightarrow 40 \times 99 = 3960$

Ex. 231×999

Step-1: Subtract 1 from the number to be multiplied.

$$231 - 1 = 230$$

Step-2: Subtract the resulting value from 999.

$$999 - 230 = 769$$

Step-3: Placed both result together $\rightarrow 230769$

i.e. $231 \times 999 = 230769$

DO IT LIKE PRO

N_1	N_2	$N_1 \times N_2$	N_1	N_2	$N_1 \times N_2$
45	99		2562	9999	
352	999		21212	9999	
257	999		345678	999999	

Calculation

Chris cross Method of Multiplication

2 Digit × 2 Digit

Ex.
$$\begin{array}{r} 37 \\ \times 14 \\ \hline \end{array}$$

Step I:
$$\begin{array}{r} 3 \quad 7 \\ 1 \quad 4 \end{array} = 28, \text{ we will write 8 \& 2 will be taken as carry}$$

Step II:
$$\begin{array}{r} 3 \quad 7 \\ 1 \quad 4 \end{array} = 12 + 7 + 2 = 21, \text{ we will write 1 \& 2 will be taken as carry}$$

Step III:
$$\begin{array}{r} 3 \quad 7 \\ 1 \quad 4 \end{array} = 3 + 2 = 5$$

Hence, $14 \times 37 = 518$

DO IT LIKE PRO

$\begin{array}{r} 23 \\ \times 13 \\ \hline \end{array}$	$\begin{array}{r} 34 \\ \times 24 \\ \hline \end{array}$
$\begin{array}{r} 57 \\ \times 74 \\ \hline \end{array}$	$\begin{array}{r} 56 \\ \times 68 \\ \hline \end{array}$
$\begin{array}{r} 79 \\ \times 86 \\ \hline \end{array}$	$\begin{array}{r} 69 \\ \times 83 \\ \hline \end{array}$

3 Digit × 3 Digit

Ex.
$$\begin{array}{r} 123 \\ \times 584 \\ \hline \end{array}$$

Step I:
$$\begin{array}{r} 1 \quad 2 \quad 3 \\ 5 \quad 8 \quad 4 \end{array} = 12, \text{ we will write 2 \& 1 will be taken as carry}$$

Step II:
$$\begin{array}{r} 1 \quad 2 \quad 3 \\ 5 \quad 8 \quad 4 \end{array} = 2 \times 4 + 8 \times 3 + 1 = 33 \text{ we will write 3 \& 3 will be taken as carry}$$

Step III:
$$\begin{array}{r} 1 \quad 2 \quad 3 \\ 5 \quad 8 \quad 4 \end{array} = 1 \times 4 + 5 \times 3 + 2 \times 8 + 3 = 38 \text{ we will write 8 \& 3 will be taken as carry}$$

Step IV:
$$\begin{array}{r} 1 \quad 2 \quad 3 \\ 5 \quad 8 \quad 4 \end{array} = 1 \times 8 + 2 \times 5 + 3 = 21 \text{ we will write 1 \& 2 will be taken as carry}$$

Step V:
$$\begin{array}{r} 1 \quad 2 \quad 3 \\ 5 \quad 8 \quad 4 \end{array} = 5 + 2 = 7$$

Hence, $123 \times 584 = 71832$

DO IT LIKE PRO

$\begin{array}{r} 123 \\ \times 414 \\ \hline \end{array}$	$\begin{array}{r} 153 \\ \times 461 \\ \hline \end{array}$
$\begin{array}{r} 307 \\ \times 782 \\ \hline \end{array}$	$\begin{array}{r} 513 \\ \times 694 \\ \hline \end{array}$

3 Digit × 2 Digit

This can also be done similar to 3 digit × 3 digit.

Just consider the second 3rd digit as 0

DO IT LIKE PRO

$\begin{array}{r} 537 \\ \times 049 \\ \hline \end{array}$	$\begin{array}{r} 197 \\ \times 47 \\ \hline \end{array}$
--	---

Practice Questions

$\begin{array}{r} 19 \\ \times 66 \\ \hline \end{array}$	$\begin{array}{r} 79 \\ \times 33 \\ \hline \end{array}$
$\begin{array}{r} 78 \\ \times 19 \\ \hline \end{array}$	$\begin{array}{r} 69 \\ \times 26 \\ \hline \end{array}$
$\begin{array}{r} 307 \\ \times 27 \\ \hline \end{array}$	$\begin{array}{r} 639 \\ \times 37 \\ \hline \end{array}$
$\begin{array}{r} 373 \\ \times 87 \\ \hline \end{array}$	$\begin{array}{r} 269 \\ \times 41 \\ \hline \end{array}$
$\begin{array}{r} 762 \\ \times 852 \\ \hline \end{array}$	$\begin{array}{r} 529 \\ \times 139 \\ \hline \end{array}$
$\begin{array}{r} 292 \\ \times 843 \\ \hline \end{array}$	$\begin{array}{r} 633 \\ \times 796 \\ \hline \end{array}$
$\begin{array}{r} 984 \\ \times 513 \\ \hline \end{array}$	$\begin{array}{r} 723 \\ \times 326 \\ \hline \end{array}$

Division

Division means splitting a number into equal parts.

CONCEPT-01

Division by 5ⁿ (5, 25, 125)

(A) **Divided by 5:** In this approach first, multiply the number by 2. Then, put a decimal point before the last digit.

Ex. $\frac{66}{5} = ?$

Step-1: Multiply the number by 2. $\rightarrow 66 \times 2 = 132$

Step-2: Now put a decimal before the last digit.

13.2

i.e. $\frac{66}{5} = 13.2$

Calculation

Ex. $\frac{4257}{5} = ?$

Step-1: Multiply the number by 2. $\rightarrow 4257 \times 2 = 8514$

Step-2: Now put a decimal before the last digit.

851.4

i.e. $\frac{4257}{5} = 851.4$

(B) **Divided by 25:** In this approach first, multiply the number by 4. Then, put a decimal point before the last 2 digit.

Ex. $\frac{76}{25} = ?$

Step-1: Multiply the number by 4. $\rightarrow 76 \times 4 = 304$

Step-2: Now put a decimal before the last 2 digit.

3.04

i.e. $\frac{76}{25} = 3.04$

Ex. $\frac{2481}{25} = ?$

Step-1: Multiply the number by 4. $\rightarrow 2481 \times 4 = 9924$

Step-2: Now put a decimal before the last 2 digit.

99.24

i.e. $\frac{2481}{25} = 99.24$

(C) **Divided by 125:** In this approach first, multiply the number by 8. Then, put a decimal point before the last 3 digit.

Ex. $\frac{387}{125} = ?$

Step-1: Multiply the number by 8. $\rightarrow 387 \times 8 = 3096$

Step-2: Now put a decimal before the last 3 digit.

3.096

i.e. $\frac{387}{125} = 3.096$

Ex. $\frac{54303}{125} = ?$

Step-1: Multiply the number by 8. $\rightarrow 54303 \times 8 = 434424$

Step-2: Now put a decimal before the last 3 digit.

434.424

i.e. $\frac{54303}{125} = 434.424$

DO IT LIKE PRO

N_1	N_2	$N_1 + N_2$	N_1	N_2	$N_1 + N_2$
33	5		701	25	
241	5		1407	25	
879	5		19813	125	
1128	25		8125	125	

Square Mirror

$$\begin{aligned}
 14^2 + 87^2 &= 78^2 + 41^2 \\
 15^2 + 75^2 &= 57^2 + 51^2 \\
 17^2 + 84^2 &= 48^2 + 71^2 \\
 26^2 + 97^2 &= 79^2 + 62^2 \\
 27^2 + 96^2 &= 69^2 + 72^2
 \end{aligned}$$

TABLE - 1 SQUARE

Number	Square	Number	Square	Number	Square	Number	Square
$(11)^2$	121	$(21)^2$	441	$(31)^2$	961	$(41)^2$	1681
$(12)^2$	144	$(22)^2$	484	$(32)^2$	1024	$(42)^2$	1764
$(13)^2$	169	$(23)^2$	529	$(33)^2$	1089	$(43)^2$	1849
$(14)^2$	196	$(24)^2$	576	$(34)^2$	1156	$(44)^2$	1936
$(15)^2$	225	$(25)^2$	625	$(35)^2$	1225	$(45)^2$	2025
$(16)^2$	256	$(26)^2$	676	$(36)^2$	1296	$(46)^2$	2116
$(17)^2$	289	$(27)^2$	729	$(37)^2$	1369	$(47)^2$	2209
$(18)^2$	324	$(28)^2$	784	$(38)^2$	1444	$(48)^2$	2304
$(19)^2$	361	$(29)^2$	841	$(39)^2$	1521	$(49)^2$	2401
$(20)^2$	400	$(30)^2$	900	$(40)^2$	1600	$(50)^2$	2500

TABLE - 2 CUBE

Number	Cube	Number	Cube	Number	Cube
$(1)^3$	1	$(11)^3$	1331	$(21)^3$	9261
$(2)^3$	8	$(12)^3$	1728	$(22)^3$	10648
$(3)^3$	27	$(13)^3$	2197	$(23)^3$	12167
$(4)^3$	64	$(14)^3$	2744	$(24)^3$	13824
$(5)^3$	125	$(15)^3$	3375	$(25)^3$	15625
$(6)^3$	216	$(16)^3$	4096	$(26)^3$	17576
$(7)^3$	343	$(17)^3$	4913	$(27)^3$	19683
$(8)^3$	512	$(18)^3$	5832	$(28)^3$	21952
$(9)^3$	729	$(19)^3$	6859	$(29)^3$	24389
$(10)^3$	1000	$(20)^3$	8000	$(30)^3$	27000

TABLE - 3 SQUARE ROOT

Number	Square Root	Number	Square Root
1	1	6	2.449
2	1.414	7	2.646
3	1.732	8	2.828
4	2	9	3
5	2.236	10	3.162

TABLE - 4

Expression	Result	Expression	Result	Expression	Result
2^1	2	3^1	3	4^1	4
2^2	4	3^2	9	4^2	16
2^3	8	3^3	27	4^3	64
2^4	16	3^4	81	4^4	256
2^5	32	3^5	243	4^5	1024
2^6	64	3^6	729	5^1	5
2^7	128	6^1	6	5^2	25
2^8	256	6^2	36	5^3	125
2^9	512	6^3	216	5^4	625
2^{10}	1024	6^4	1296	5^5	3125

TABLE - 5

	011	012	013	014	015	016	017	018	019	020
0×1	11	12	13	14	15	16	17	18	19	20
0×2	22	24	26	28	30	32	34	36	38	40
0×3	33	36	39	42	45	48	51	54	57	60
0×4	44	48	52	56	60	64	68	72	76	80
0×5	55	60	65	70	75	80	85	90	95	100
0×6	66	72	78	84	90	96	102	108	114	120
0×7	77	84	91	98	105	112	119	126	133	140
0×8	88	96	104	112	120	128	136	144	152	160
0×9	99	108	117	126	135	144	153	162	171	180
×10	110	120	130	140	150	160	170	180	190	200
×11	121	132	143	154	165	176	187	198	209	220
×12	132	144	156	168	180	192	204	216	228	240
×13	143	156	169	182	195	208	221	234	247	260
×14	154	168	182	196	210	224	238	252	266	280
×15	165	180	195	210	225	240	255	270	285	300
×16	176	192	208	224	240	256	272	288	304	320
×17	187	204	221	238	255	272	289	306	323	340
×18	198	216	234	252	270	288	306	324	342	360
×19	209	228	247	266	285	304	323	342	361	380
×20	220	240	260	280	300	320	340	360	380	400



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